



Chapter 5 — Passive and Active Current Mirrors

Design of Analog CMOS
Integrated circuits - Behzad
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CHAPTER OVERVIEW

Concepts covered -

1. Basic Current Mirrors
2. Cascode Current Mirrors
3. Active Current Mirrors
 - a. Large Signal Analysis
 - b. Small Signal Analysis
 - c. Common - Mode Properties



Additional Points

- **Cadence Virtuoso** is used for schematic design and simulation of the circuits.
- The **GPDK180 library** is used for all circuit simulations.
- Initially, some simulations were performed using **VDD = 3 V** as specified in the textbook.
- Later, based on the recommended operating conditions of the **GPDK180 library**, the supply voltage was changed to **VDD = 1.8 V** for subsequent simulations to ensure compatibility with the technology model.
- The change in supply voltage may result in slight differences between the textbook results and simulation results.

Problem 5.2

5.2. Consider the circuit of Fig. 5.7. Assuming I_{REF} is *ideal*, sketch I_{out} versus V_{DD} as V_{DD} varies from 0 to 3 V.

Diagram -

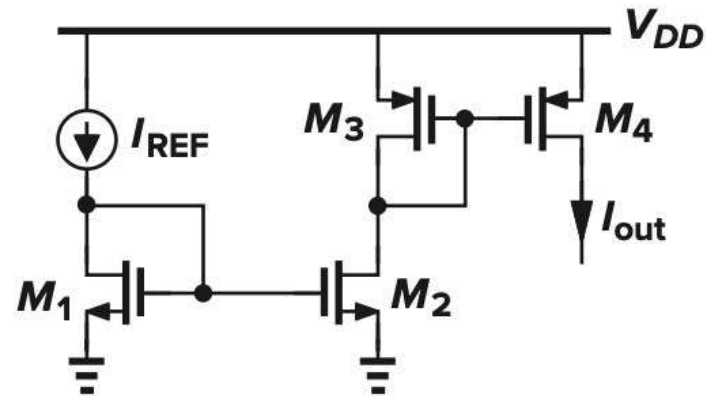
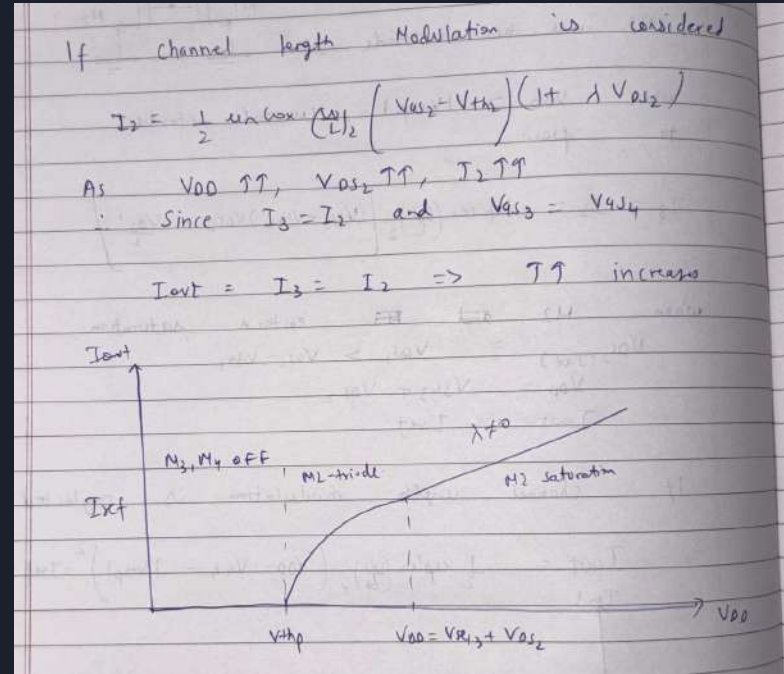
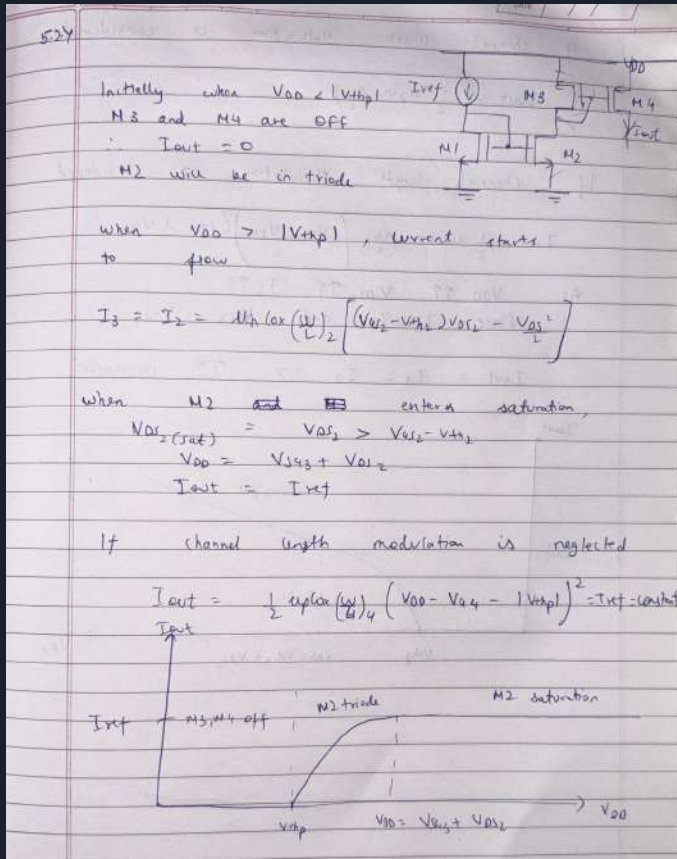
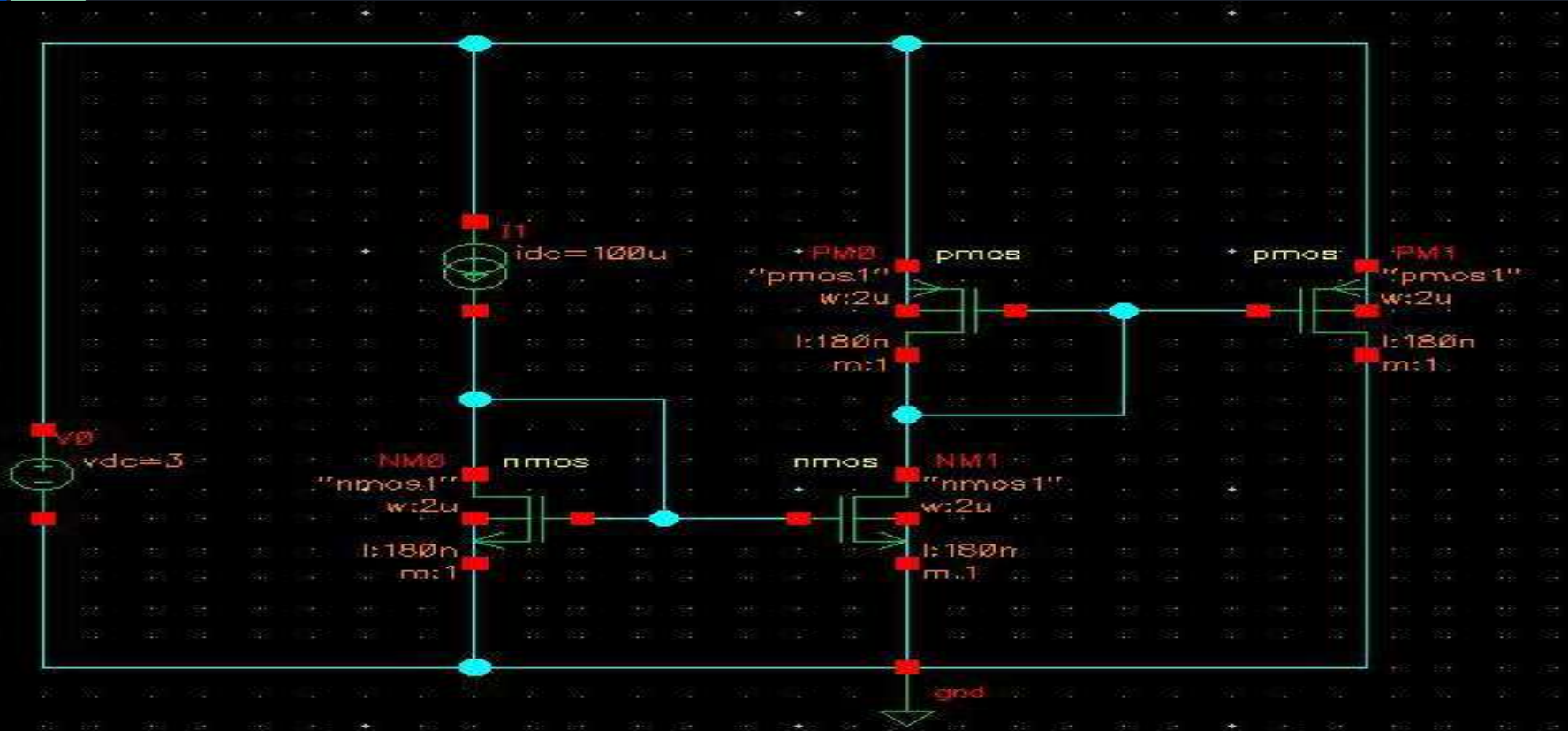


Figure 5.7

Handwritten Solution



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Iout

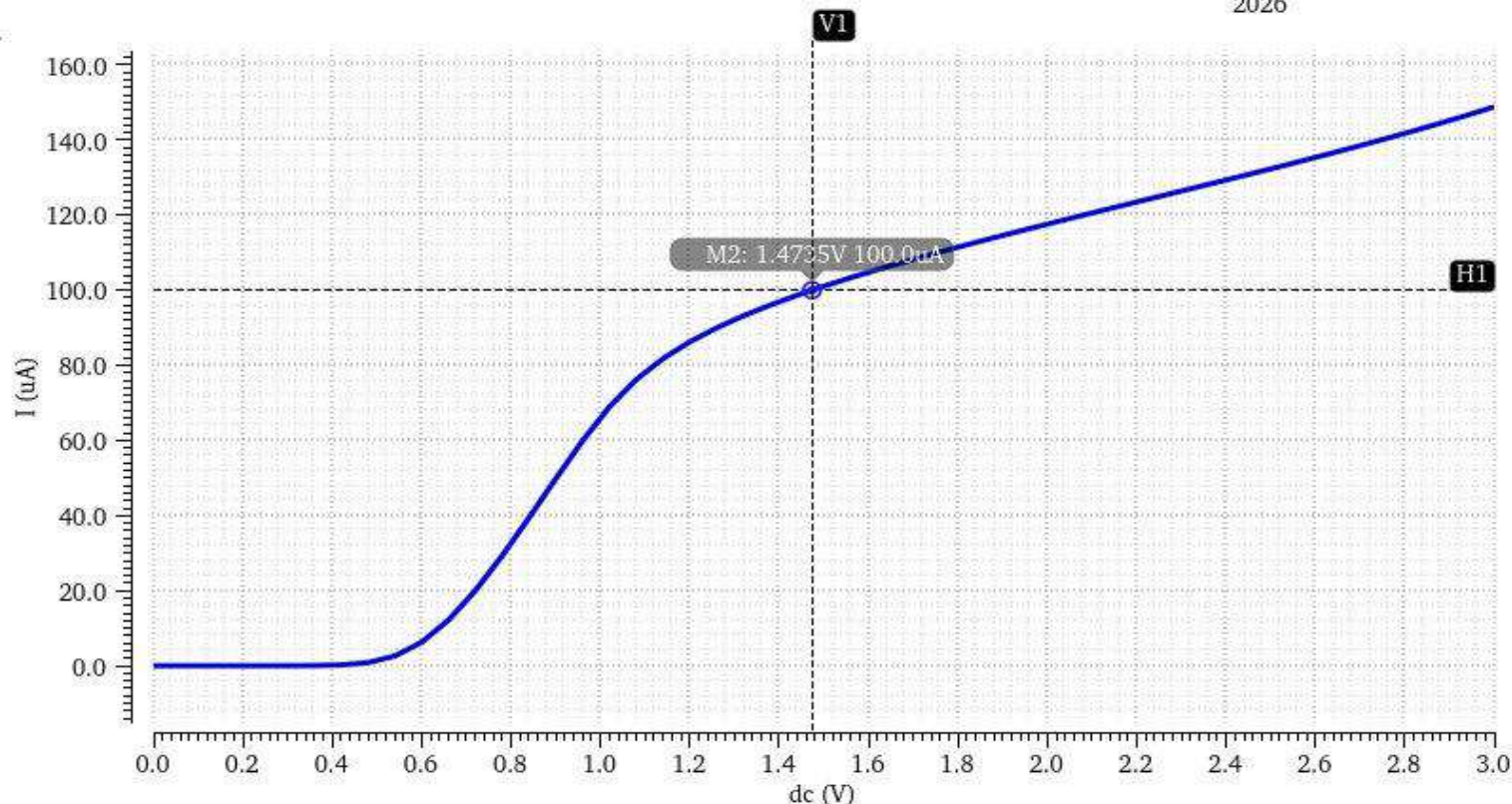
DC Response

Tue Jun 23 16:23:53

1

Name ... V1

■ /PM1/S ... 0.0uA



Problem 5.4

5.4. In the circuit of Fig. 5.11, sketch V_{out} versus V_{DD} as V_{DD} varies from 0 to 3 V.

Diagram -

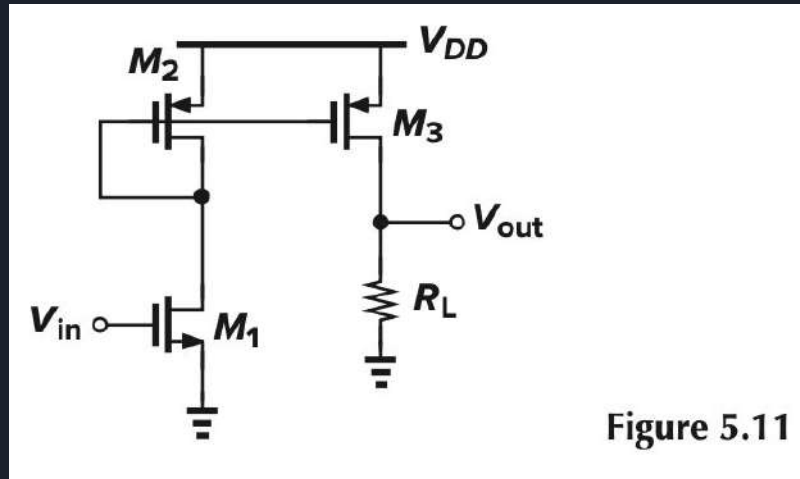


Figure 5.11

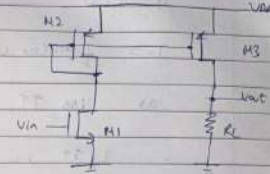
Handwritten Solution

5.47

Initially when $V_{DD} < |V_{thp}|$

M2 and M3 are OFF

$V_{out} = 0$



Once $V_{DD} > |V_{thp}|$, Assuming V_{in} is fixed

M1 first enters triode region

$$I_1 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 \left[(V_{in} - V_{thn}) V_{ds1} - \frac{V_{ds1}^2}{2} \right]$$

$$I_3 = I_2 = I_1$$

$$V_{out} = I_2 R_L = I_1 R_L$$

as V_{DD} increases $V_{ds1} \uparrow \uparrow$, $I_1 \uparrow \uparrow$, $I_2 \uparrow \uparrow$, $V_{out} \uparrow \uparrow$

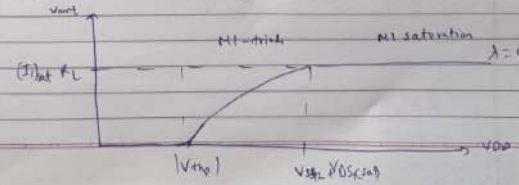
$V_{DD} = V_{ds2} + V_{ds1}(V_{out})$

After $V_{ds1} = V_{ds1}(V_{out})$, M1 enters saturation region

Considering $\lambda = 0$

$$I_1 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 \left[(V_{in} - V_{thn})^2 \right]$$

$I_2 = I_1 = \text{constant}$, $V_{out} = I_2 R_L = \text{constant}$

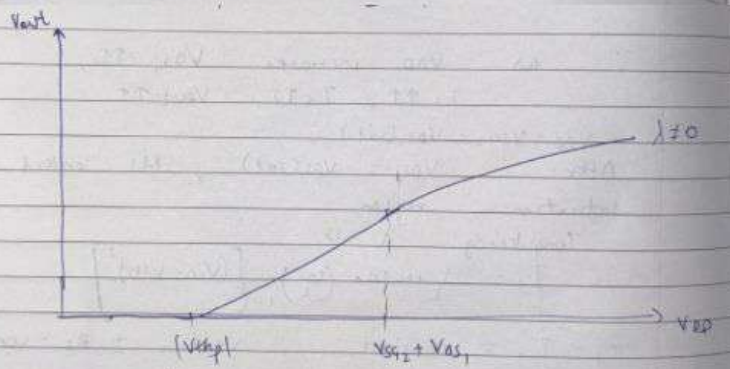


If $\lambda \neq 0$

$$I_1 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{in} - V_{thn})^2 (1 + \lambda V_{ds1})$$

as $V_{DD} \uparrow \uparrow$, $V_{ds1} \uparrow \uparrow$, $I_1 \uparrow \uparrow$

$I_2 \uparrow \uparrow$, $\therefore V_{out} \uparrow \uparrow$

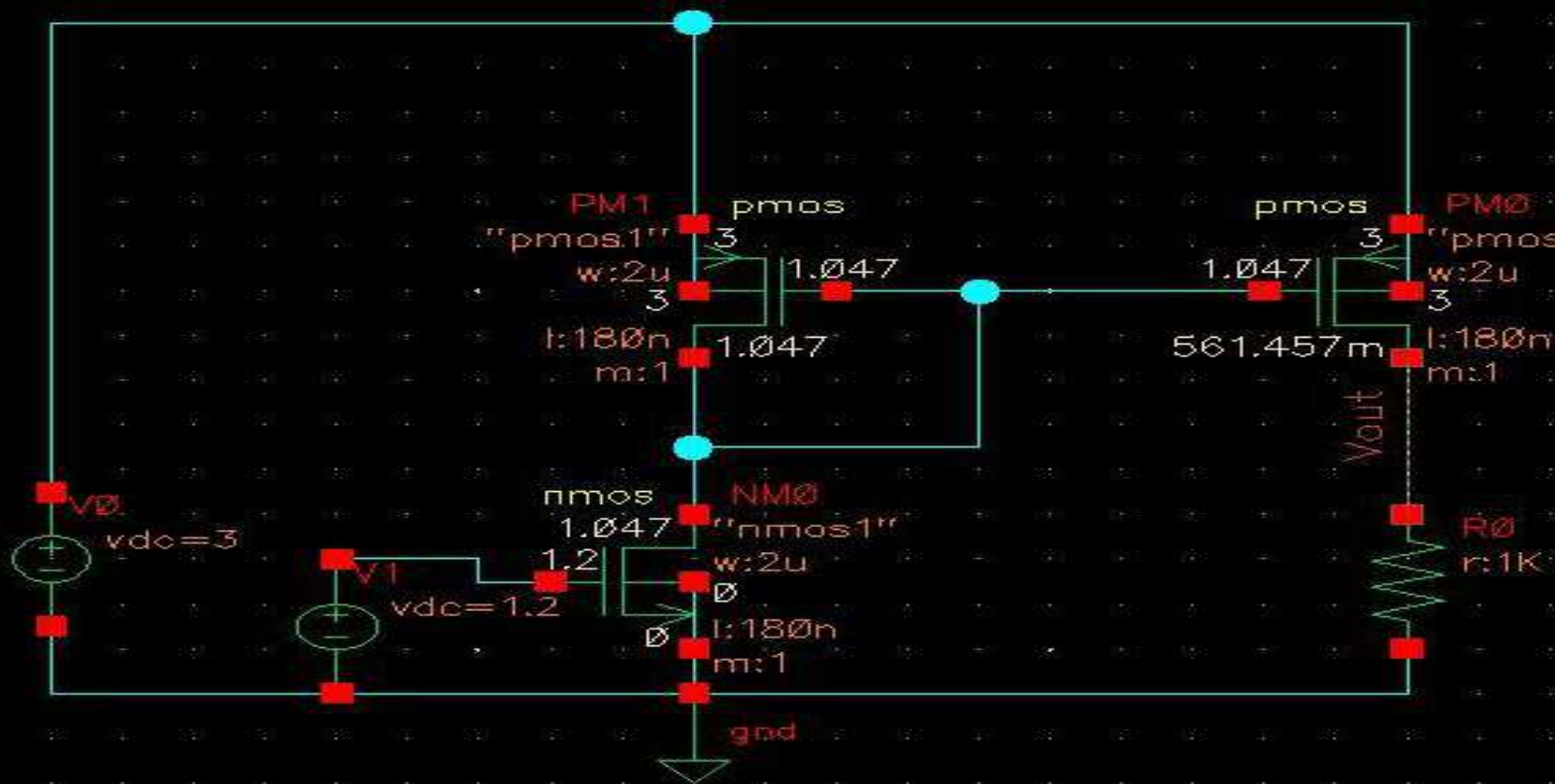




Why V_{DS1} increases as V_{DD} increases ?

- Since M1 and M2 are connected in series, the same current flows through both transistors.
- Also by KVL, $V_{DD} = V_{SG2} + V_{DS1}$
- As M2 is diode-connected, its gate-source voltage adjusts to accommodate the required current with a relatively small voltage variation.
- Therefore, when V_{DD} increases, the additional voltage mainly appears across M1, causing V_{DS1} to increase.

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vout

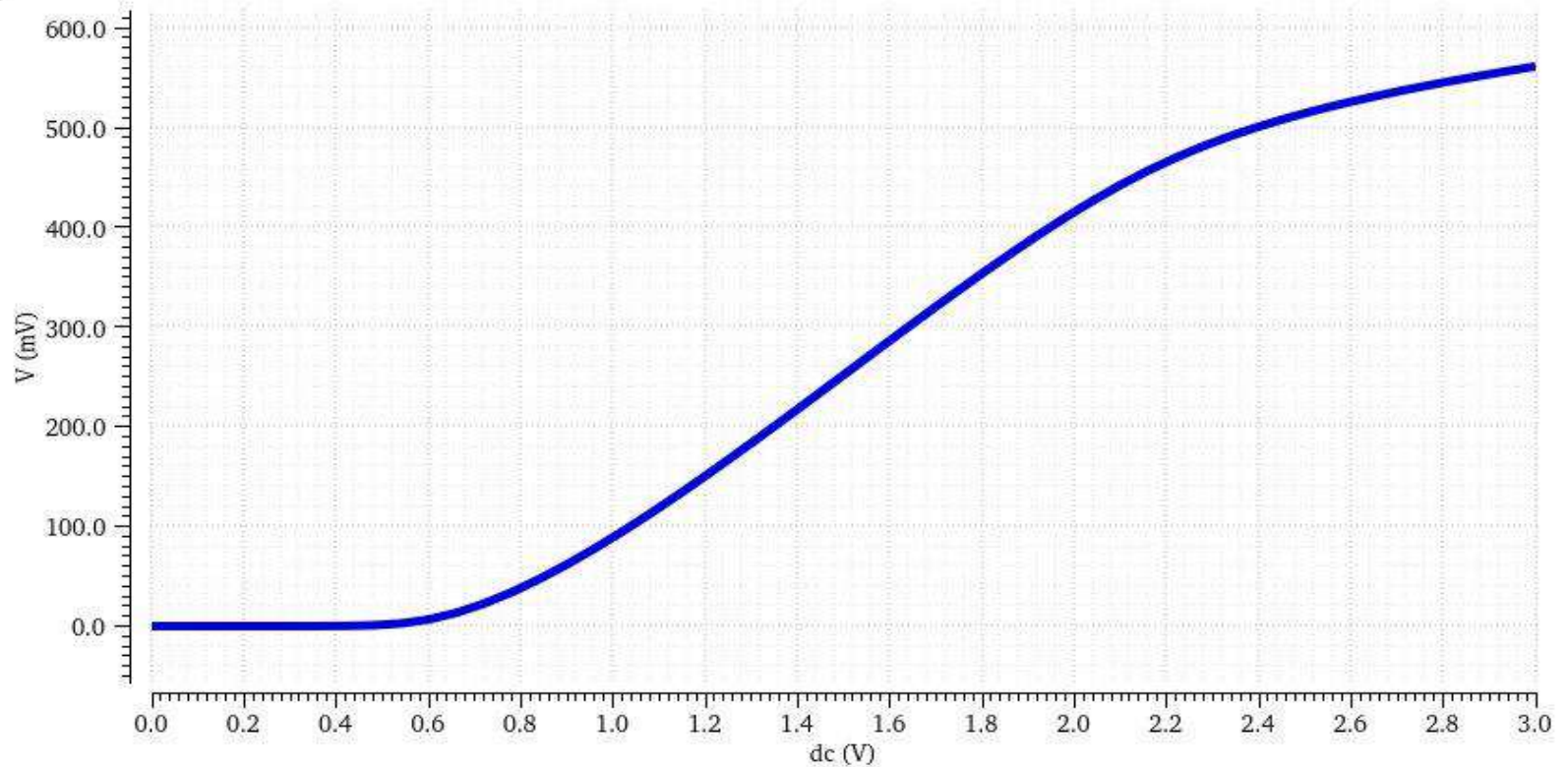
DC Response

Tue Jun 23 16:12:30
2026

1

Name Vis

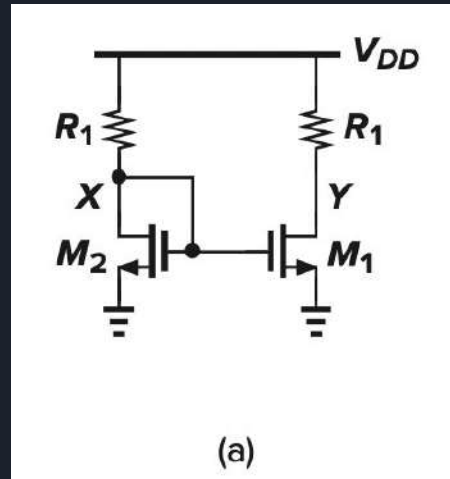
Vout



Problem 5.9-a

5.9. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.52. Assume the transistors in each circuit are identical.

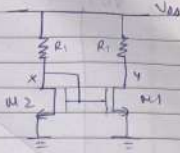
Diagram -



Handwritten Solution

5.9.4a

When $V_{DD} < V_{th}$, M1 and M2 are OFF, $\therefore$ no current flows



$$\therefore V_x = V_y = V_{DD}$$

Once $V_{DD} > V_{th}$, current starts to flow, since $V_{GS1} = V_{GS2}$, $I_1 = I_2$

$$V_{GS2} = V_{DD} - I_2 R_1$$

$$V_{GS2} = V_{DD} - \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{GS2} - V_{th})^2 R_1$$

$$\therefore \text{as } V_{DD} \uparrow, V_{GS2} \uparrow, I_2 = I_1 \uparrow$$

$\therefore$ Since currents are same, drop across resistors would also be equal

$$\therefore V_x = V_y \text{ increase}$$

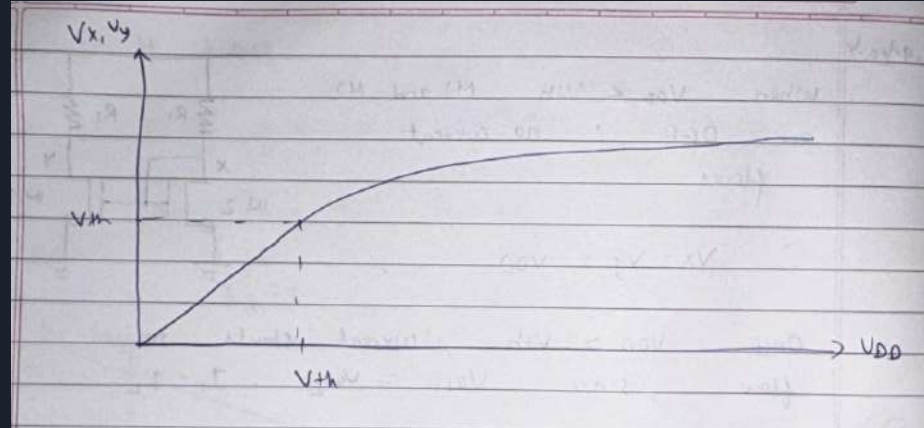
As V_{DD} increases, it tries to force more current into the branch

$$\therefore V_x = V_{GS2} = V_{GS1}$$

if current through M2 has to increase V_{GS} has to increase relatively less as $I \propto (V_{GS} - V_{th})^2$

$\therefore V_x$ increases slowly

and as $V_{GS1} = V_{GS2}$, I_1 increases and thus drop across R_1 in branch 1 and branch 2 is equal $\therefore V_y = V_x$

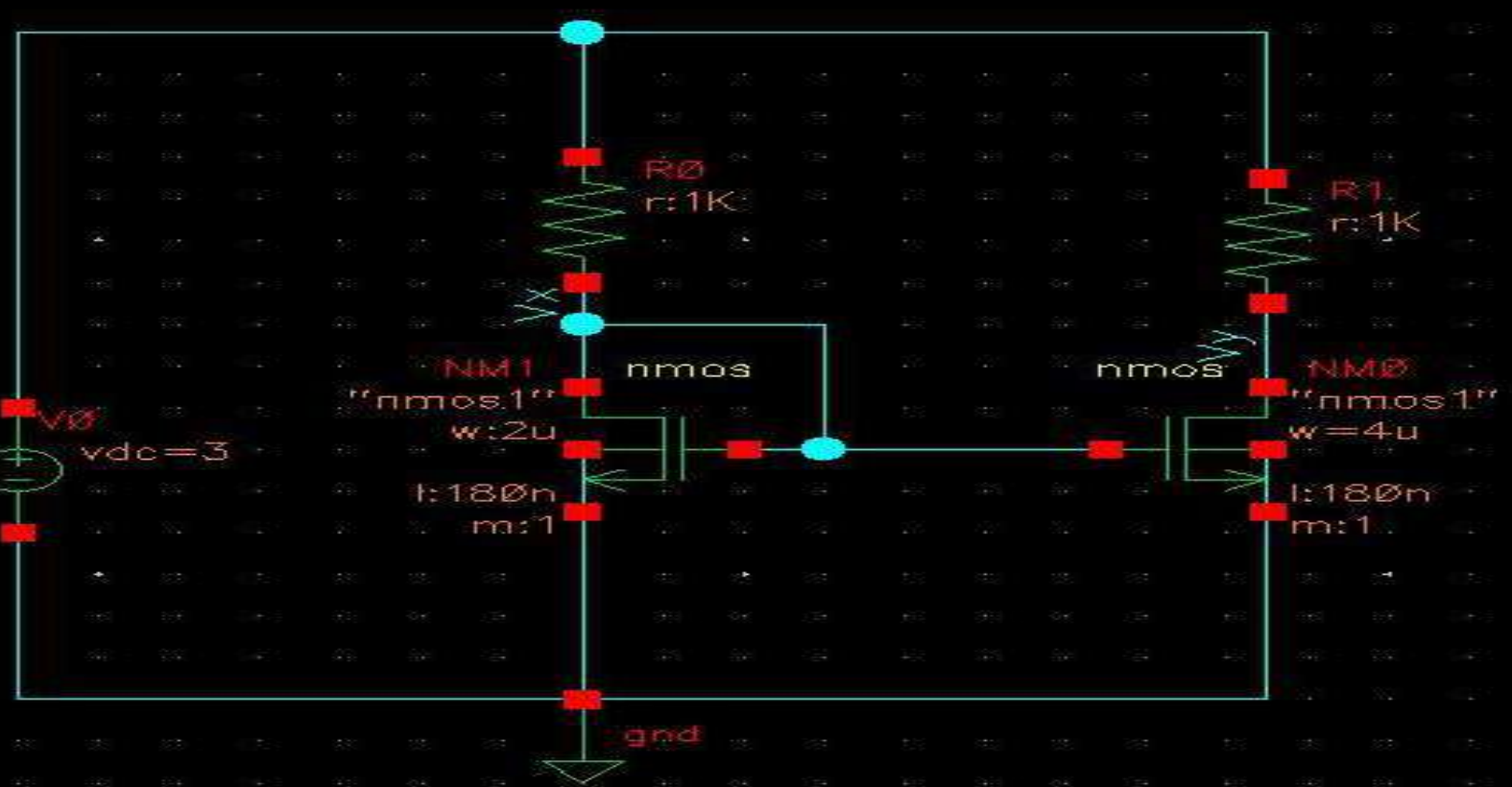




Observations

- As V_{DD} increases, V_X increases slowly because the diode-connected transistor M_2 requires only a small change in V_{GS2} to support the increased current.
- Due to the symmetry of the circuit, V_Y follows the same behavior as V_X , maintaining $V_Y = V_X$.
- Most of the increase in V_{DD} appears across the resistors, resulting in an increase in the circuit current.
- Since $V_{DS1} = V_{GS1} - V_{th}$, transistor M_1 remains in the saturation region throughout the operation.

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

DC Response

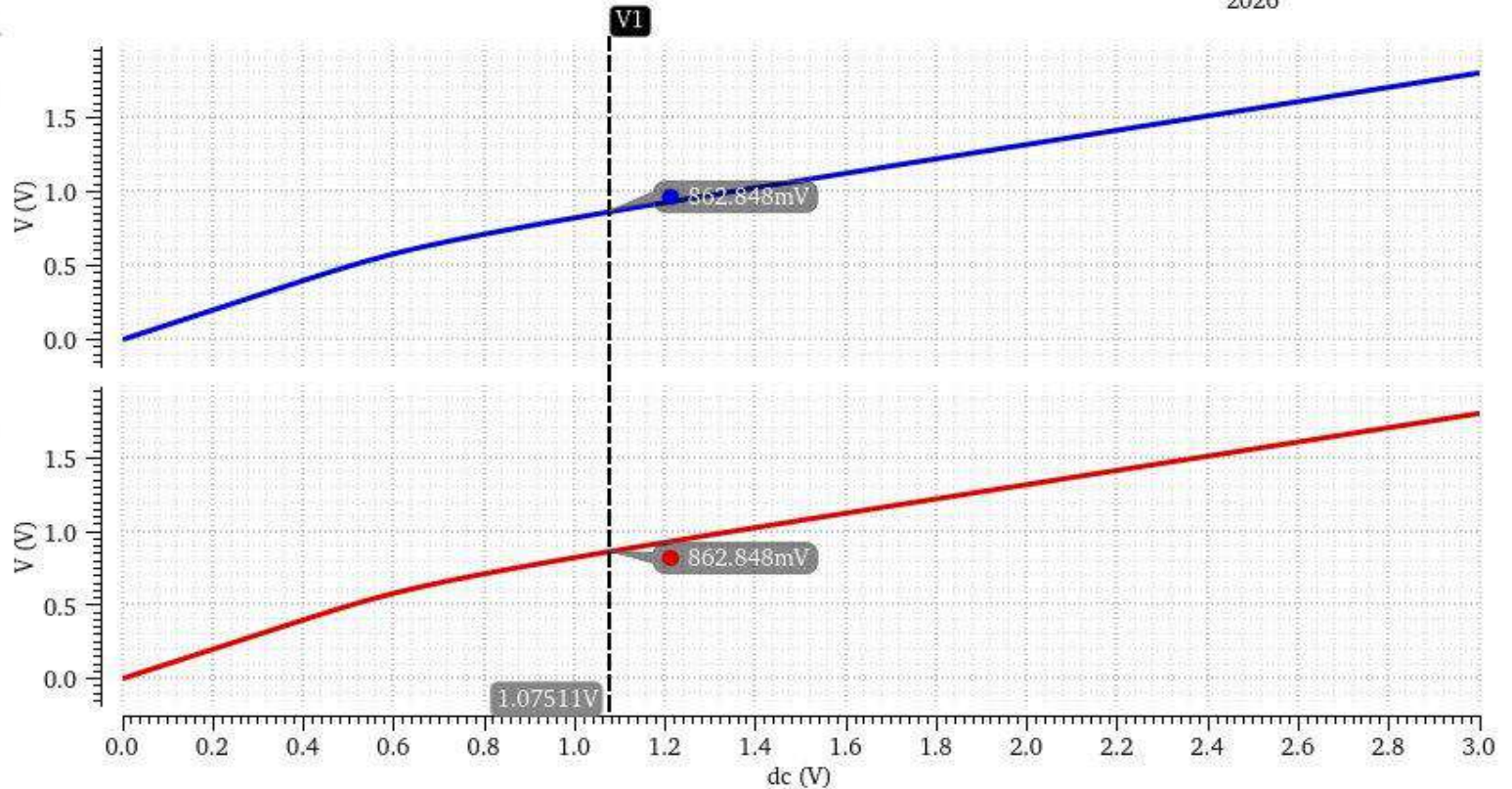
Tue Jun 23 16:35:43
2026

1

Name V1

☒ /Vy ☒ ...848mV

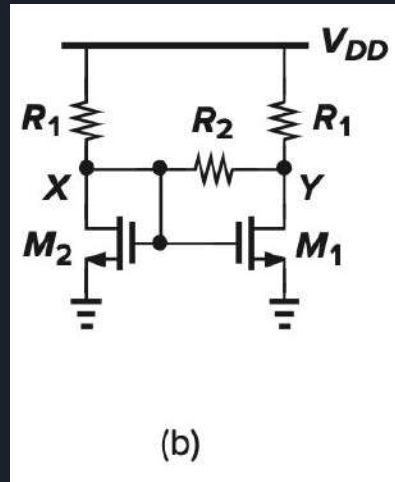
☒ /Vx ☒ ...848mV



Problem 5.9-b

5.9. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.52. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

59.66V

As we saw in the previous question

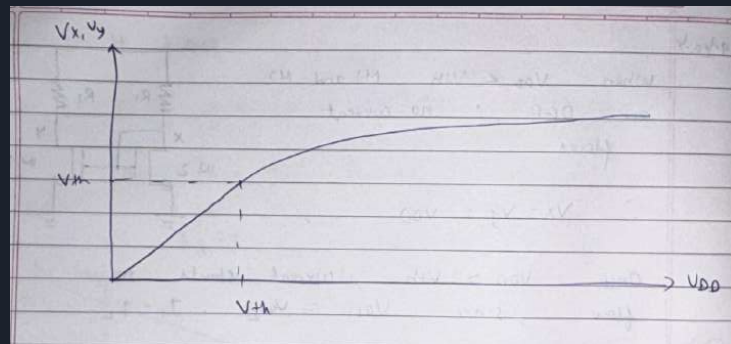
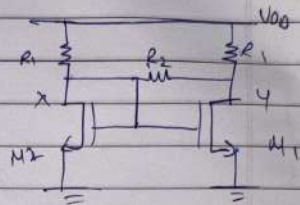
$$\therefore V_{GS2} = V_{GS1}$$

$$I_2 = I_1$$

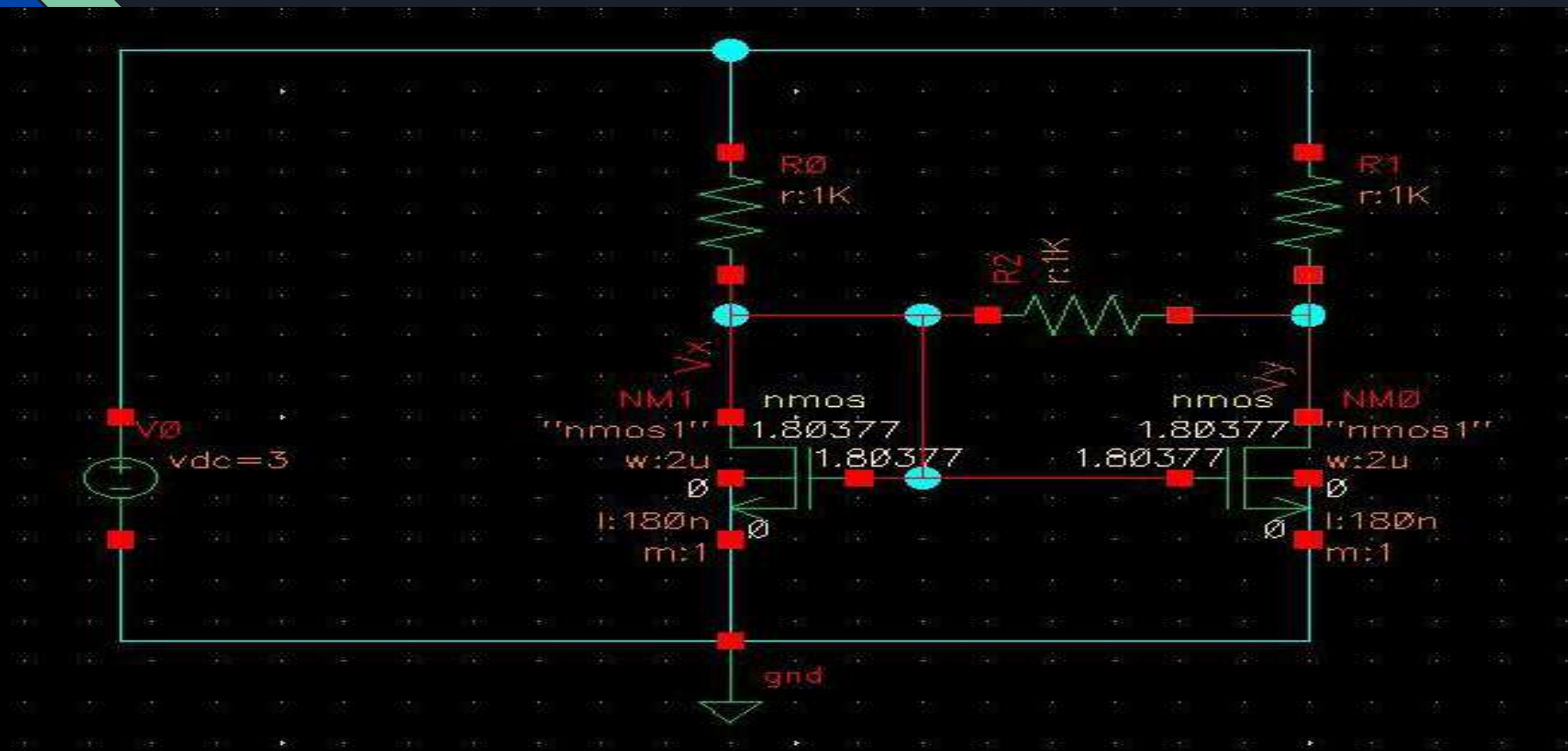
$$\therefore V_X = V_Y$$

$\therefore$ no current flows through R_2

and plot of V_X and V_Y v/s V_{DD} is exactly the same as previous question



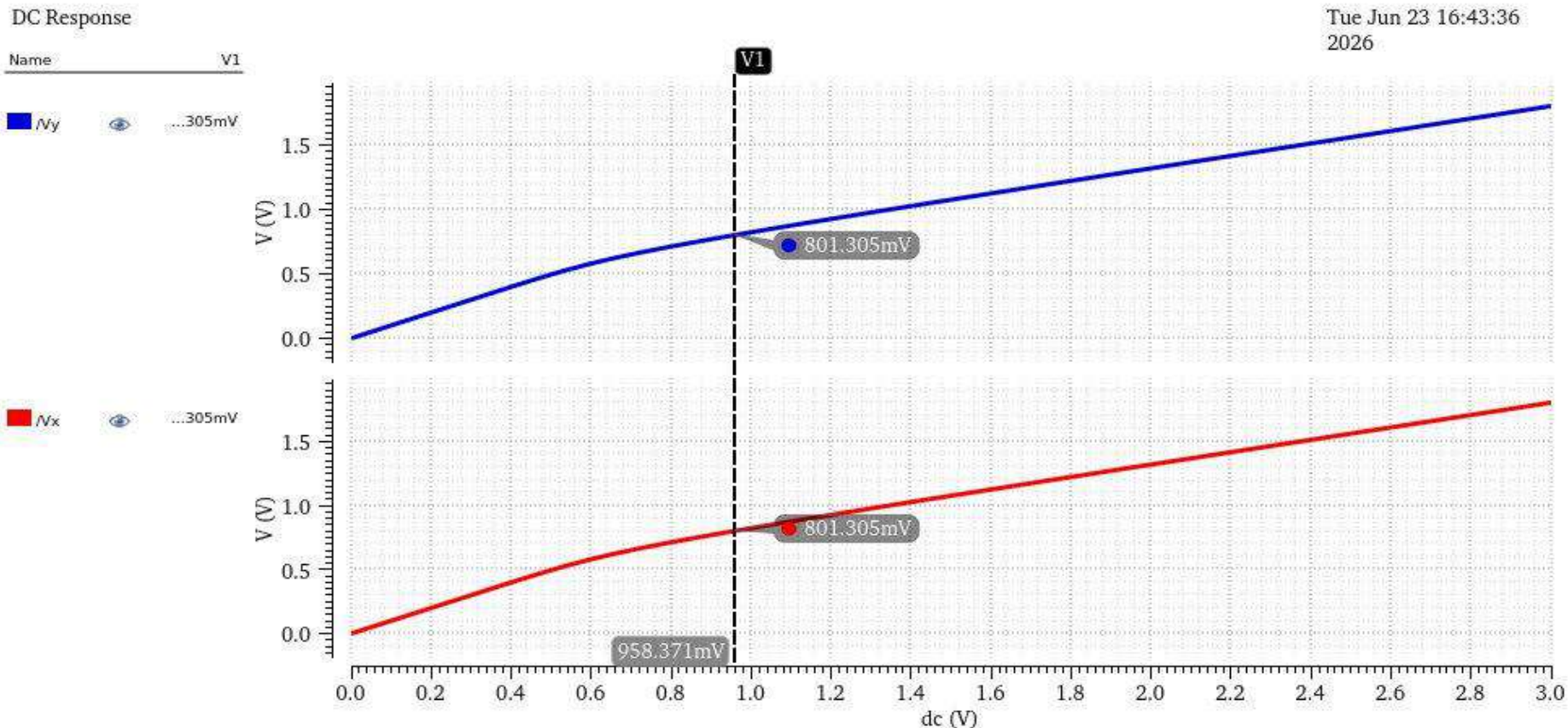
Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

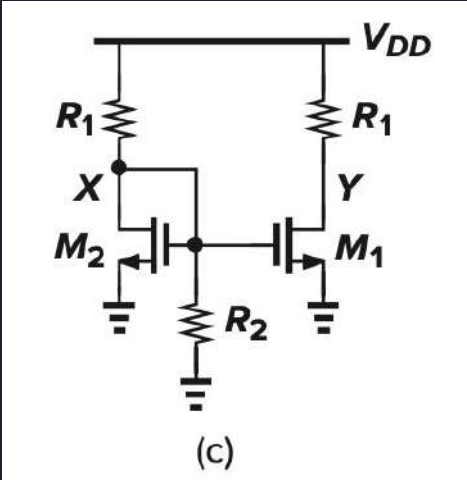
Tue Jun 23 16:43:36
2026

1



5.9. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.52. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

594414

Before $V_{DD} = V_{th}$

Both M_1 and M_2 are off,

$\therefore V_x = V_{DD}$

$$V_x = \frac{R_2}{R_1 + R_2} V_{DD}$$

Once $V_{DD} > V_{th}$

$I_{D1} = I_{D2}$, but at node X

$$I_{R12} = I_{D2} + I_{R2}$$

$\therefore (I_{R12}) > I_{D1}$

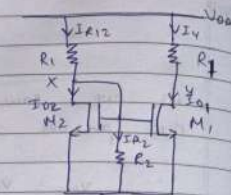
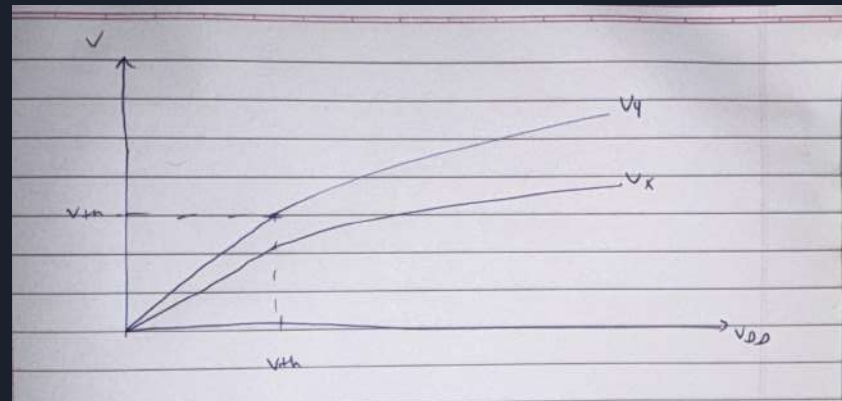
Left branch Right branch

$\therefore$ Voltage Drop in left branch is more

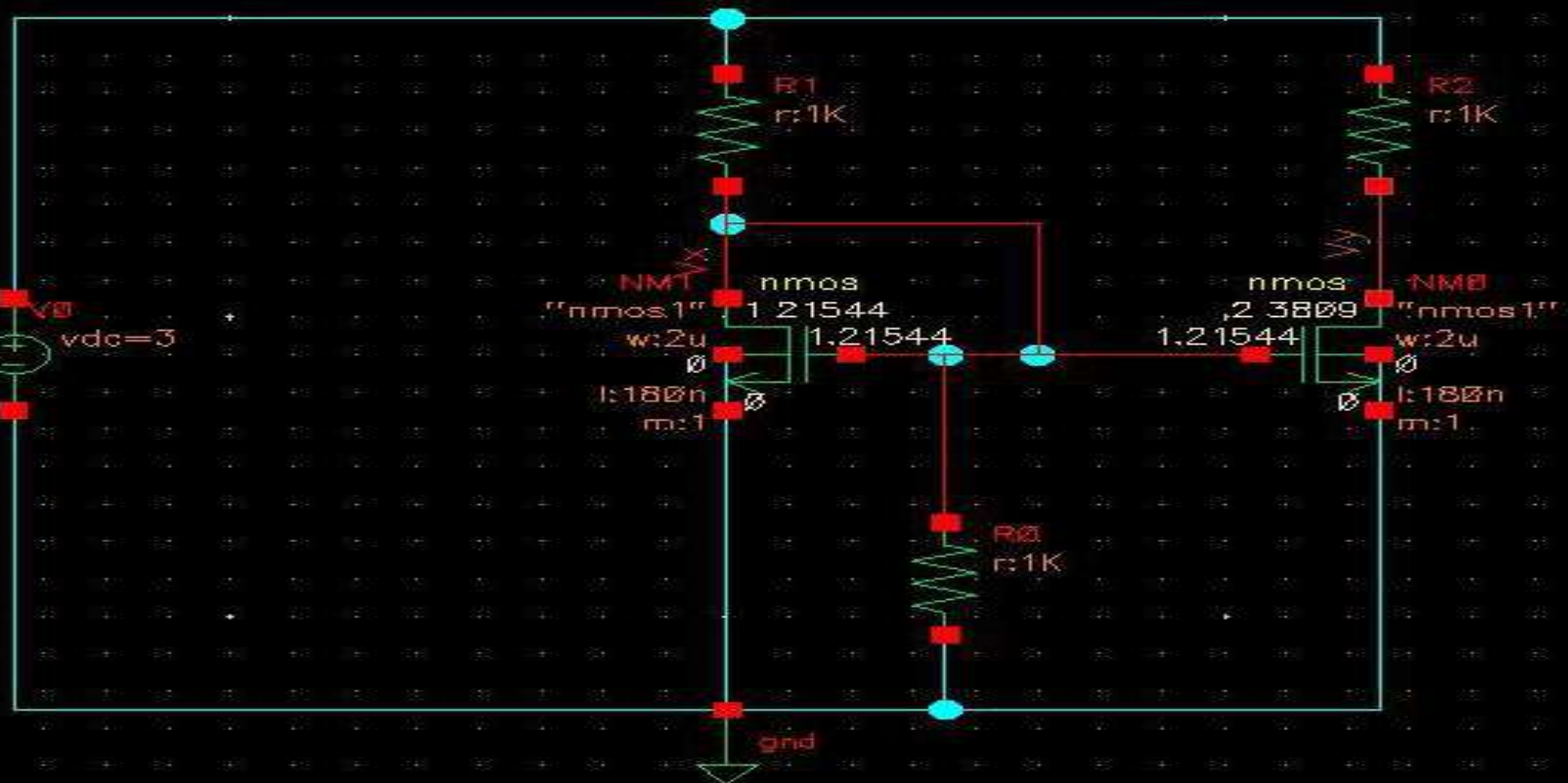
$\therefore V_x < V_y$

But as $V_{DD} \uparrow$, I_{D1} and $I_{D2} \uparrow$

$\therefore V_x$ and $V_y \uparrow$

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

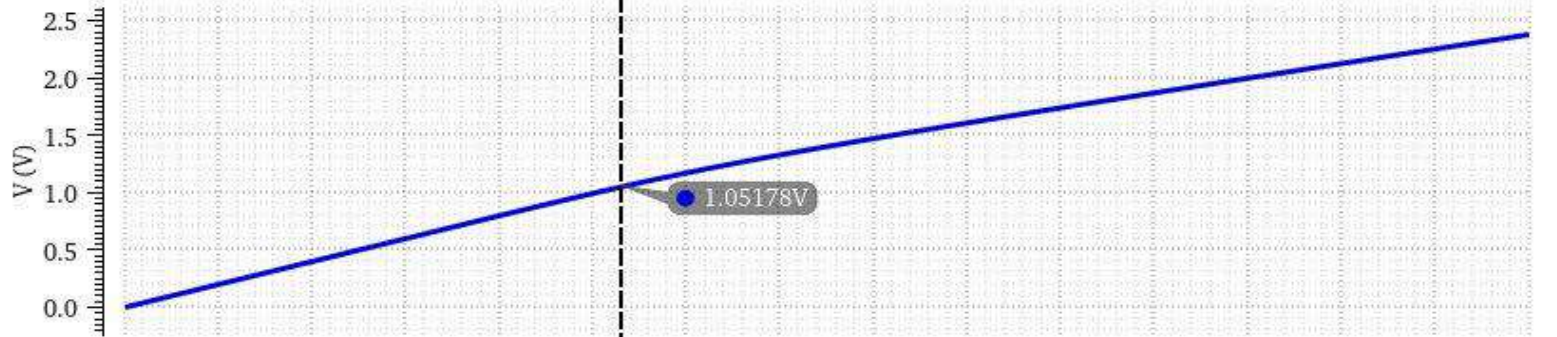
DC Response

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2026

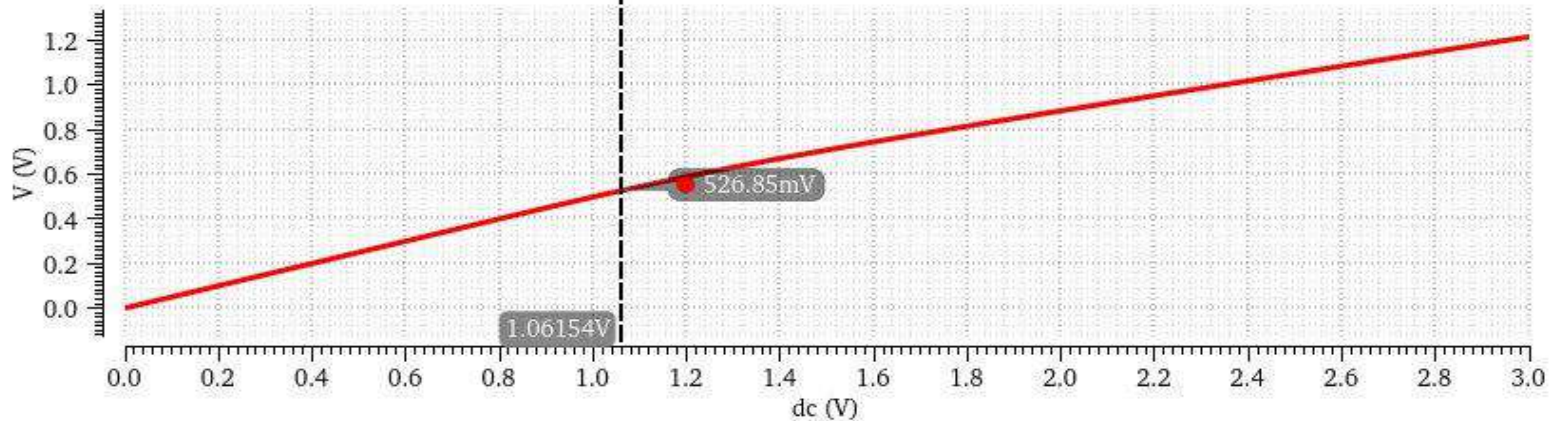
1

Name V1

V_y ...5178V



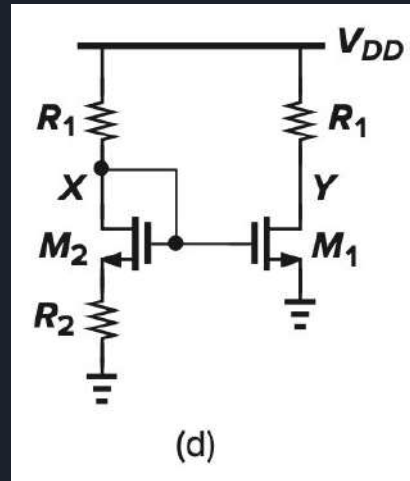
V_x ...85mV



Problem 5.9-d

5.9. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.52. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

5.9.4d

Initially when $V_{DD} < V_{th}$
 M_1 and M_2 are OFF

$$\therefore V_x = V_y = V_{DD}$$

Once $V_{DD} > V_{th}$

$V_{GS2} < V_{GS1}$ due to K_2

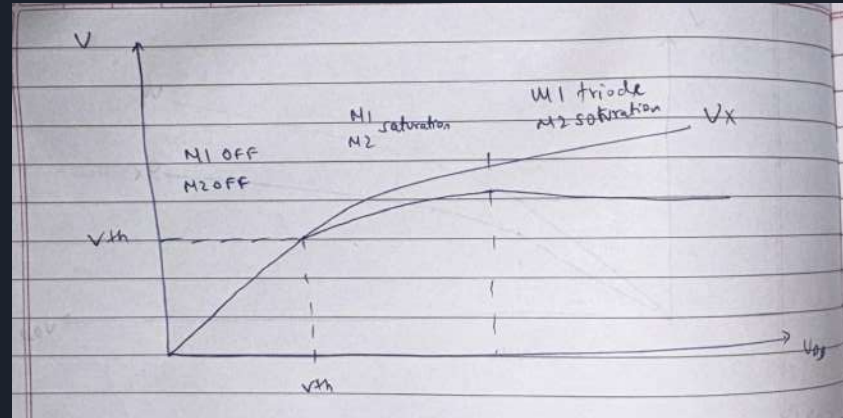
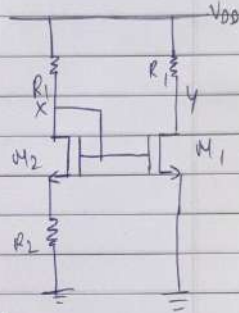
$\therefore I_2$ is less than I_1

$$\therefore V_x > V_y$$

As $V_{DD} \uparrow$, I_1 and I_2 increase

$\therefore V_x$ and V_y increase

As $V_{DD} \uparrow$, at some point V_{GS1} becomes large enough so that M_1 enters triode region



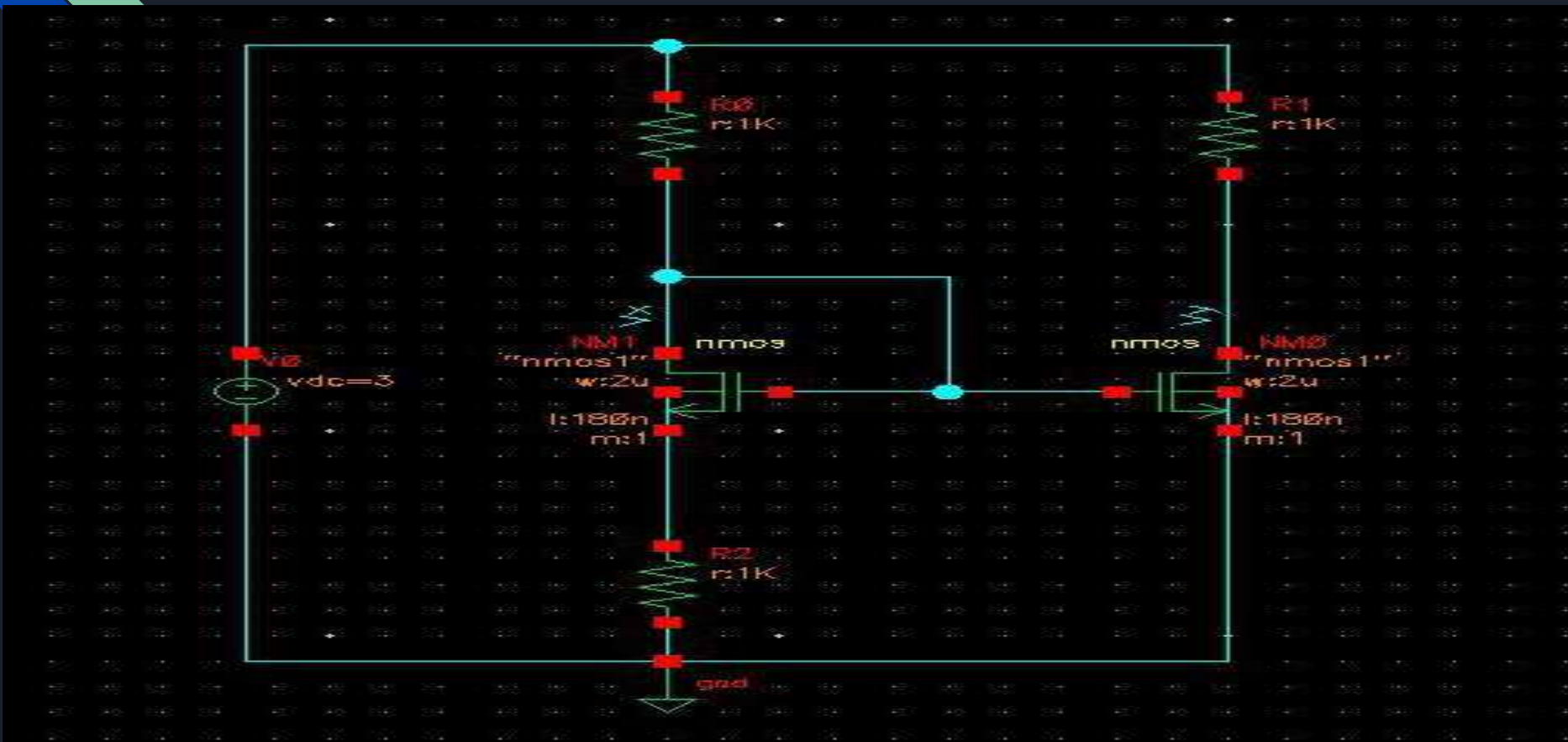


Observations

When M1 enters the triode region:

- M1 no longer behaves as a constant current source and starts acting like a voltage-controlled resistor, changing the behavior of V_Y .
- As V_{DD} increases, V_X also increases, increasing the gate voltage of M1 and reducing its channel resistance R_{on1} .
- The decreasing R_{on1} strengthens the pull-down effect, opposing the natural rise of V_Y due to increasing V_{DD} .
- As a result, the increase in V_Y becomes very slow, causing the curve to flatten and approach a near-constant value.

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

Tue Jun 23 17:08:01
2026

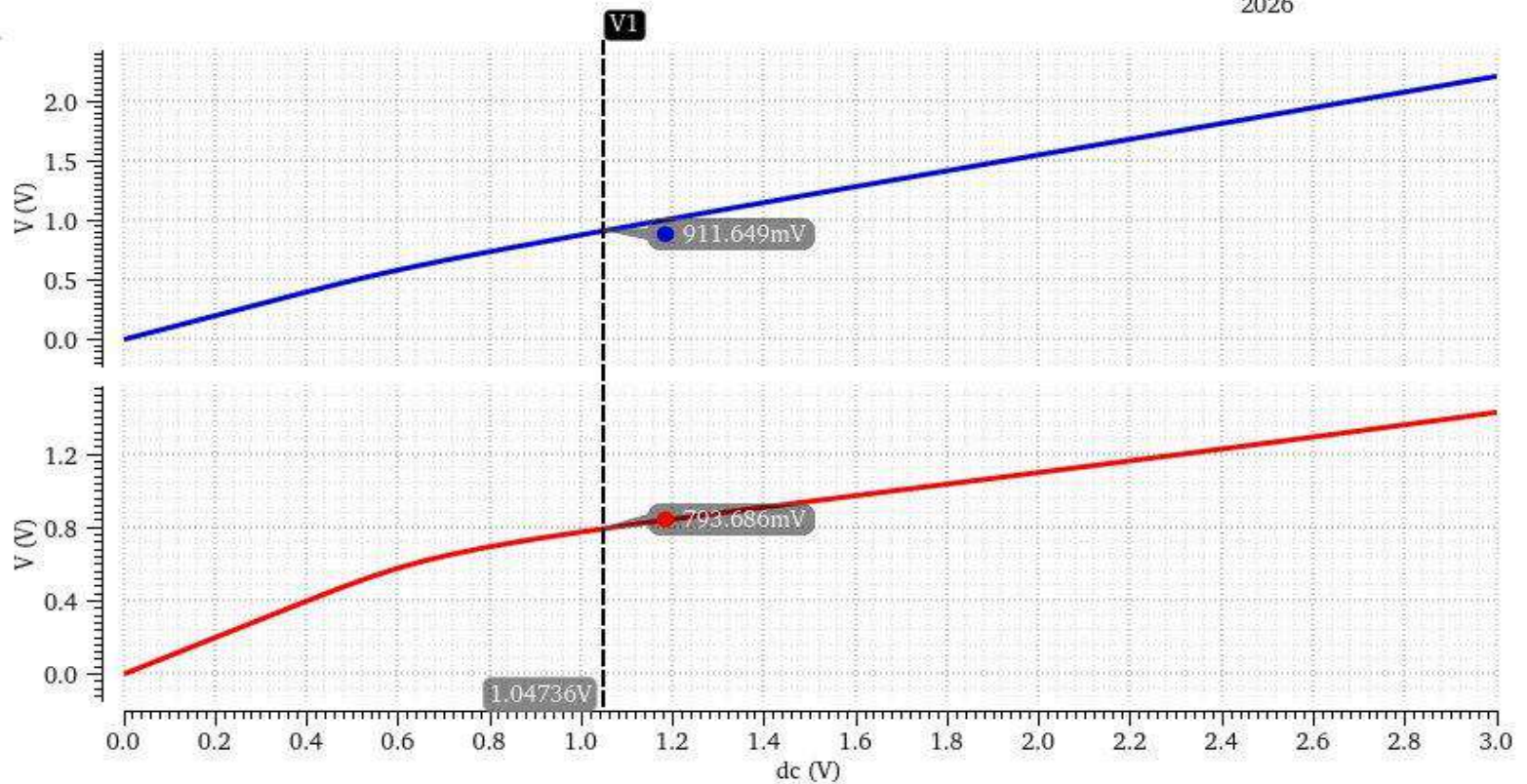
1

DC Response

Name V1

Vx ...649mV

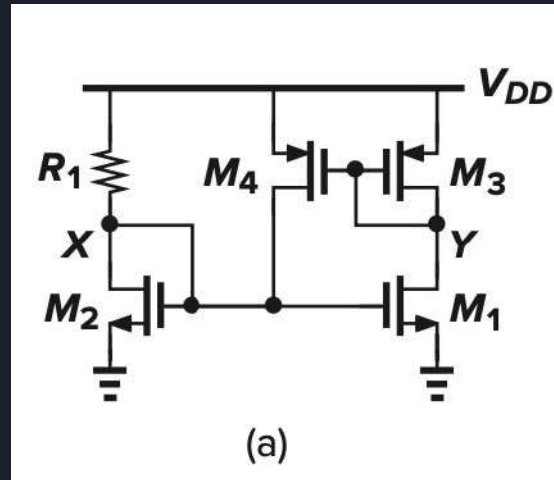
Vy ...686mV



Problem 5.10-a

5.10. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.53. Assume the transistors in each circuit are identical.

Diagram -



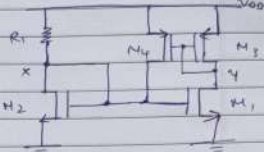
Handwritten Solution

5.10 Ya.

When $V_{DD} < V_{th1,2}$

For NMOS to turn on
 $V_{GS} > V_{thn}$
 $V_G - V_S > V_{thn}$
 $\downarrow$
 grounded
 $V_G > V_{thn}$, but V_G can be between V_{DD} and 0, it can't be more than V_{DD}
 Hence $M1$ and $M2$ are OFF
 $V_X = V_{DD}$

Similarly for PMOS
 $V_{SG} > |V_{thp}|$
 $V_{DD} - V_G > |V_{thp}|$
 Since $M1$ is off, $V_Y = V_{G3,4}$ is floating
 between 0 and V_{DD}
 $\therefore$ even in best case if $V_{G3,4} = 0$
 $V_{DD} > |V_{thp}|$, $M3$ and $M4$ are OFF



Once $V_{DD} > V_{thn}$, initially on NMOS turn on,
 $\therefore I_{M2} = I_{R1}$

I_{R1} increases as V_{DD} increases
 $\therefore V_{GS2}$ has to increase

PMOS turns ON almost immediately and then
 $I_{M2} = I_{R1} + I_{M4}$

$\therefore$ As V_{DD} increases, $I_{R1} \uparrow \uparrow$ and $I_{M4} \uparrow \uparrow$
 I_{M2} increase even more
 V_{GS2} increases $V_X \uparrow \uparrow$

as V_X increases $V_{GS1} \uparrow \uparrow$, which turns on $M1$ strongly and tries to pull V_Y to ground, also since $V_{DD} \uparrow \uparrow$, I_{M2} tries to increase which also tries to reduce V_Y around that point

Handwritten Solution

Also, V_x is high, V_y is low

$\therefore M_1 = V_{GS1} \leq V_{GS1} - V_{th1} \rightarrow \text{triode}$ } throughout the sweep

$\therefore M_2 = V_{GS2} < V_{GS2} - |V_{th2}| \rightarrow \text{triode}$

$\therefore$ There is a +ve feedback loop

As $V_{DD} \uparrow$, $I_{R1} + I_{M1}$ increase, $\therefore V_x \uparrow$

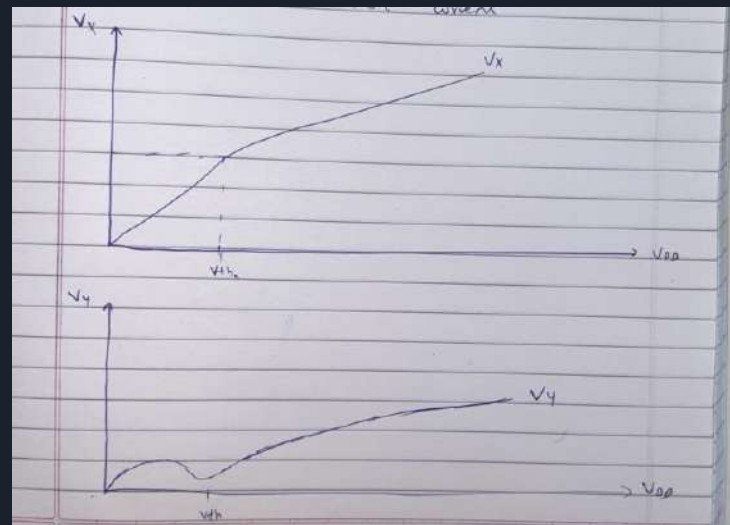
As $V_x \uparrow$, $I_{D2} = I_{M1} \uparrow$, $\therefore V_y$ decreases

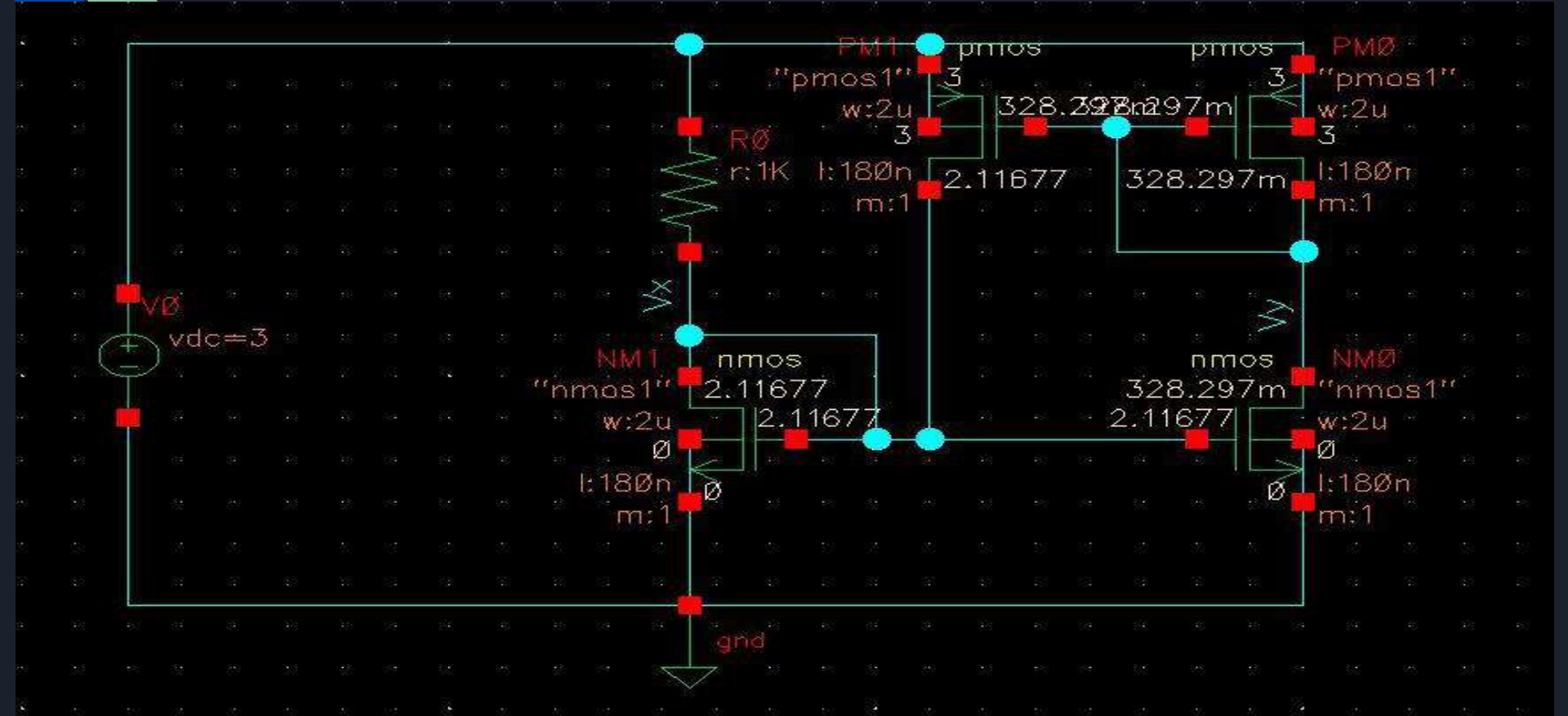
As $V_y \downarrow$, M_4 turns on harder (V_{GS4} increases)

$\therefore I_{M4}$ increases $\therefore I_{M2} = I_{R1} + I_{M4}$ increases, resulting in increase in V_x

And since M_1 is in triode region,

I_{M1} depends on $V_{GS1} = V_y$, As $V_{DD} \uparrow$ current through M_1 increases $\therefore V_y$ must rise to allow that current





Cadence ADE-L DC Simulation for Vx and Vy

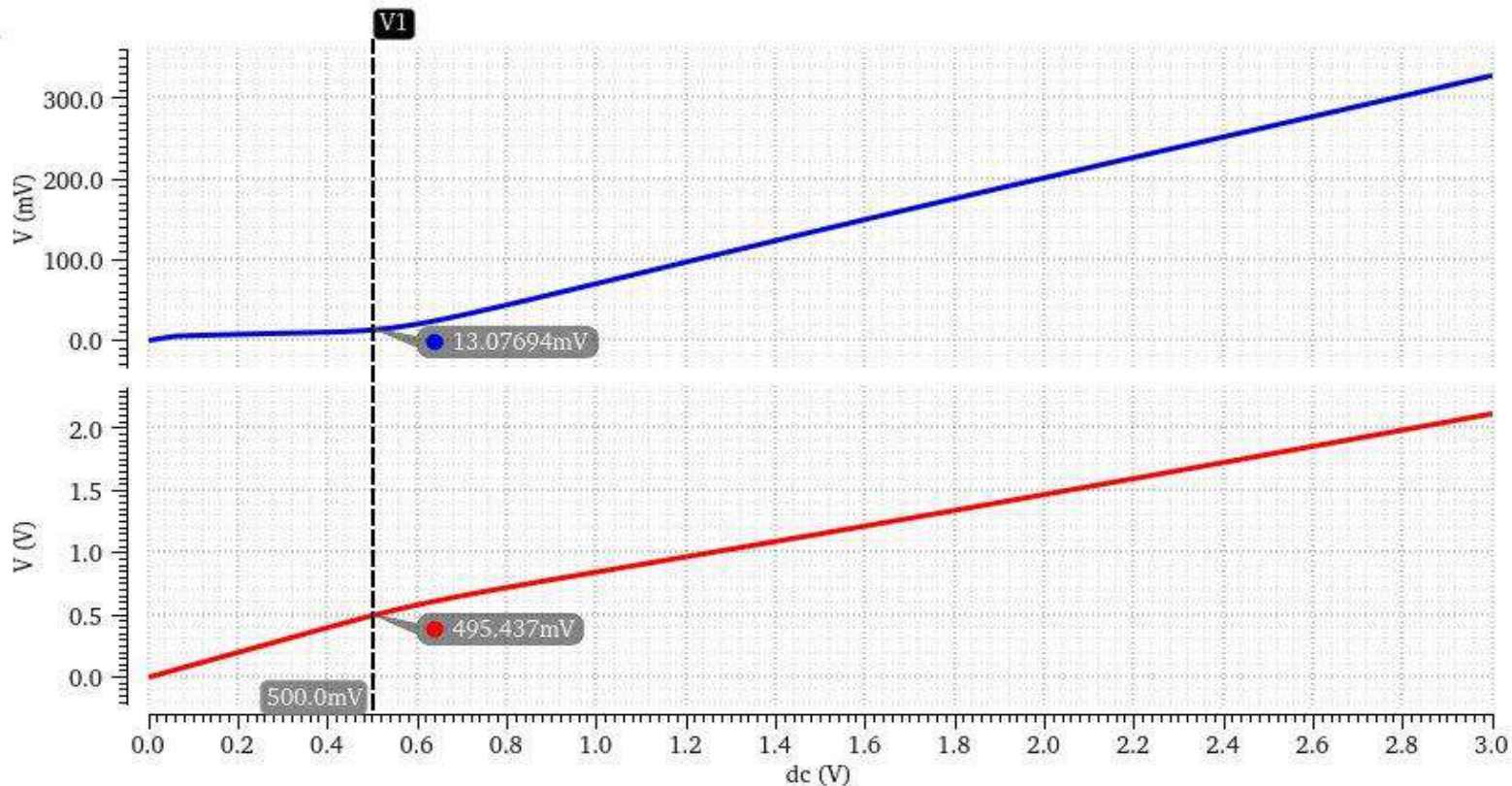
DC Response

Wed Jun 24 14:03:23 2026 1

Name V1

■ /Vy ...694mV

■ /Vx ...437mV





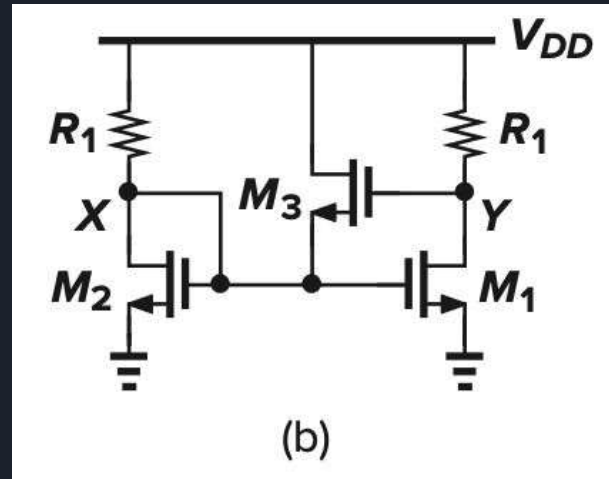
Observations

- Theoretical analysis predicts a small initial rise and dip in V_Y around V_{th} due to the transition of M1 from OFF to ON.
- In the Cadence simulation, this variation appears almost flat because the change in V_Y is very small.
- Since V_Y is already close to ground, M1 cannot pull it significantly lower after turning ON.
- After this point, V_Y begins to increase as M1 enters the triode region.

Problem 5.10-b

5.10. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.53. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

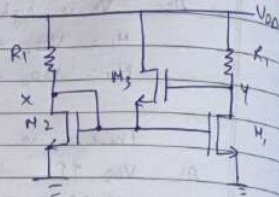
5.10.4.4

When $V_{DD} < V_{thn}$

M_1, M_2 and M_3 are OFF

$$V_X = V_Y = V_{DD}$$

Once $V_{DD} > V_{thn}$



Current starts to flow through M_1 and M_2

$$\therefore V_{GS2} = V_{GS1}, I_{M2} = I_{M1}$$

$\therefore$ drop across resistors w/U be same

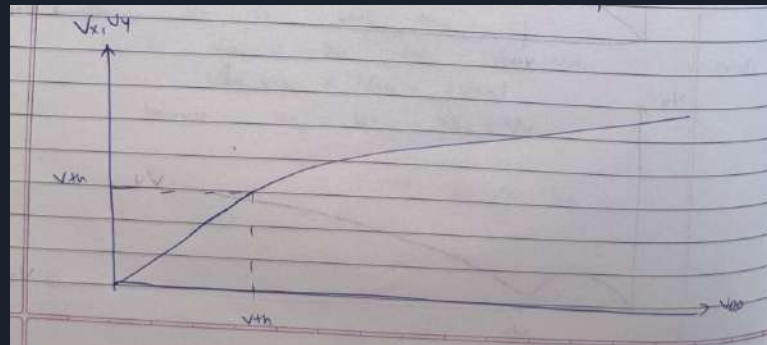
and $V_X \approx V_Y$

$$\therefore \text{for } M_3, V_{GS3} = V_Y - V_X \approx 0$$

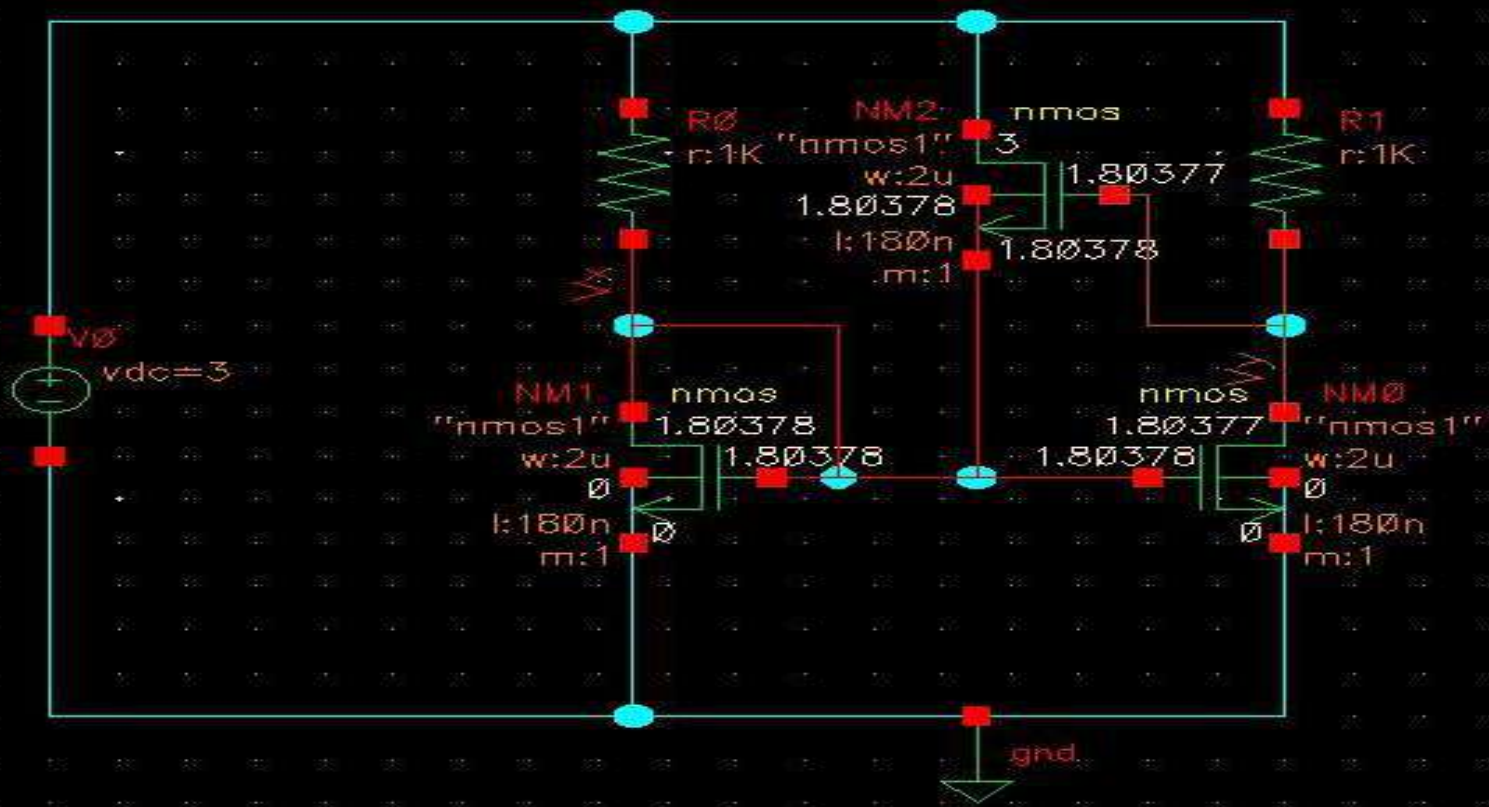
$\therefore M_3$ never turns ON

$\therefore$ As V_{DD} increases, $I_{M2} = I_{M1}$ increase

resulting in increase in $V_X = V_Y$



Cadence Virtuoso Schematic



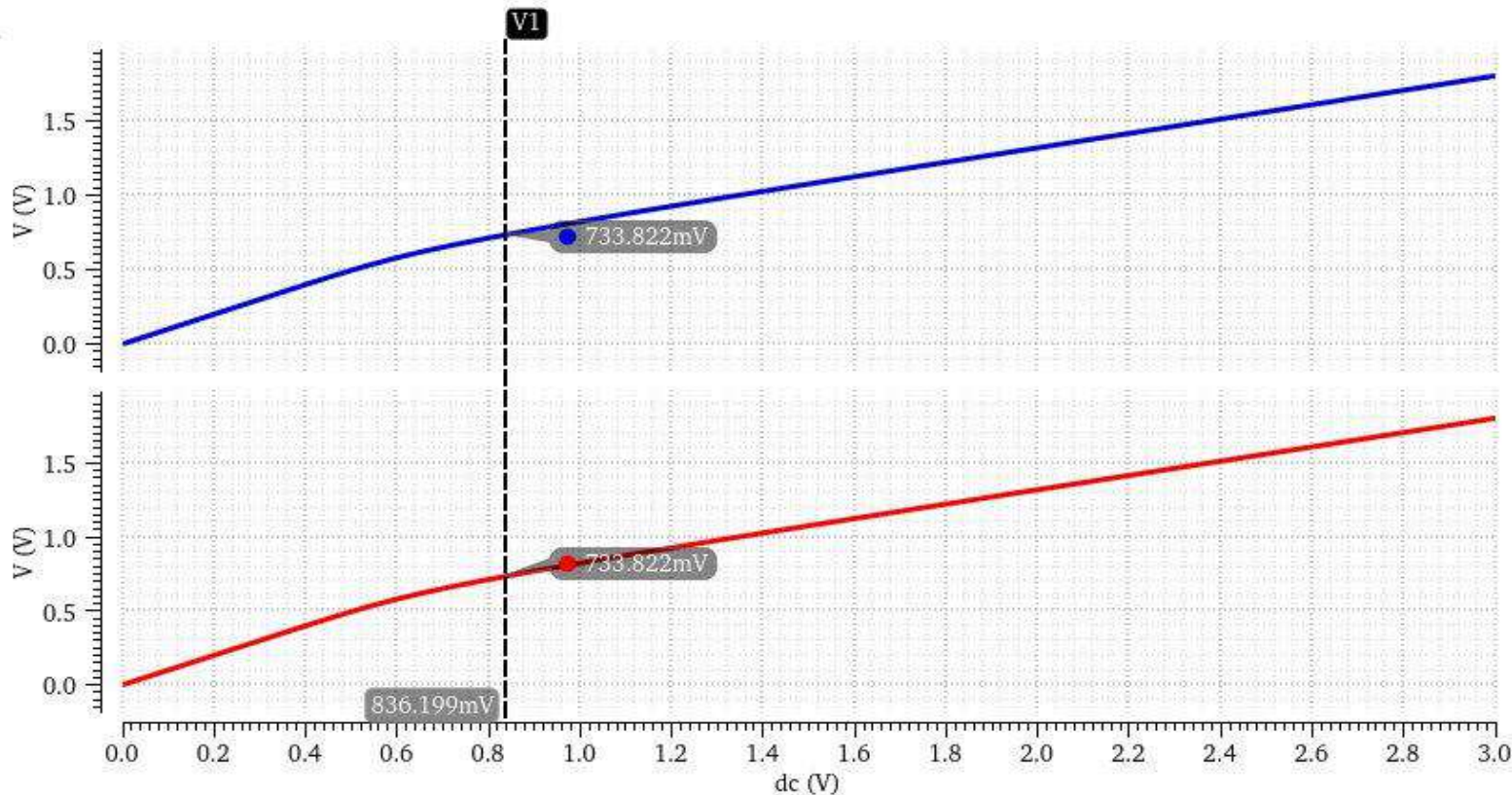
Cadence ADE-L DC Simulation for Vx and Vy

DC Response

Wed Jun 24 14:12:32 2026 1

Name V1

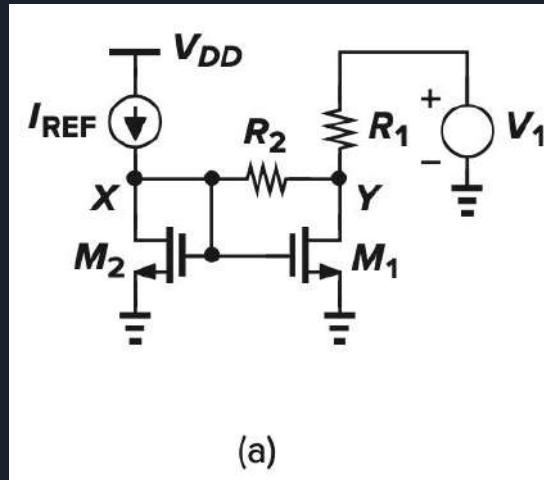
Vy ...822mV



Problem 5.11-a

5.11. For each circuit in Fig. 5.54, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

5.11.2016

At node X

$$I_{M2} = I_{ref} - I_{R2} \quad \text{--- (1)}$$

At node Y

$$I_{M1} = I_{R2} - I_{R1} \quad \text{--- (2)}$$

Also we know that, $I_{M1} = I_{M2} = I$

$\therefore$ using $I_{R2} = I_{M1} + I_{R1}$ in eqn (1)

$$2I_{M2} = I_{ref} - I_{R1}$$

$$2I = I_{ref} - \frac{V_X - V_Y}{R_1} \quad \text{--- (3)}$$

$$I = \frac{V_X - V_Y}{R_2} = \frac{V_Y - V_1}{R_1} \quad \text{--- (4)}$$

$$\therefore V_Y = \frac{2R_1 V_X + R_2 V_1}{2R_1 + R_2} - I_{ref} R_1 R_2$$

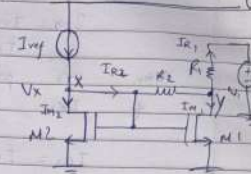
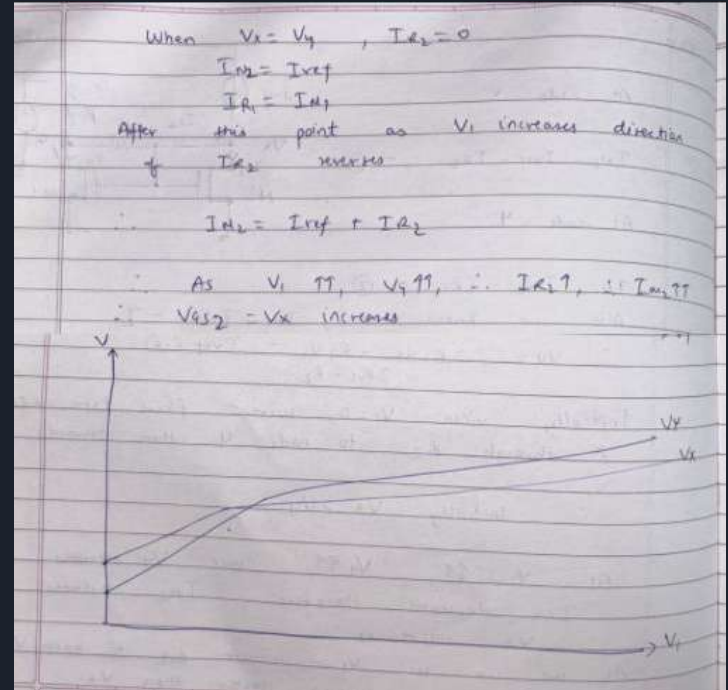
Initially when $V_1 = 0$, current flows from node X through R_2 to node Y then towards V_1 .

$\therefore$ Initially $V_X > V_Y$

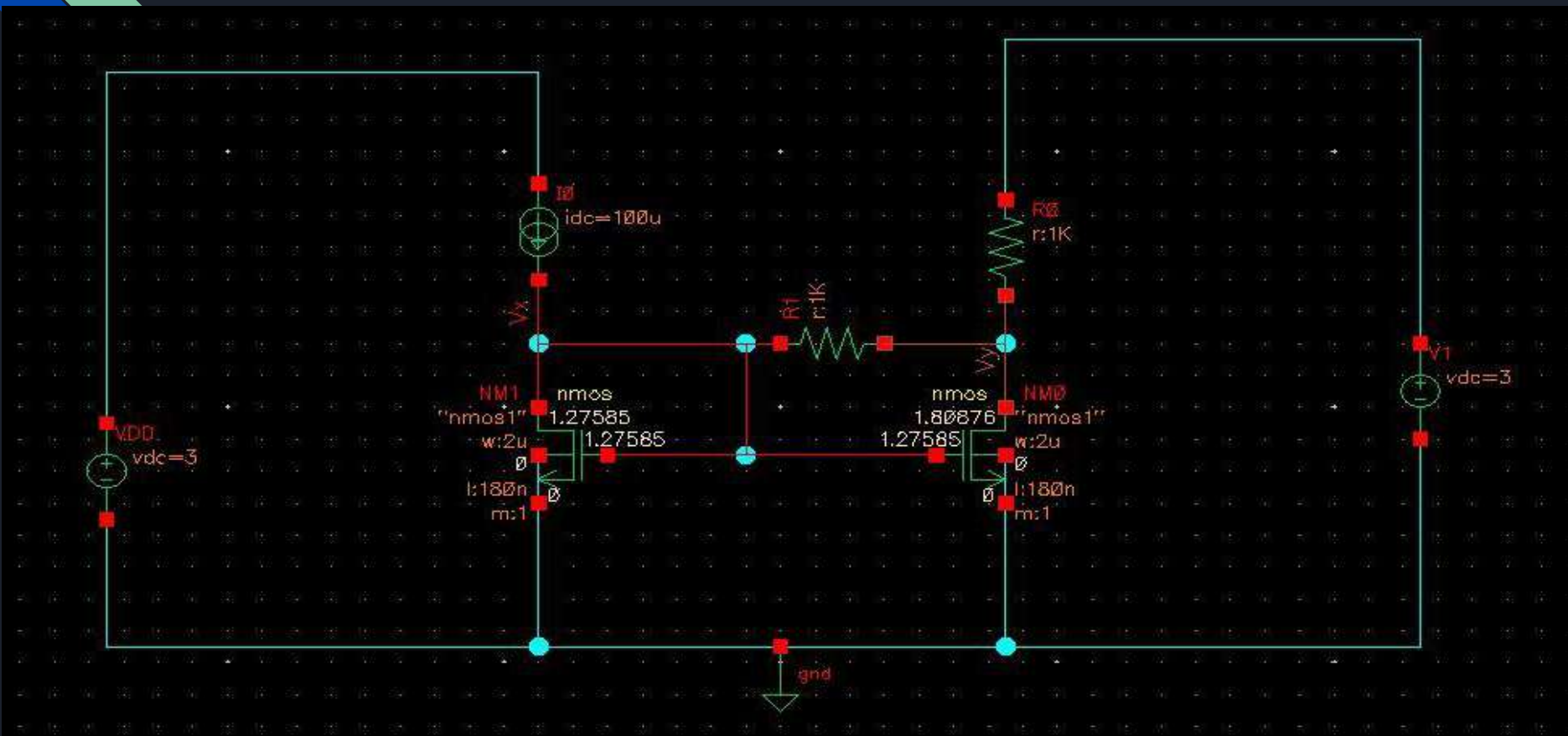
As $V_1 \uparrow \uparrow$, $V_Y \uparrow \uparrow$, since V_Y increases I_{R2} decreases, therefore, I_{M2} increases.

V_X increases

As we can see V_Y increases due to both V_X and V_1 , $\therefore V_Y$ increases faster than V_X .

Cadence Virtuoso Schematic



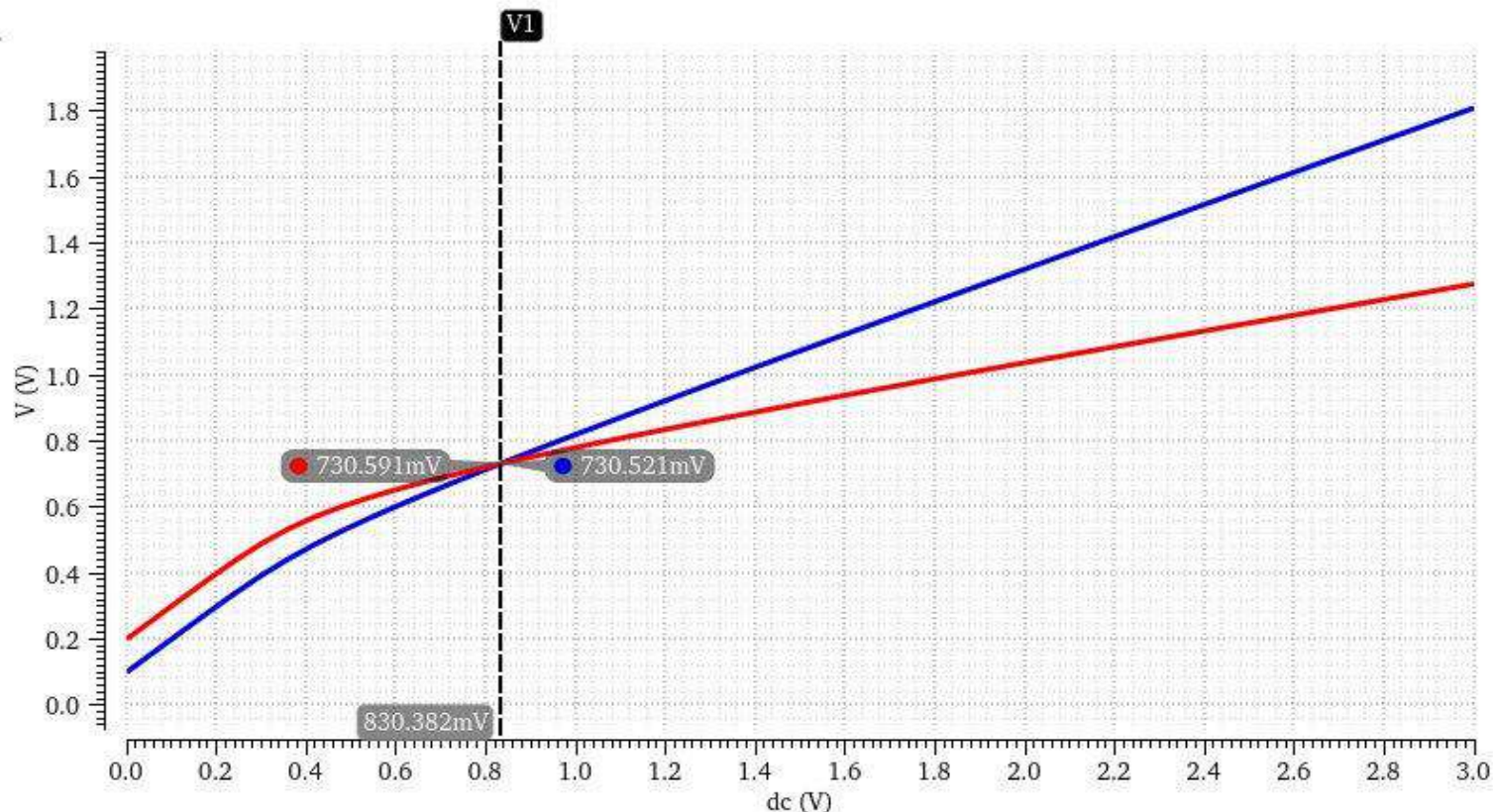
Cadence ADE-L DC Simulation for Vx and Vy

DC Response

Wed Jun 24 14:21:44 2026 1

Name V1

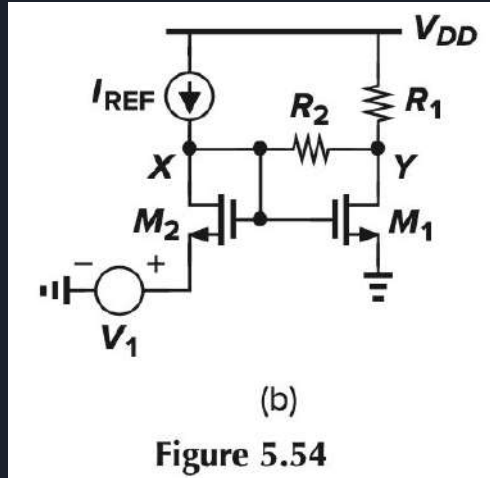
■ /Vy	730.521mV
■ /Vx	730.591mV



Problem 5.11-b

5.11. For each circuit in Fig. 5.54, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

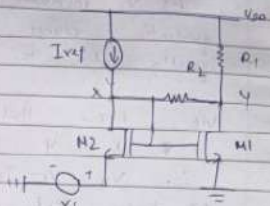
Diagram -



Handwritten Solution

Initially, when $V_i = 0$
 $I_{M1} = I_{M2} = I_{ref}$

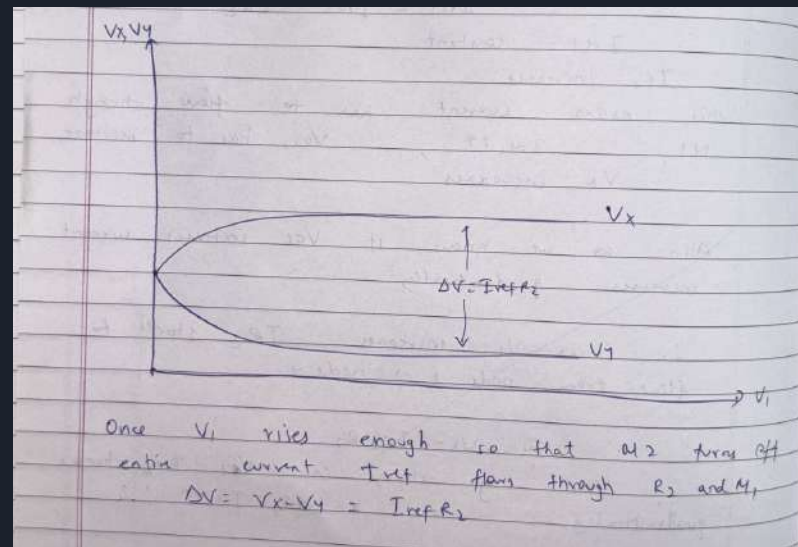
$I_{R2} = 0$
 $V_x \approx V_y$



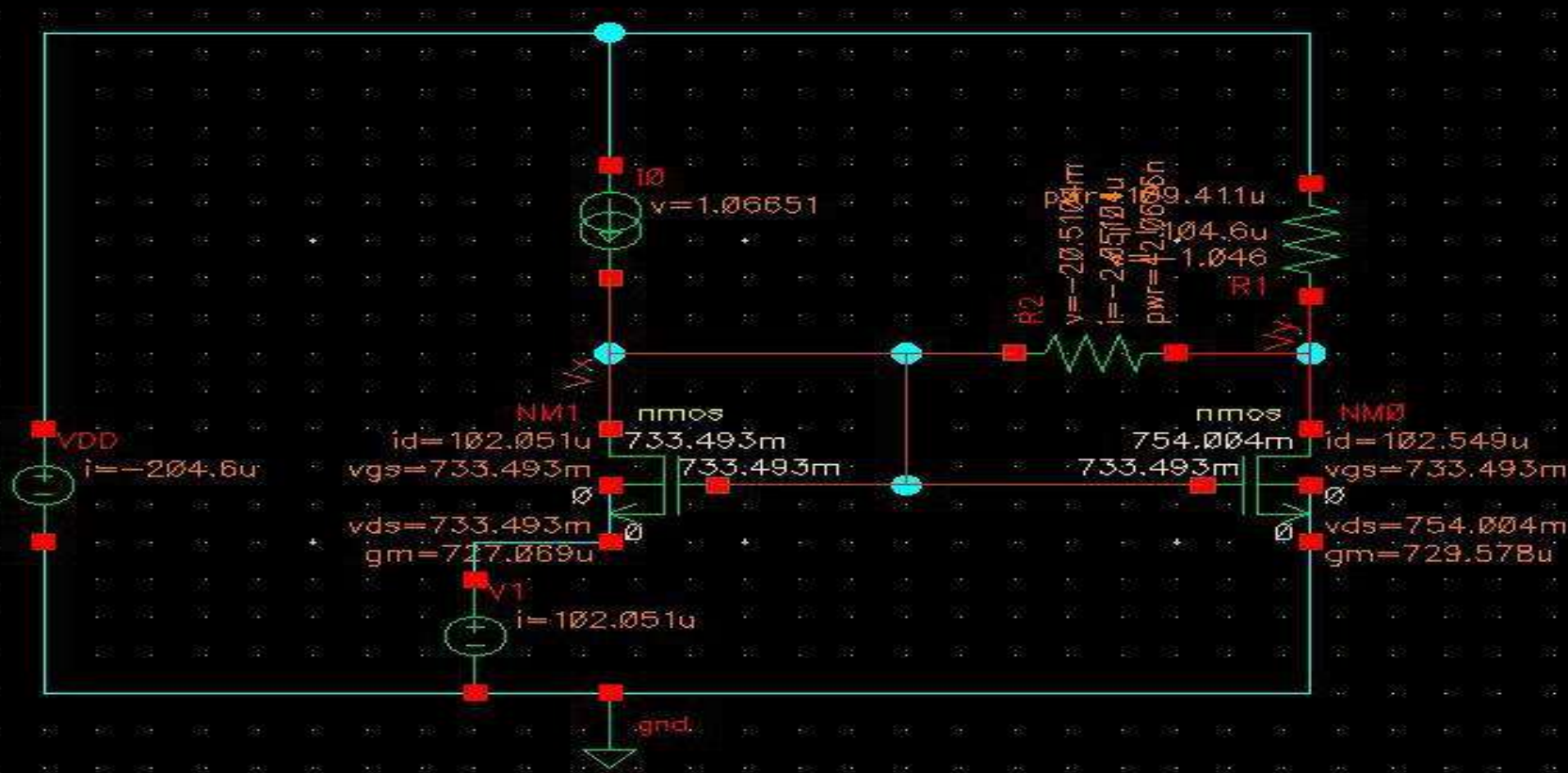
When V_i starts increasing, V_{GS2} decreases
 I_{M2} decreases
 At node X
 $I_{ref} = I_{M1} + I_{M2}$
 $I_{ref} = \text{constant}$
 I_{M2} increases
 this extra current has to flow through $M1$ $I_{M1} \uparrow$, V_{GS1} has to increase
 V_x increases

As V_i increases, small change in V_{GS1} causes quadratic change in I_{M1} , I_{M2} also increases greatly, due to this drop against R_2 increases

We know that current flows from
 $X \rightarrow Y$
 $V_y = V_x - I_{M2} R_2$
 we know that V_x is increasing, but V_y has to increase less ~~other~~ for I_{M2} to rise, $\therefore I_{M2} R_2$ dominates and V_y decreases



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

DC Response

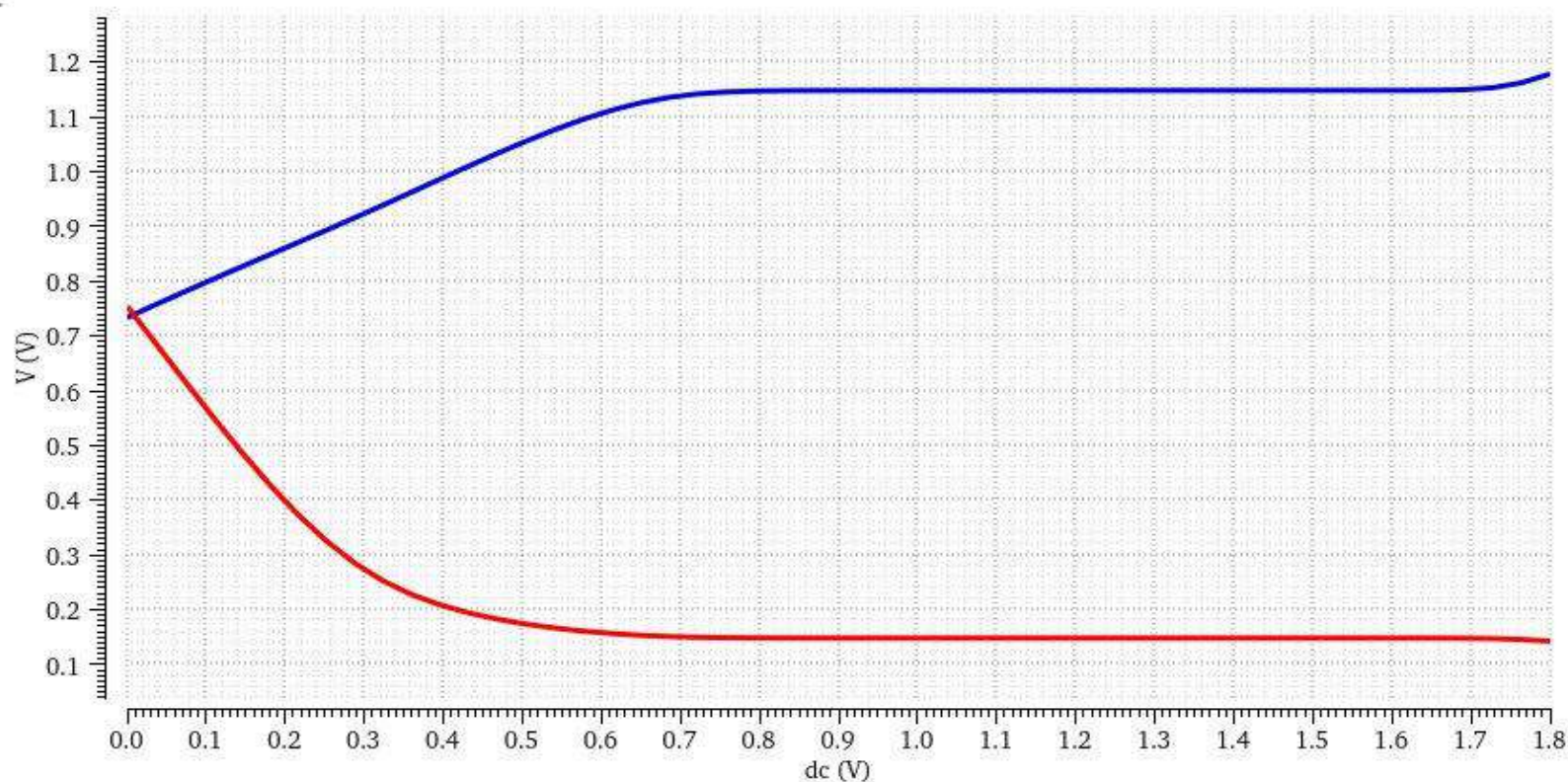
Wed Jun 24 15:22:06 2026 1

Name	Vis
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☒ /Vx



☒ /Vy





Observations

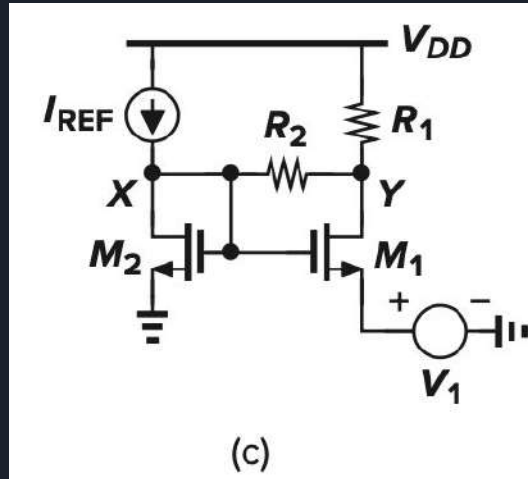
Why does V_X start to increase at the end in Cadence ADE-L simulation?

- At the end of the sweep, the decrease in V_Y reduces the drain-source voltage V_{DS1} of M1.
- When V_{DS1} becomes insufficient to satisfy the saturation condition, M1 enters the triode region.
- In the triode region, M1 loses its ideal current-source behavior and requires a higher V_{GS1} to maintain the required current.
- Since $V_{GS1} = V_X$, the gate voltage V_X increases to compensate for the reduced drain headroom.

Problem 5.11-c

5.11. For each circuit in Fig. 5.54, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

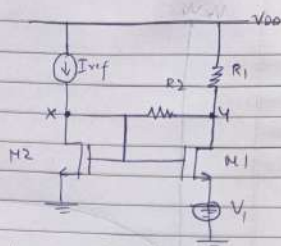
SILVY

When $V_i = 0$,

$$I_{M1} = I_{M2} = I_{ref}$$

$$\therefore I_{R2} = 0$$

$$\therefore V_x = V_y$$



Once V_i starts increasing, V_{gs1} ↓↓

$\therefore I_{M1}$ ↓↓

$\therefore I_{R2}$ starts flowing from $y \rightarrow x$

$$\therefore I_{M1} = I_{R2} + I_{ref}$$

as $V_i \uparrow \uparrow$, $I_{M1} \uparrow \uparrow$, $\therefore V_x$ increases slowly

$$\text{Also } V_y = V_x + I_{R2} R_2$$

$\therefore$ As $V_i \uparrow \uparrow$, $V_x \uparrow \uparrow$, also $I_{R2} \uparrow \uparrow$

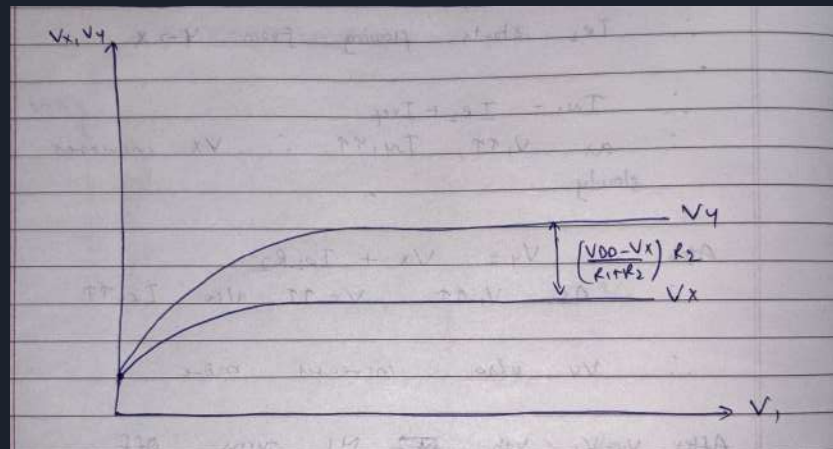
$\therefore V_y$ also increases more

After $V_y - V_i < V_{th}$, ~~M1~~ M1 turns off

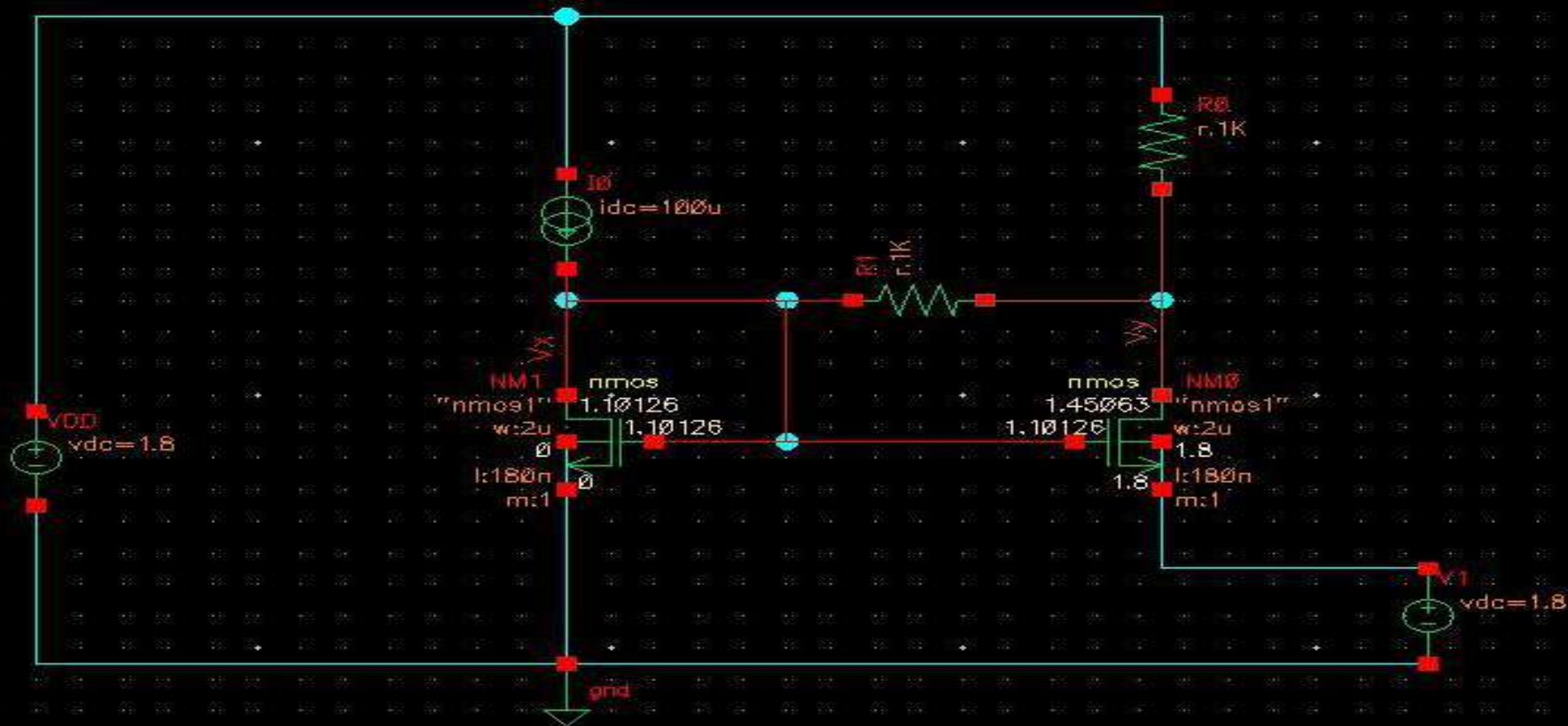
$$\therefore I_{M2} = \text{fixed} = I_{R2} + I_{ref}$$

$$\therefore V_y - V_x = I_{R2} R_2$$

$$V_y - V_i = \left(\frac{V_{DD} - V_x}{R_1 + R_2} \right) R_2$$



Cadence Virtuoso Schematic



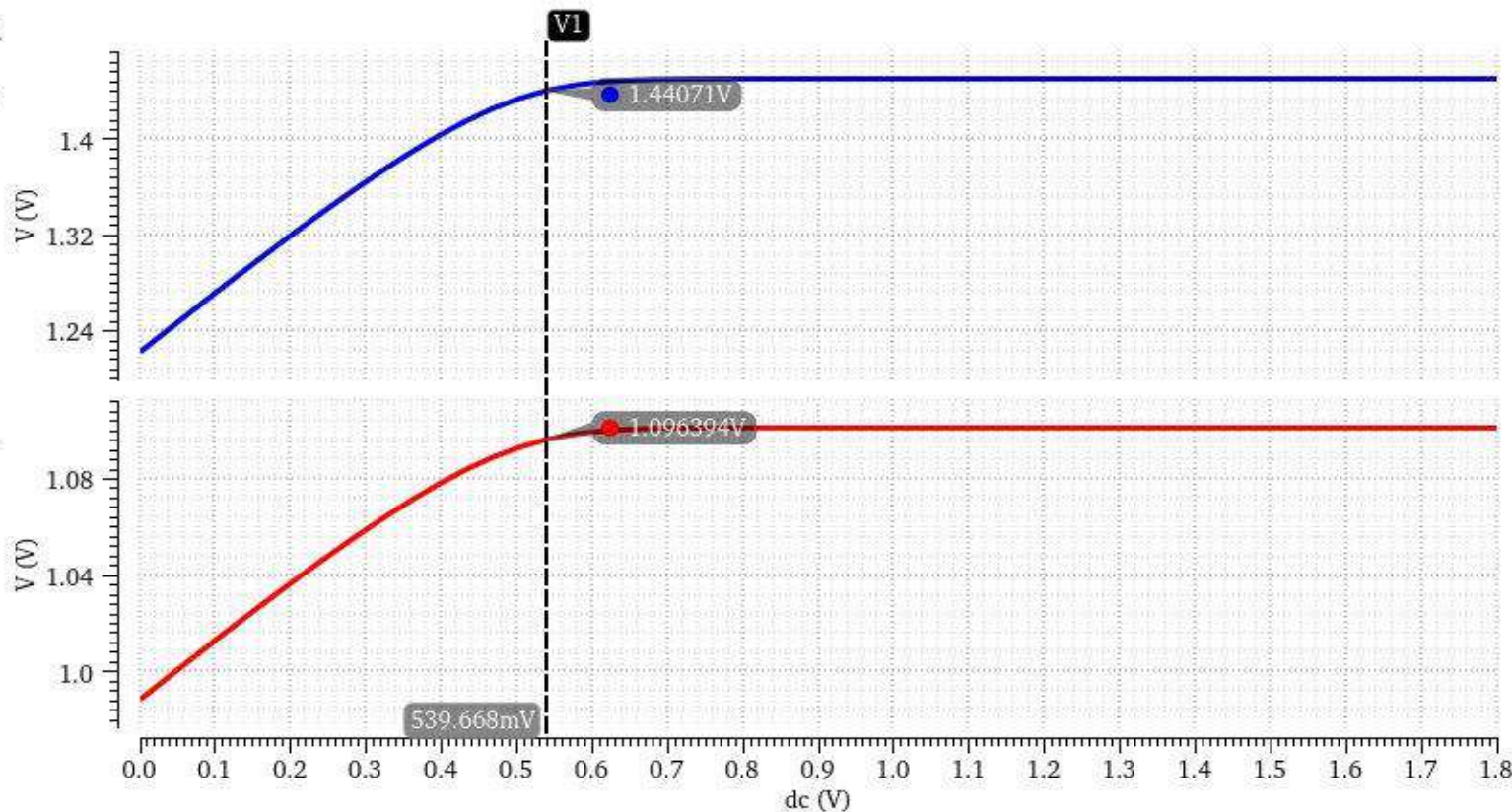
Cadence ADE-L DC Simulation for Vx and Vy

DC Response

Wed Jun 24 15:44:32 2026 1

Name V1

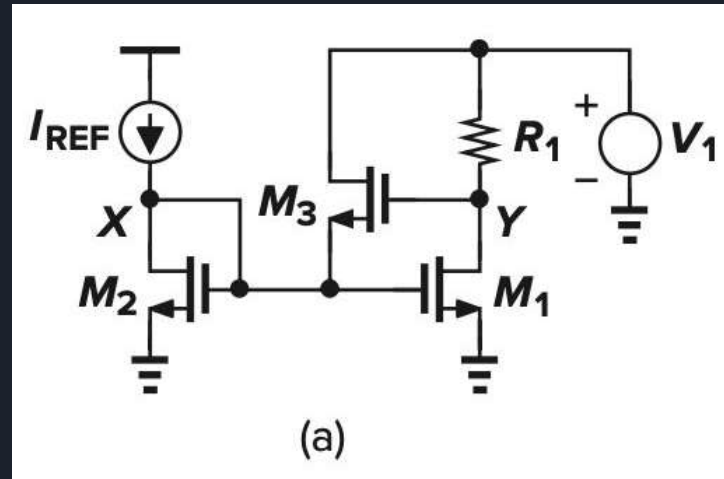
V_y 1.44071V



Problem 5.12-a

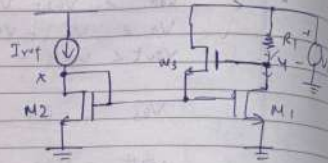
5.12. For each circuit in Fig. 5.55, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

5/12/2016



For V_x

V_x does not depend upon V_1

$\therefore V_x$ remains constant for the entire sweep of V_1 , until V_1 reaches high enough so as to it turns M_3 ON ($V_4 - V_x > V_{th}$)

Initially when V_1 is low, M_1 is strongly on ($V_{ds1} = V_{ds2}$), V_4 is pulled to ground and M_1 is in triode

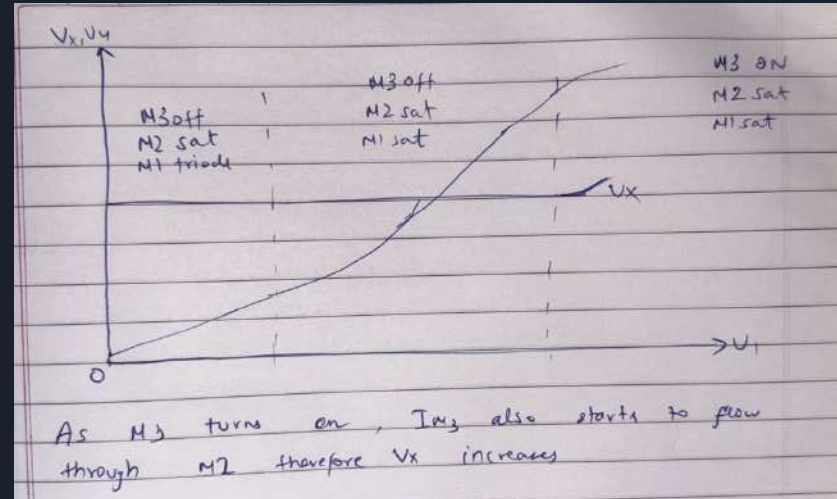
almost as V_1 increases, V_4 increases linearly as M_1 acts as a voltage controlled resistor

Once $V_4 > V_x - V_{th}$, M_1 enters saturation and $V_4 = V_1 - I_{ref} R_1$

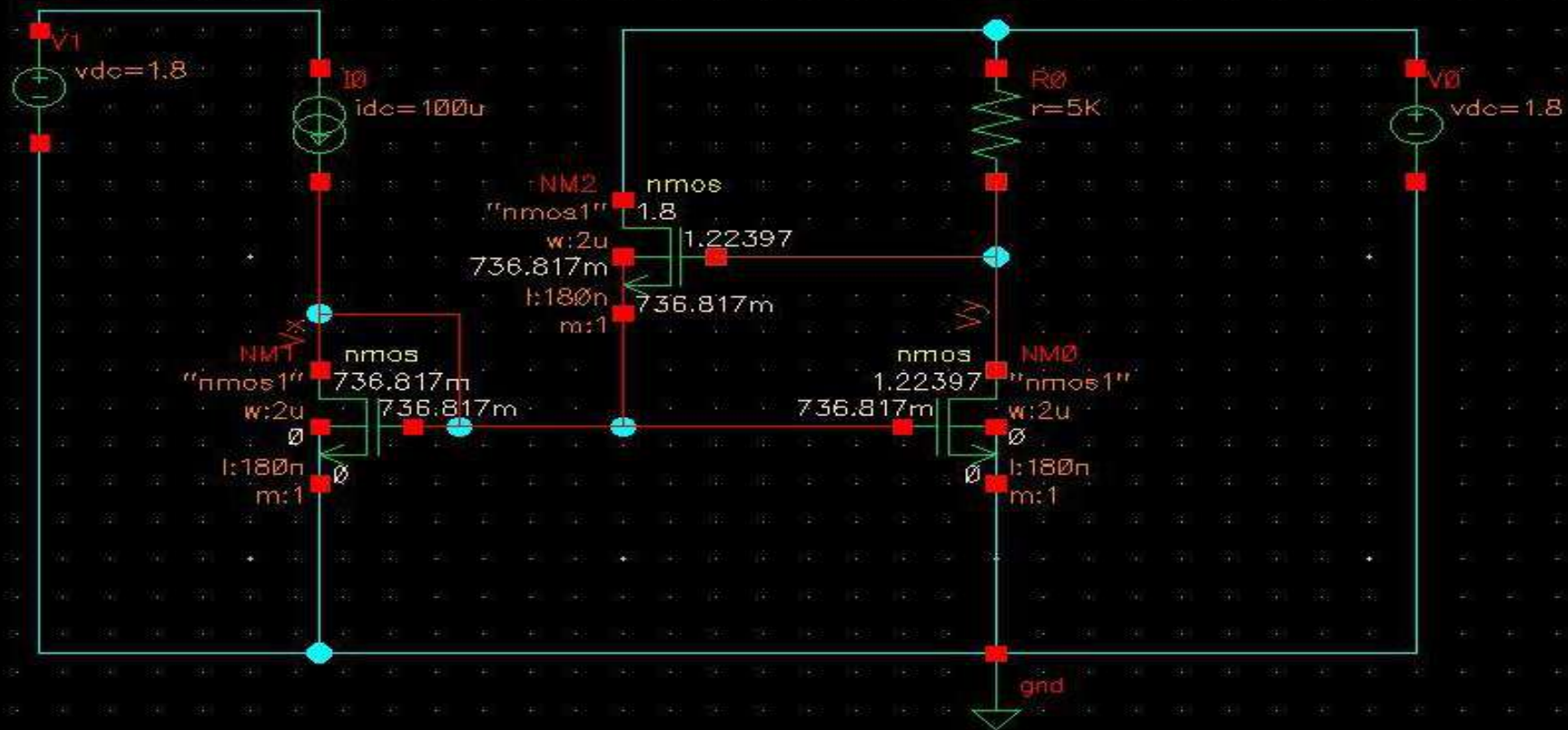
also since $V_{ds1} = V_{ds2}$, $I_{d1} = I_{ref}$

$\therefore V_4 = V_1 - I_{ref} R_1$

as V_1 increases further, V_4 increases also due to channel length modulation, I_{d1} also increases as V_1 increases, but V_1 is the dominating factor hence as V_1 increases further, V_4 increases



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for V_x and V_y

Wed Jun 24 16:44:09 2026 1

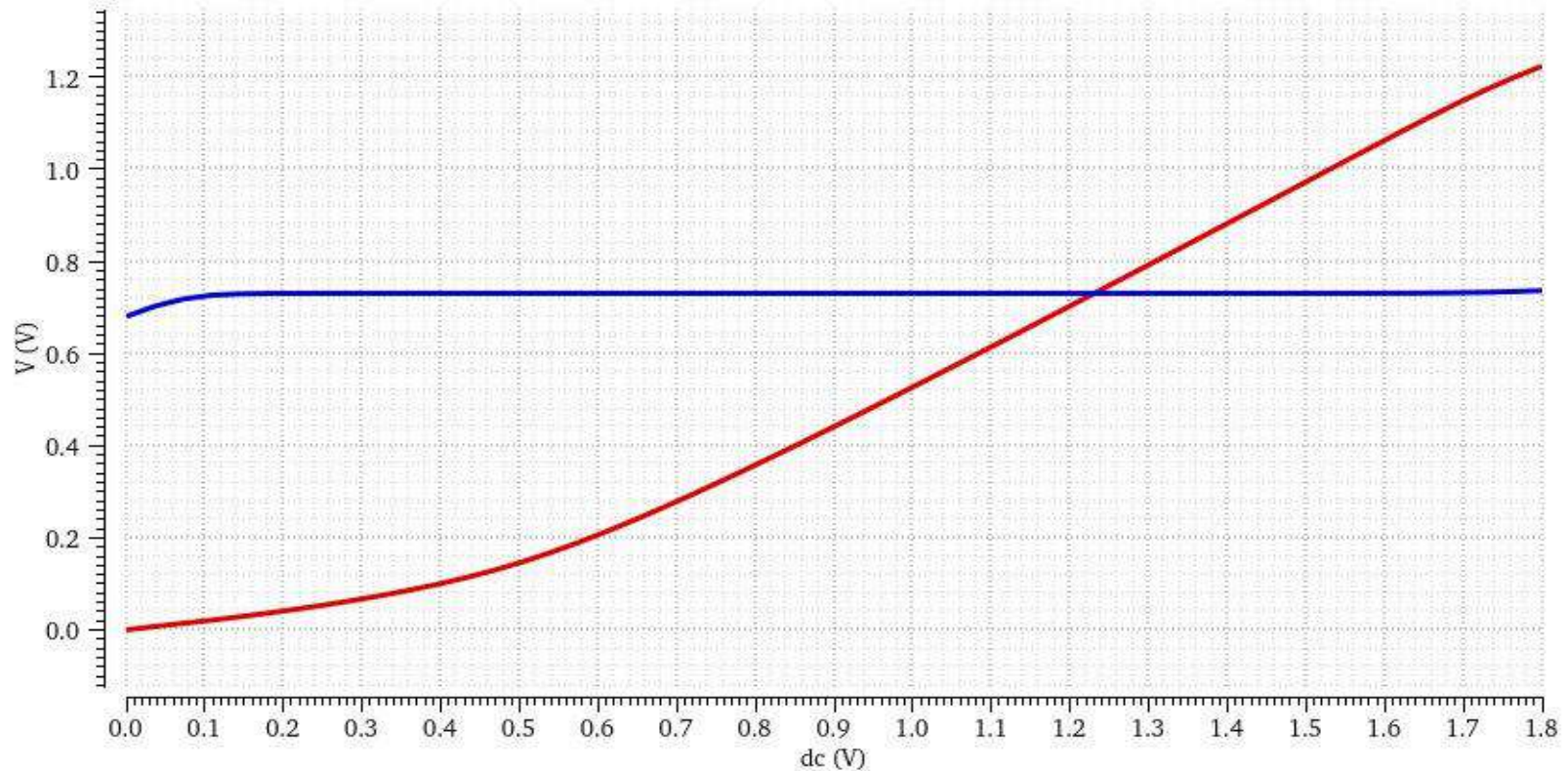
DC Response

Name	Vis
------	-----

V_y



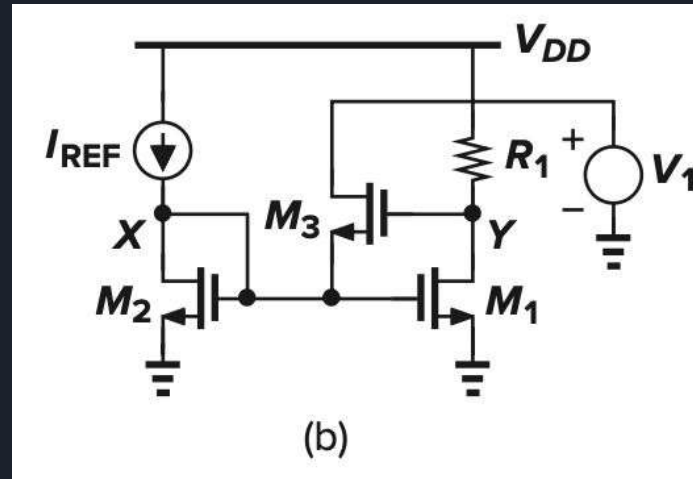
V_x



Problem 5.12-b

5.12. For each circuit in Fig. 5.55, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

Diagram -

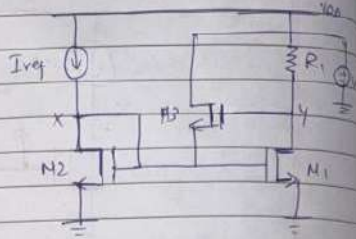


Handwritten Solution

5/24/64

Initially we assume $M1$ and $M2$ to be off when $V_i = 0V$

and Source and Drain of $M3$ are Interchanged.

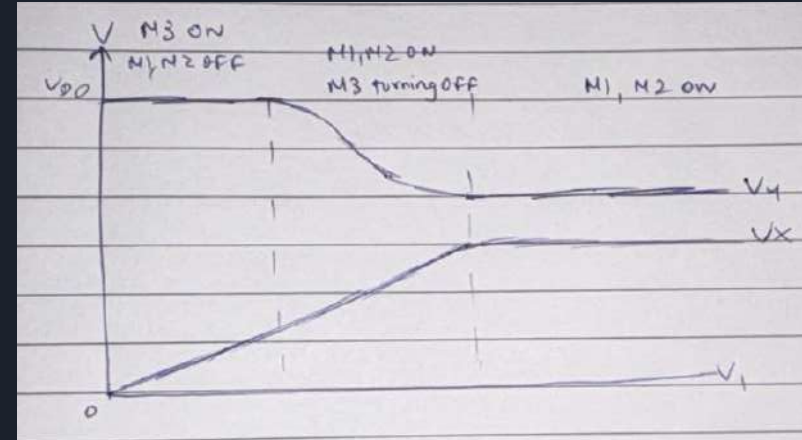


when $V_i = 0$, $V_x = V_{DD}$, $\therefore M3$ is turned on strongly $\therefore$ it pulls V_x down. V_x is very low, hence the assumption is true. ($M1$ & $M2$ are OFF)

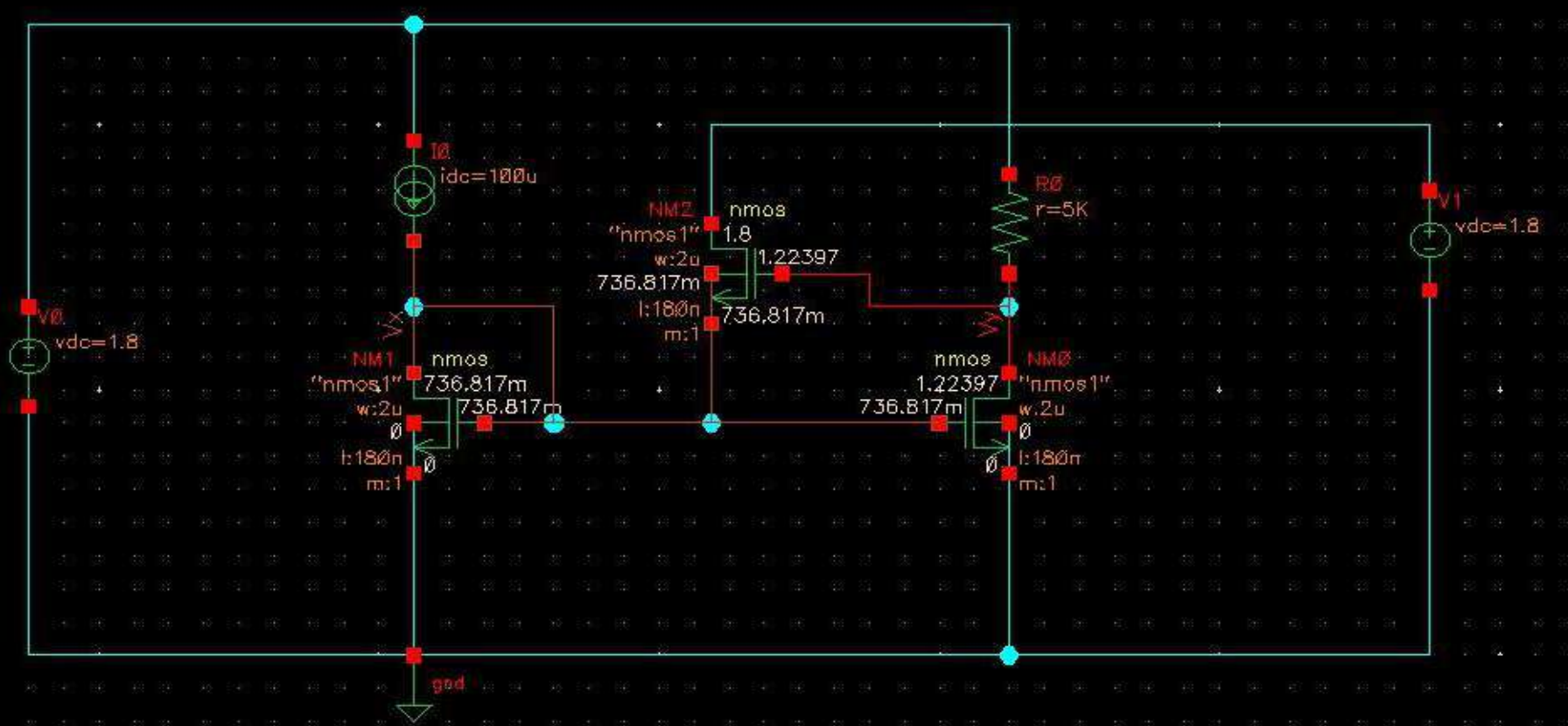
As V_i increases V_x increases ($V_x = V_i + V_{DS3}$) and $I_{M3} = I_{ref}$, $\therefore$ Once $V_x > V_{th}$ $M1$ and $M3$ turn ON and $I_{ref} = I_{M1} + I_{M3}$

Once $M1$ is ON, I_{M3} starts flowing and V_y decreases, and eventually $M3$ turns OFF

Once $M3$ is OFF, $I_{M1} = I_{M3} = I_{ref}$ $\therefore V_x$ and V_y become constant.



Cadence Virtuoso Schematic



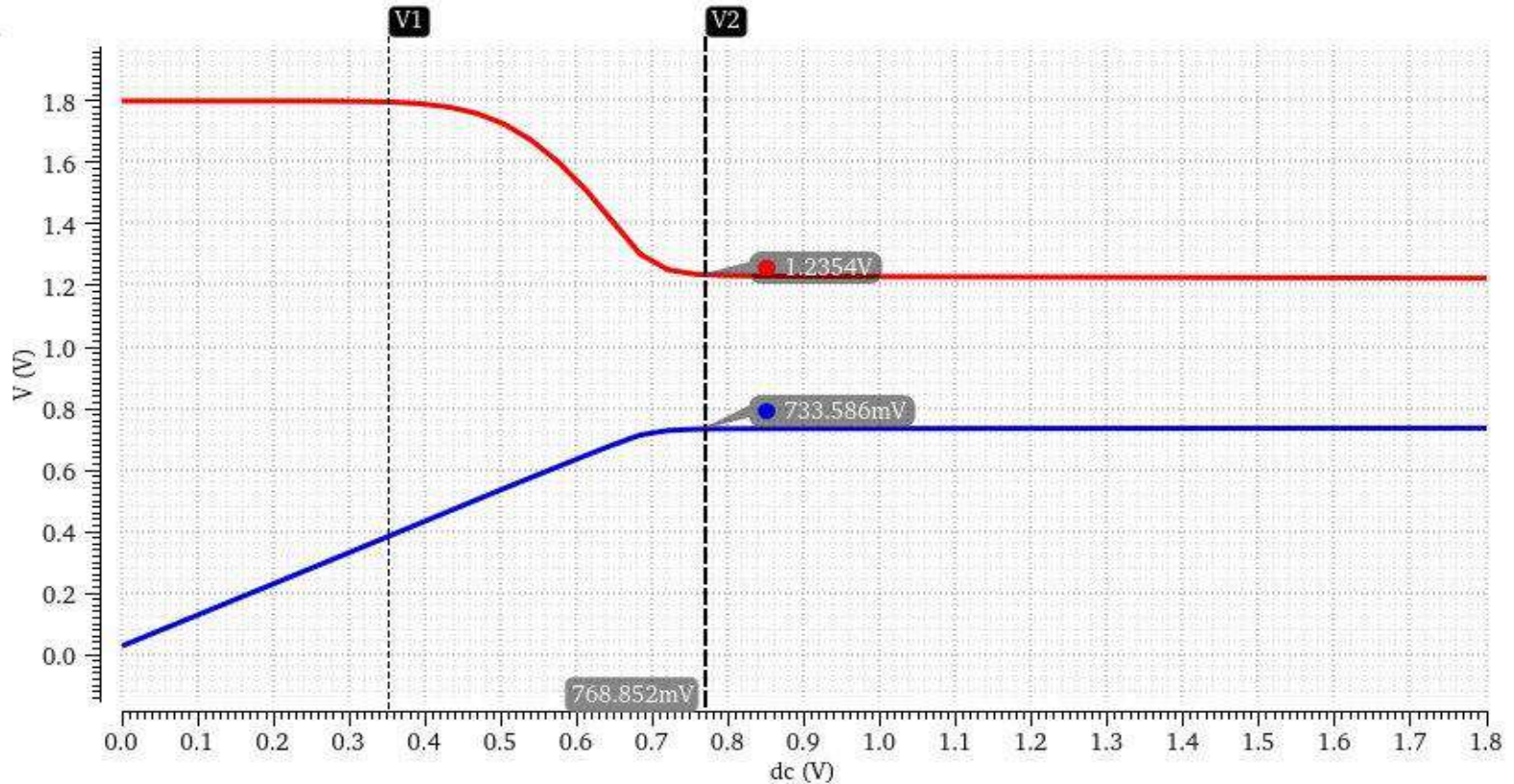
Cadence ADE-L DC Simulation for Vx and Vy

DC Response

Wed Jun 24 16:23:34 2026 1

Name V2

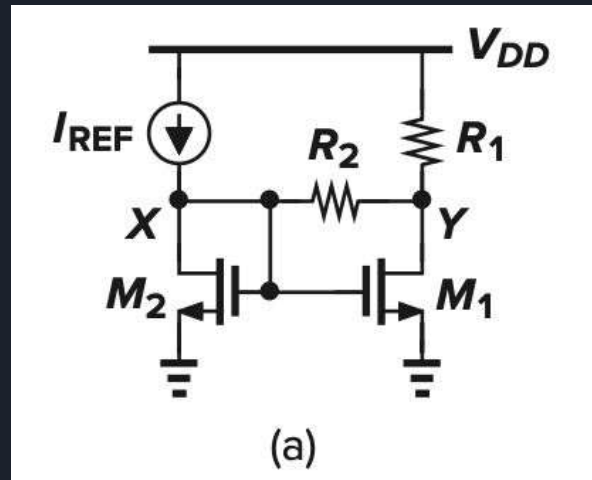
■ /Vy 1.2354V
■ /Vx 733.586mV



Problem 5.13-a

5.13. For each circuit in Fig. 5.56, sketch V_X and V_Y as a function of I_{REF} .

Diagram -



Handwritten Solution

5.13 Ya-y

When $I_{ref} = 0$

Current flows from VDD through $M1$ and $M2$

$V_{GS1} = V_{GS2}$
 $I_{M1} = I_{M2}$

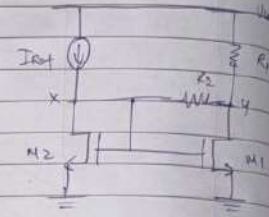
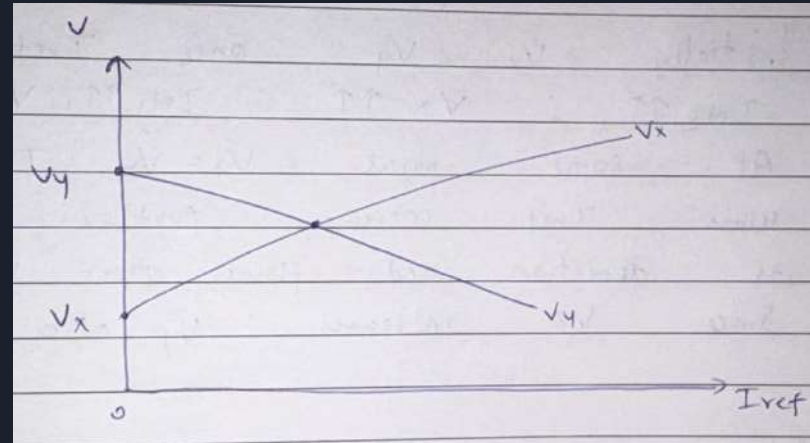
$I_{R1} = I_{R2} + I_{M1}$, $I_{R2} = I_{M2}$
 $I_{R1} = 2I_{M1} = 2I$

$V_X = V_{GS1} = V_{GS2} = V_{DD} - I_{R1}R_1$
 $V_X = V_{DD} - I(2R_1 + R_2)$

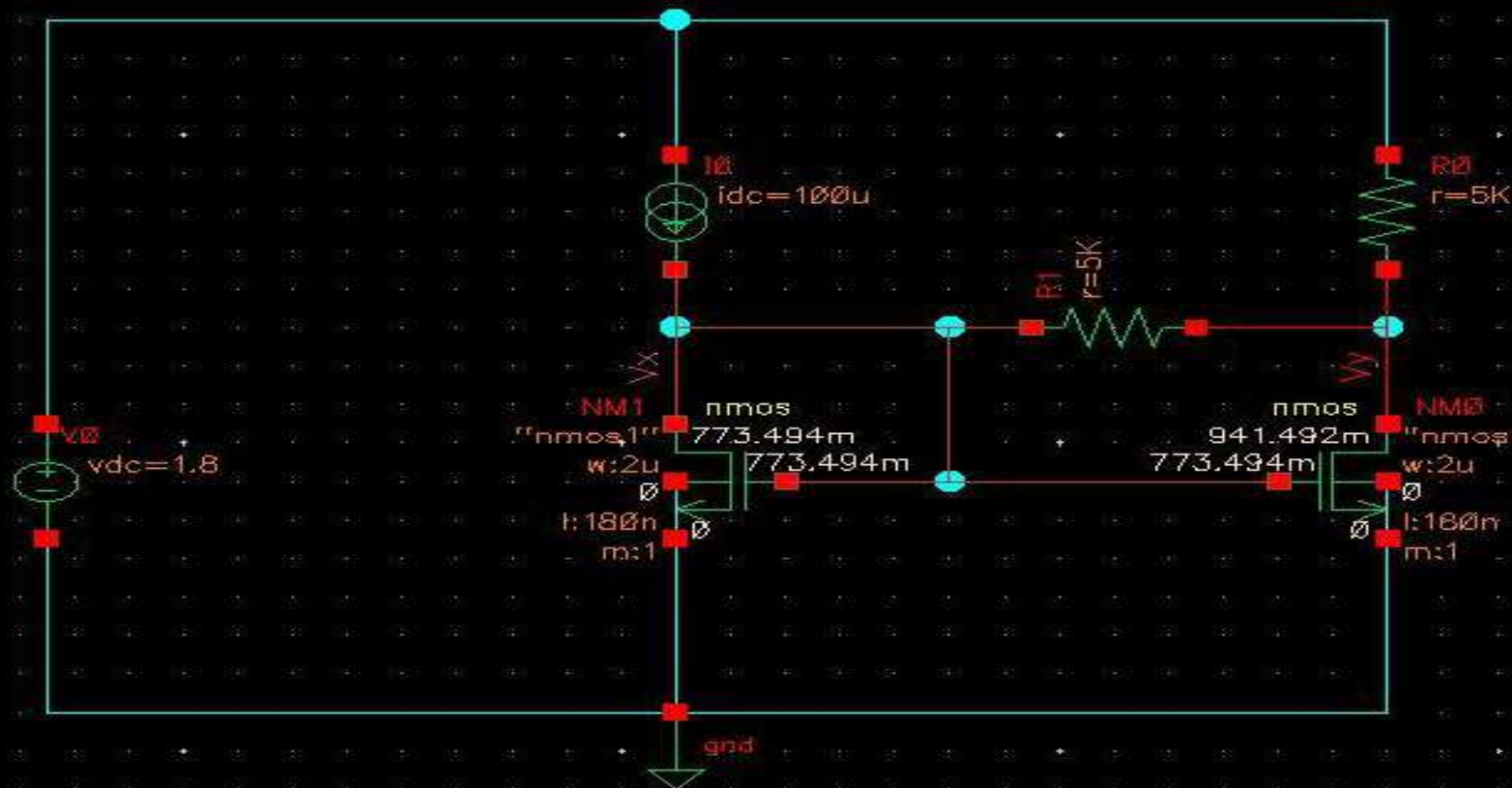
$V_Y = V_X - I_{R2}R_2$

$V_Y = V_X + \left(\frac{V_{DD} - V_X}{2R_1 + R_2} \right) R_2$

Initially $V_X < V_Y$, once $I_{ref} \uparrow$,
 $I_{M2} \uparrow$, $\therefore V_X \uparrow$, $\therefore I_{M1} \uparrow$, $V_Y \downarrow$
 At some point $V_X = V_Y$, $I_{R2} = 0$
 When I_{ref} increases further I_{R2} changes
 its direction and flows from X to Y
 Since V_X increases, V_Y decreases linearly.

Cadence Virtuoso Schematic



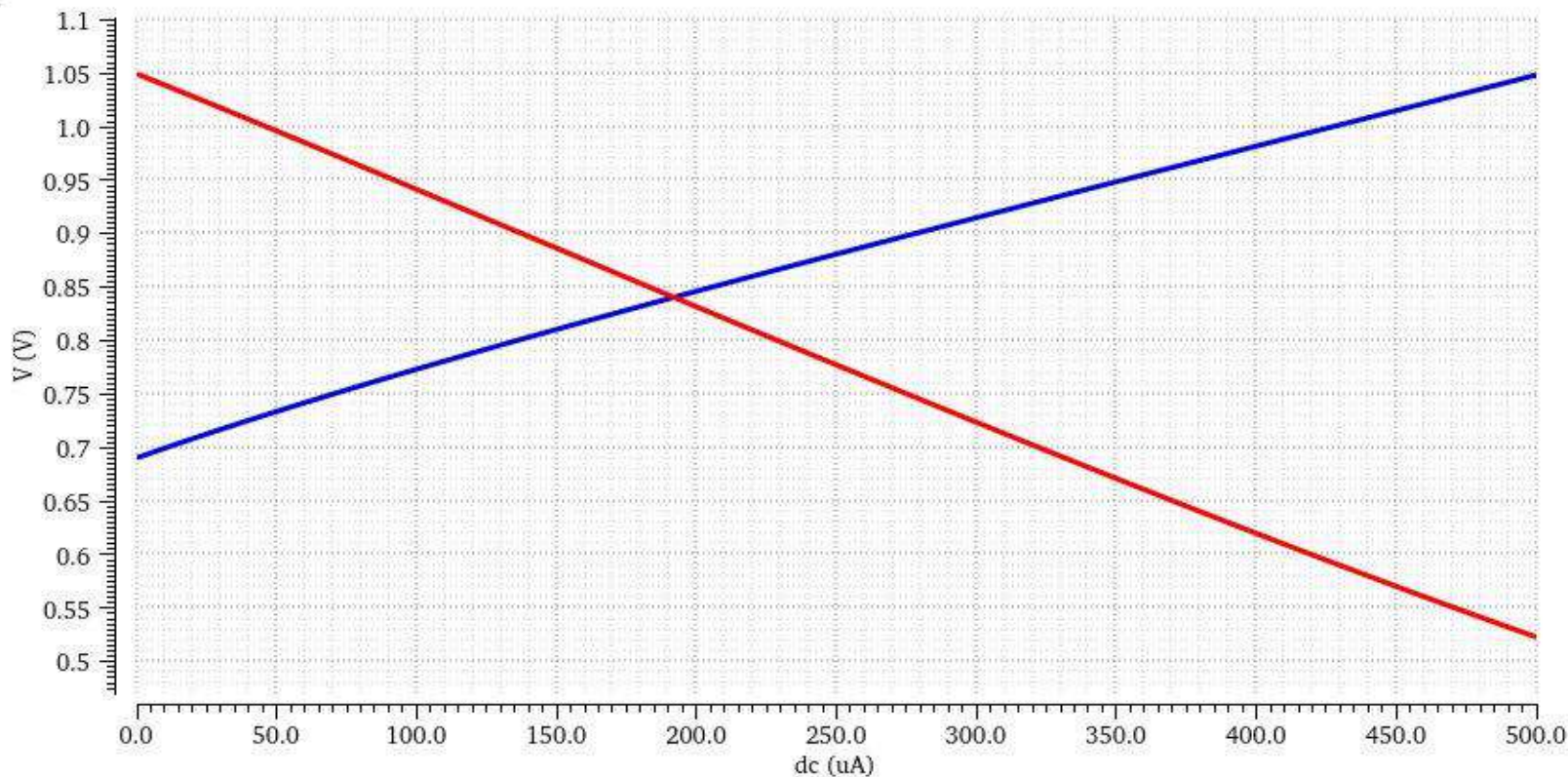
Cadence ADE-L DC Simulation for V_x and V_y

Wed Jun 24 16:41:34 2026 1

DC Response

Name	Vis
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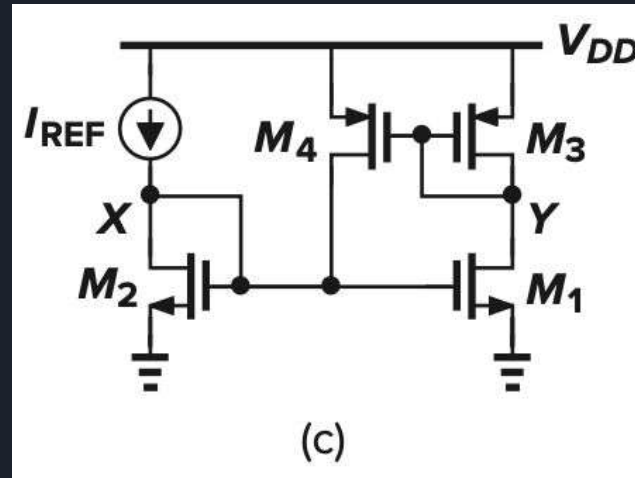
V_x	☒
V_y	☒



Problem 5.13-c

5.13. For each circuit in Fig. 5.56, sketch V_X and V_Y as a function of I_{REF} .

Diagram -



Handwritten Solution

5-13.4 cY

Initially when $I_{ref} = 0$

$I_{M4} = I_{M2}$

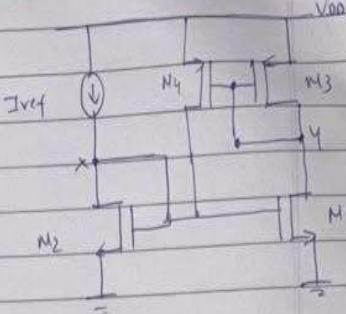
$\therefore V_X = V_Y$

As I_{ref} increases

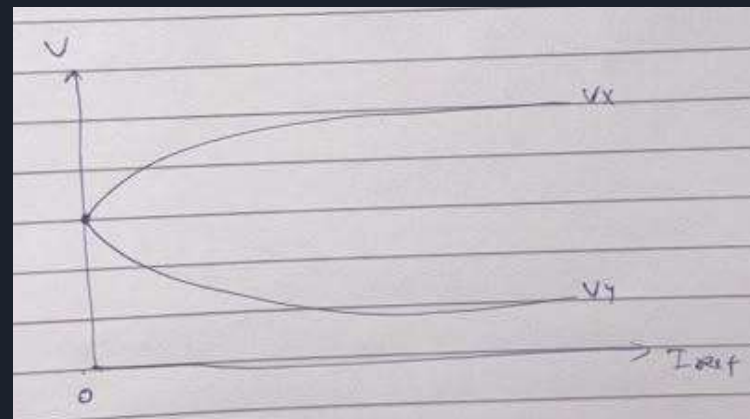
$I_{M2} = I_{ref} + I_{M4}$, $\therefore I_{M2} \uparrow$, $\therefore V_X \uparrow$

$\therefore V_X \uparrow$, $V_{GS12} \uparrow$, $\therefore I_{M1}$ also increases and V_Y gets pulled down.

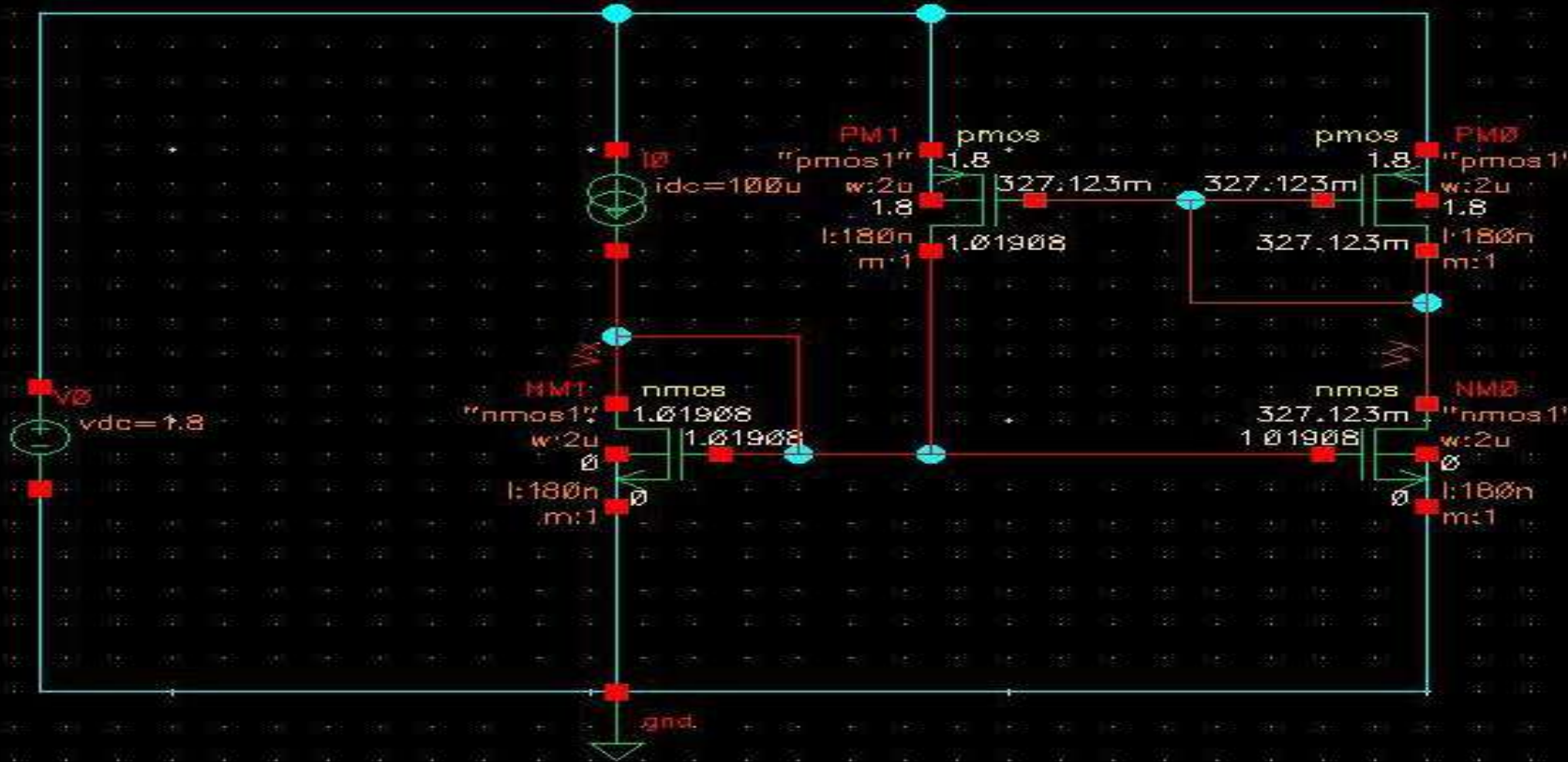
Also as $I_{ref} \uparrow$, $V_{Y12}, M1$ will enter triode region and $V_X \uparrow$, $\therefore M4$ will enter triode region



The circuit diagram shows a differential pair of NMOS transistors, M1 and M2, with their sources connected to ground. Their gates are connected to a common node X. A PMOS transistor M4 is connected between node X and VDD, with its gate also connected to node X. A PMOS transistor M3 is connected between VDD and node Y, with its gate connected to VDD. A current source Iref is connected between VDD and node X. Node Y is also connected to ground through a PMOS transistor M1. The output voltages are VX at node X and VY at node Y.



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for V_x and V_y

DC Response

Wed Jun 24 16:51:24 2026 1

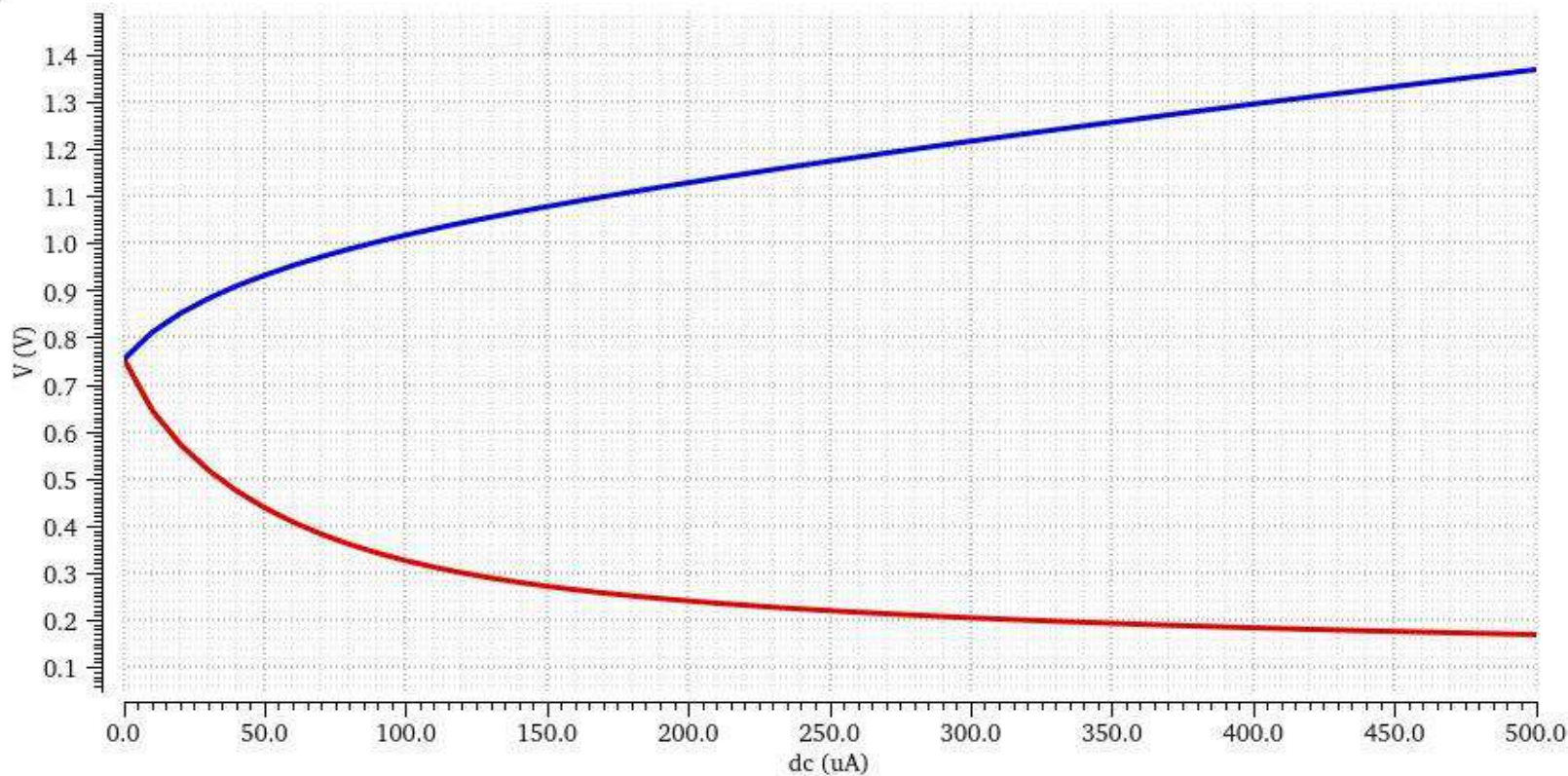
Name Vis

V_y

V_x

V_y

V_x





Observations

- As I_{ref} increases, V_X increases and V_Y decreases.
- The increase in V_X makes M1 pull V_Y down more strongly. As V_Y decreases, M3 turns ON harder and opposes this by pulling V_Y toward VDD.
- Since M1 is actively driven by the externally controlled voltage V_X , whereas M3 is diode-connected and only reacts to changes in V_Y , the NMOS pull-down action dominates. Therefore, V_Y decreases until $I_{M1} = I_{M3}$.

Handwritten Solution

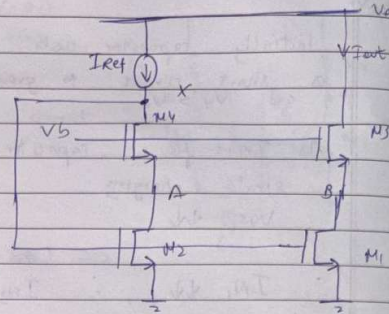
5.14-y

When $I_{ref} = 0$,

V_x will be 0V (all residual charge will flow to ground).

$\therefore$ M_2 and M_1 will be OFF, V_A and V_B will float

$\therefore I_{out} = 0$



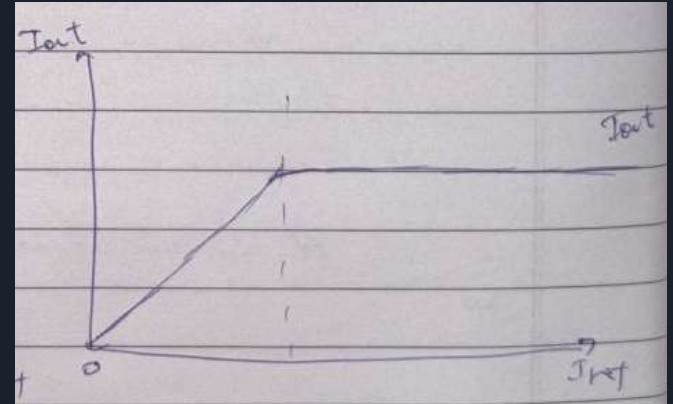
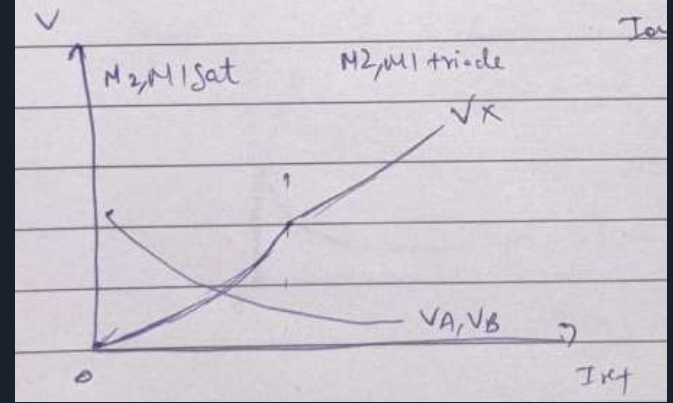
Once I_{ref} increases, V_x increases M_1 and M_2 turn ON,

$I_{M2} = I_{M1}$, $V_A = V_B$ (initially)

As I_{ref} increases further, $V_x \uparrow$

M_1 and M_2 turn ON strongly and V_A and V_B get pulled down.

If we assume I_{ref} to be ideal, after a point I_{M2} and I_{M1} cannot increase as (V_{GS} has a max value) $\therefore$ due to the current source injecting current into node X, V_x crosses V_{DD} , and I_{out} becomes constant

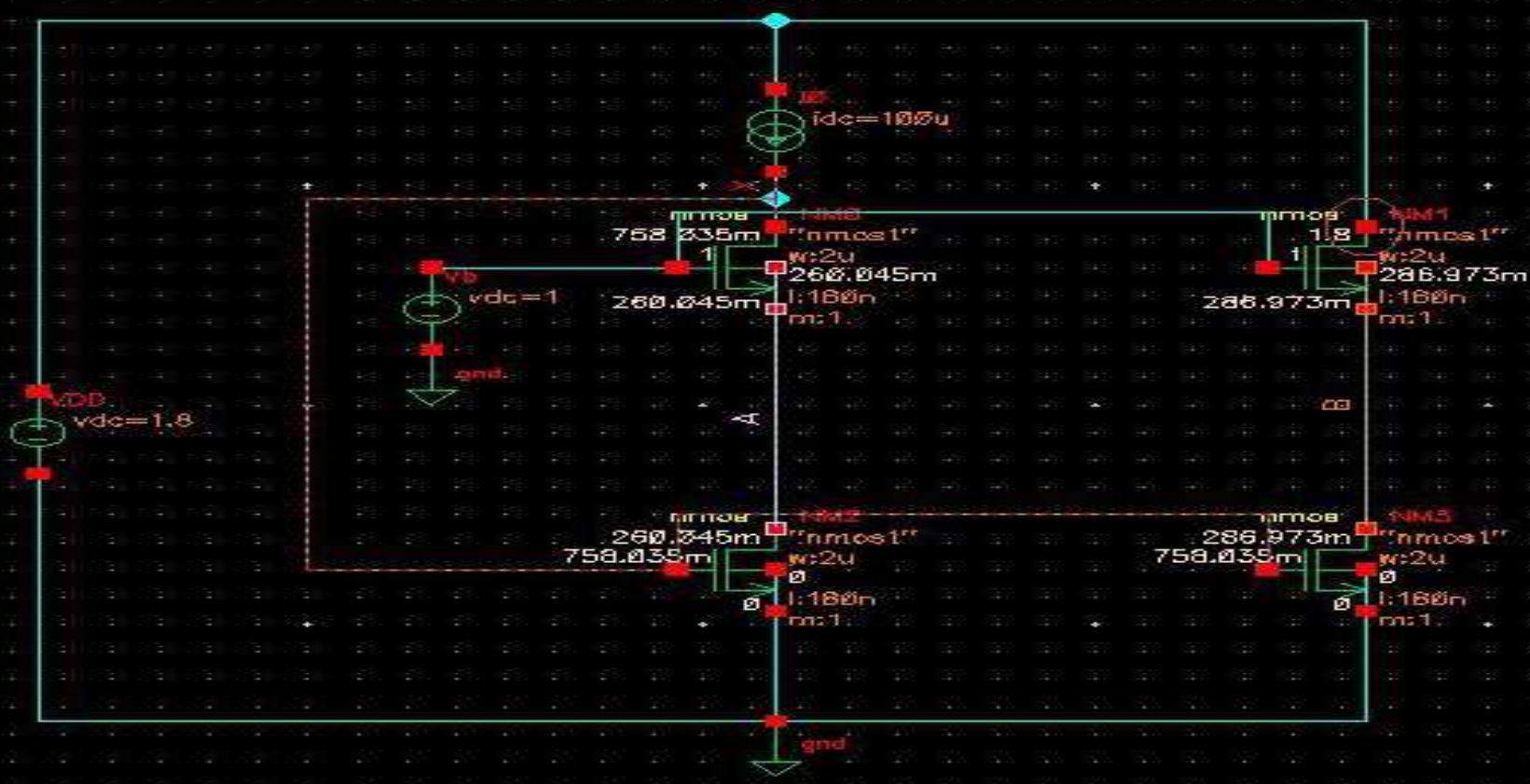




Observations

- The voltage at VX depends on the saturation current of M2, which is set by the reference current I_{ref} . Therefore, the exact variation of VX cannot be predicted analytically while M2 remains in saturation.
- Since the maximum current that M4 can conduct is limited, any increase in I_{ref} injects excess current into Node VX. This causes VX to rise above the supply voltage VDD.

Cadence Virtuoso Schematic



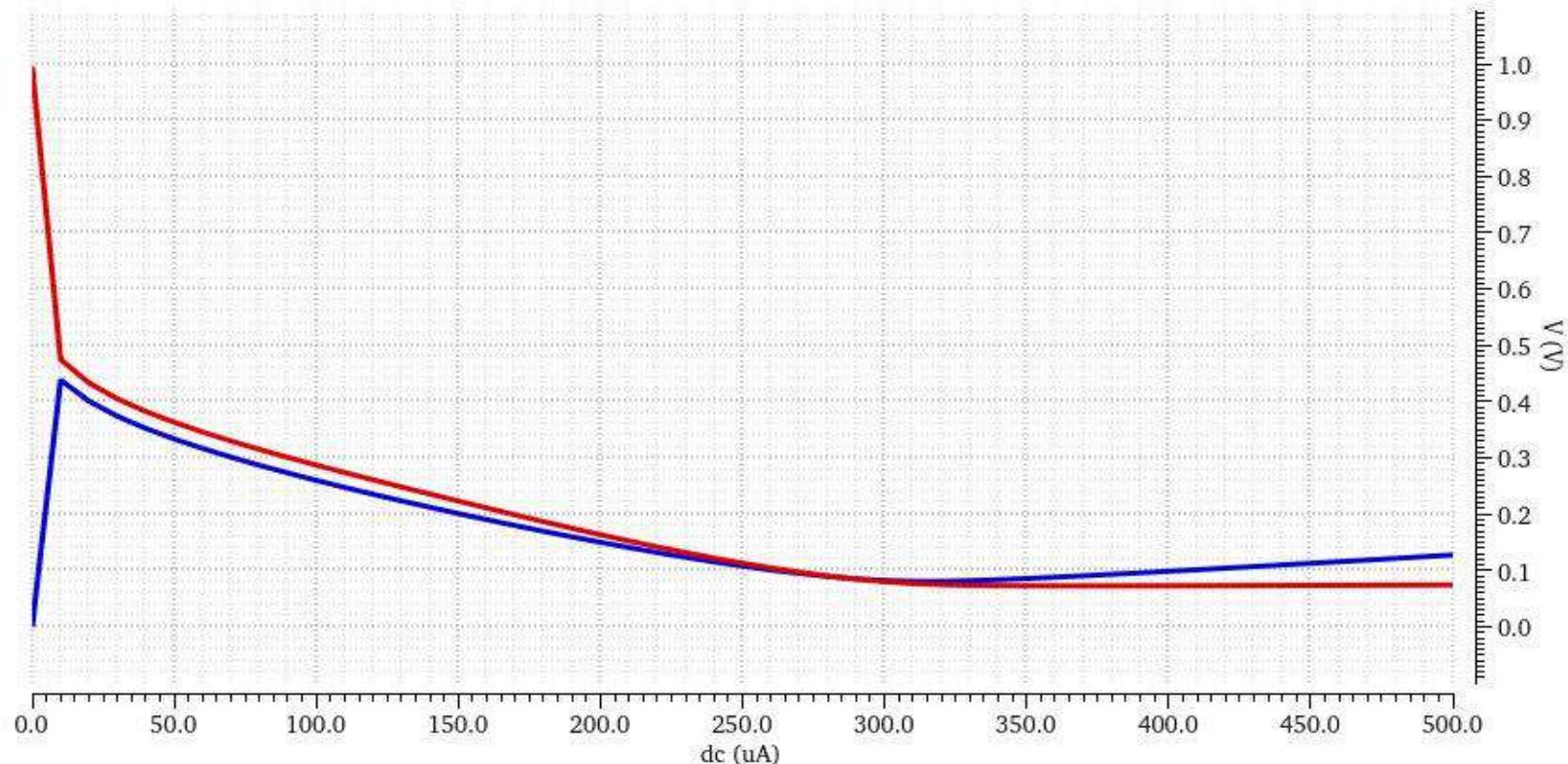
Cadence ADE-L DC Simulation for VA & VB

Wed Jun 24 17:11:21 2026 1

DC Response

Name Vis

☒ /A
☒ /B



Cadence ADE-L DC Simulation for Iout & VX

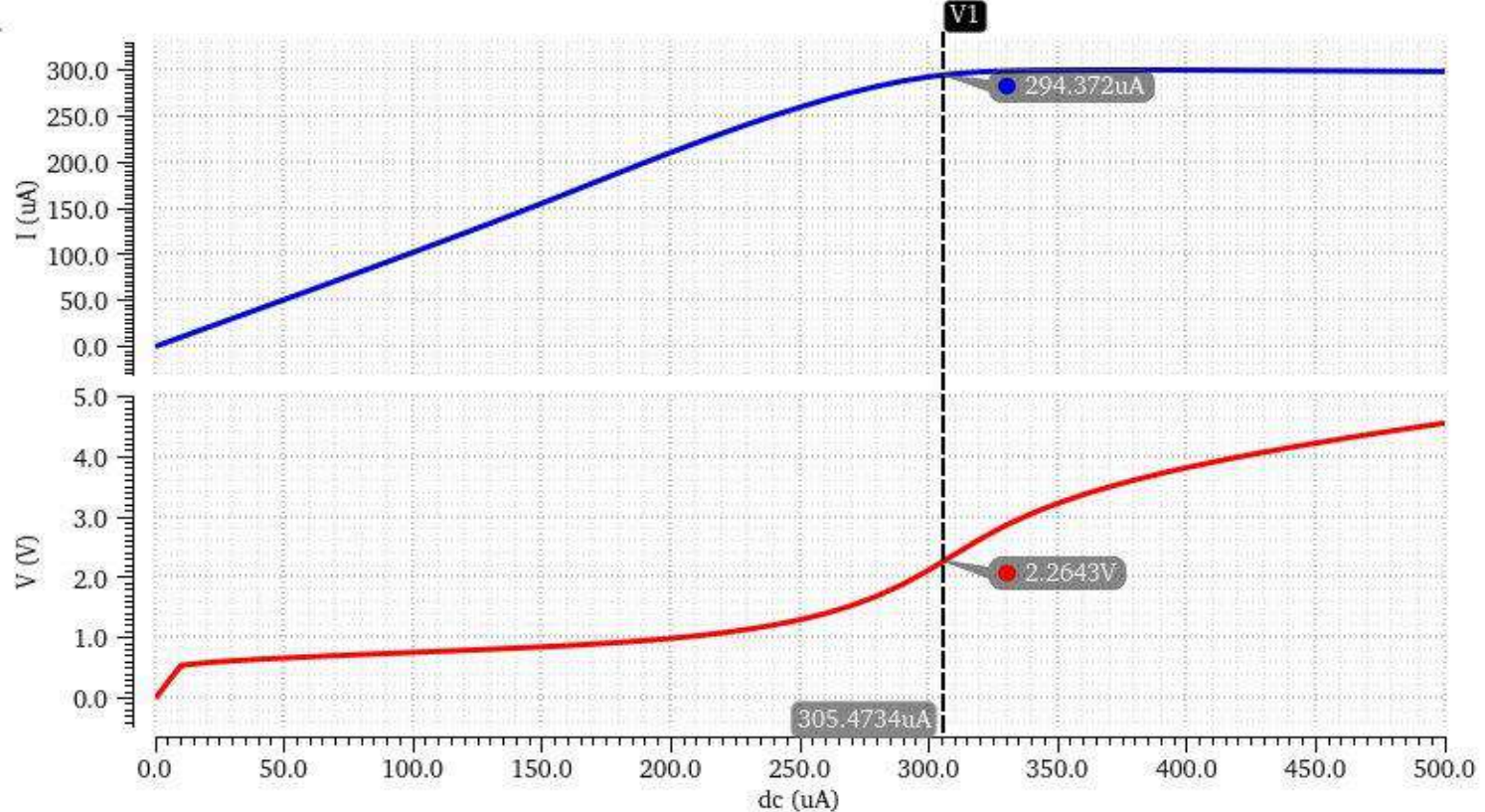
Wed Jun 24 17:12:12 2026 1

DC Response

Name ... V1

■ /NM1/D 294.372uA

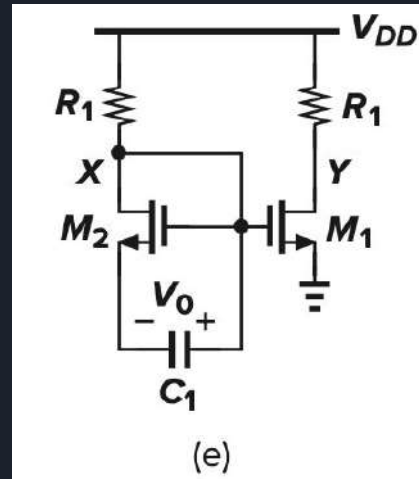
■ /X 2.2643V



Problem 5.16-e

5.16. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.59. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

Before (circuit switched on)

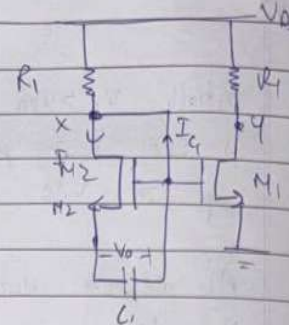
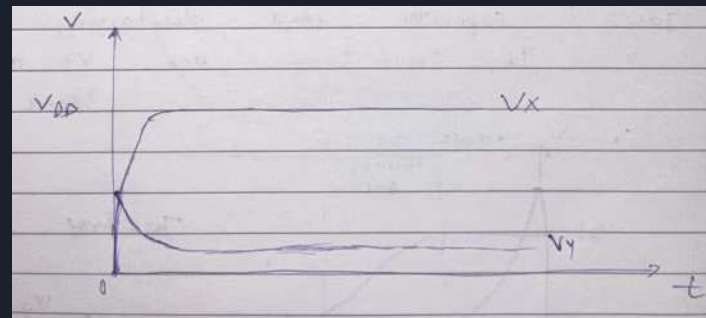
5.16.4 e.p. Initially, V_x will be low ($\sim 0V$)
 V_y will be finite value

$\therefore$ There is no path
 to ground for the current
 I_{M2} ,

$I_{M2} = I_{M1}$
 and I_{R1} (left branch) ≈ 0

$\therefore V_x$ gets pulled to V_{DD}

$\therefore$ As $V_x \uparrow$, V_{GS1} also increases
 $\therefore I_{M1} \uparrow$, V_y decreases
 (M_1 gets turned on strongly and pulls V_y down)

Cadence Virtuoso Schematic

Cadence ADE-L DC Simulation for V_x and V_y

Thu Jun 25 16:33:34
2026

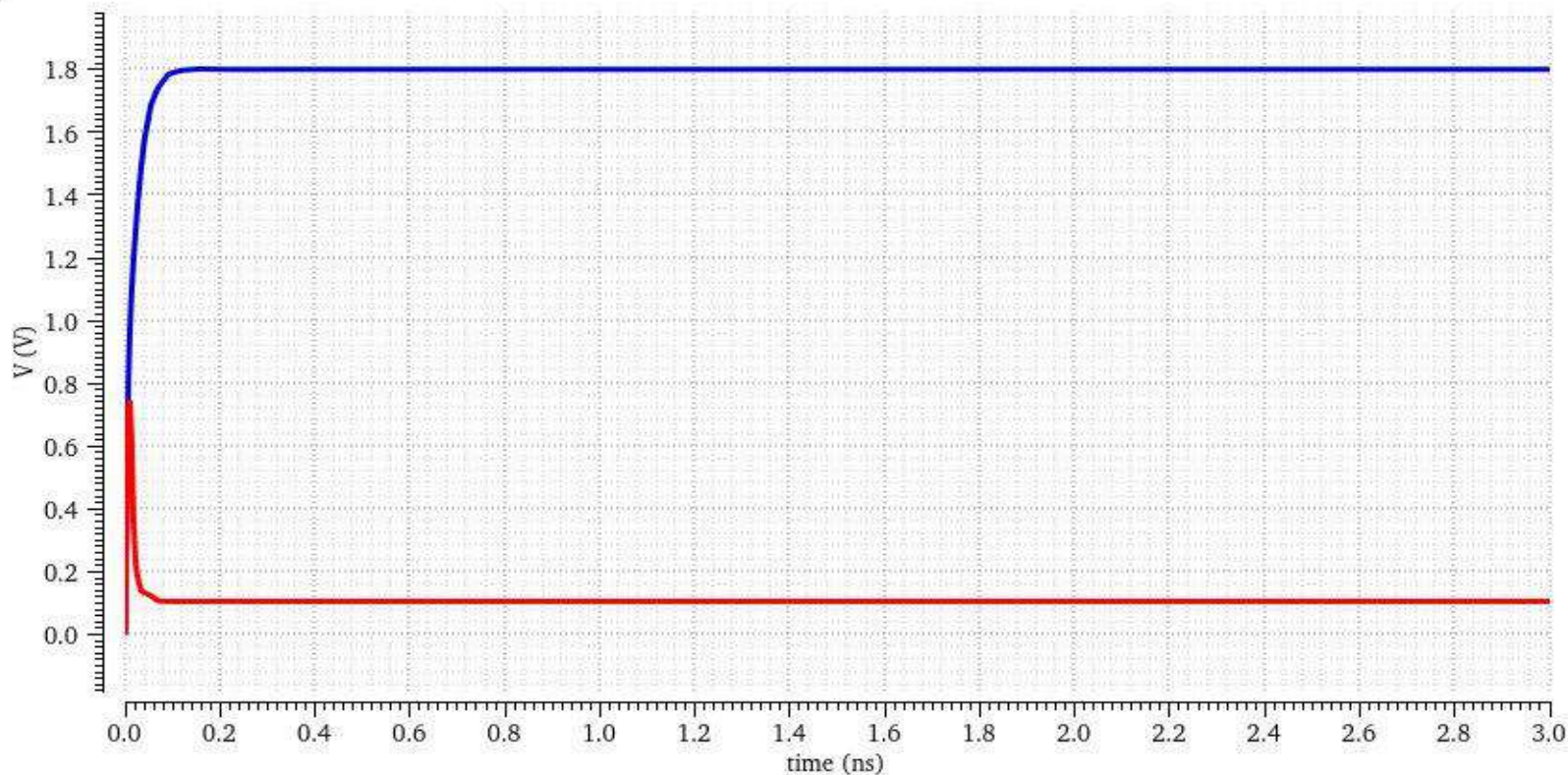
1

Transient Response

Name	Vis
V_x	☒
V_y	☒

V_x

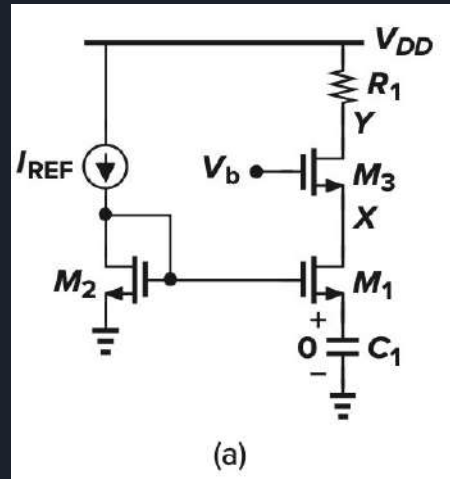
V_y



Problem 5.17-a

5.17. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.60. Assume the transistors in each circuit are identical.

Diagram -



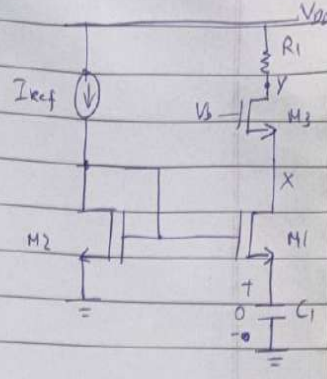
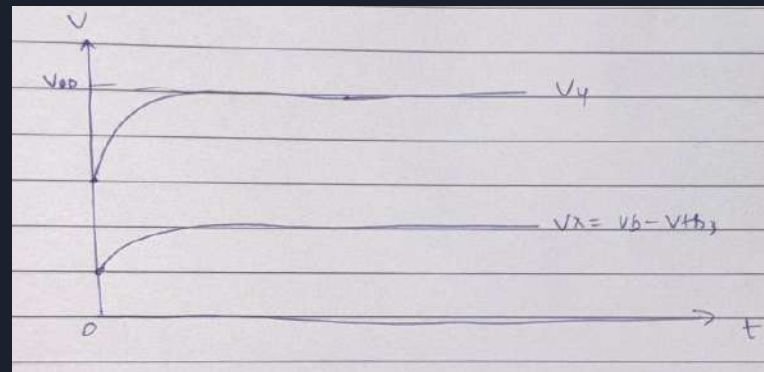
Handwritten Solution

Initially capacitor acts as a short circuit to ground and $V_Y > V_X$

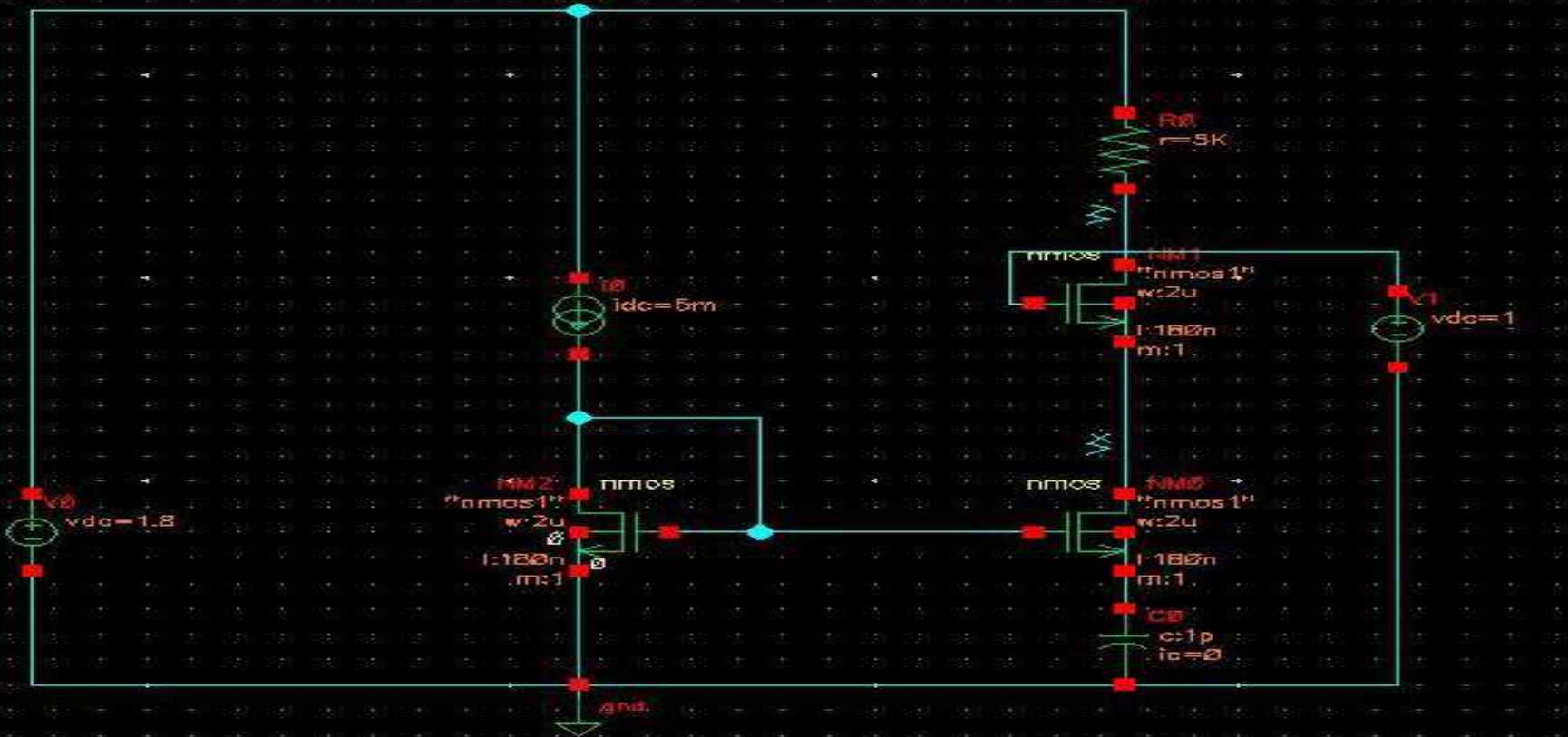
As I_{M1} flows, capacitor starts charging $V_{GS1} \downarrow$

$\therefore I_{M1} \downarrow, \therefore I_{M3} \downarrow$
 $\therefore V_Y$ increases and V_X increases

Once V_X increases so as to turn $M3$ off
 $V_Y = V_{DD}$ and $V_X = V_B - V_{th3}$

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

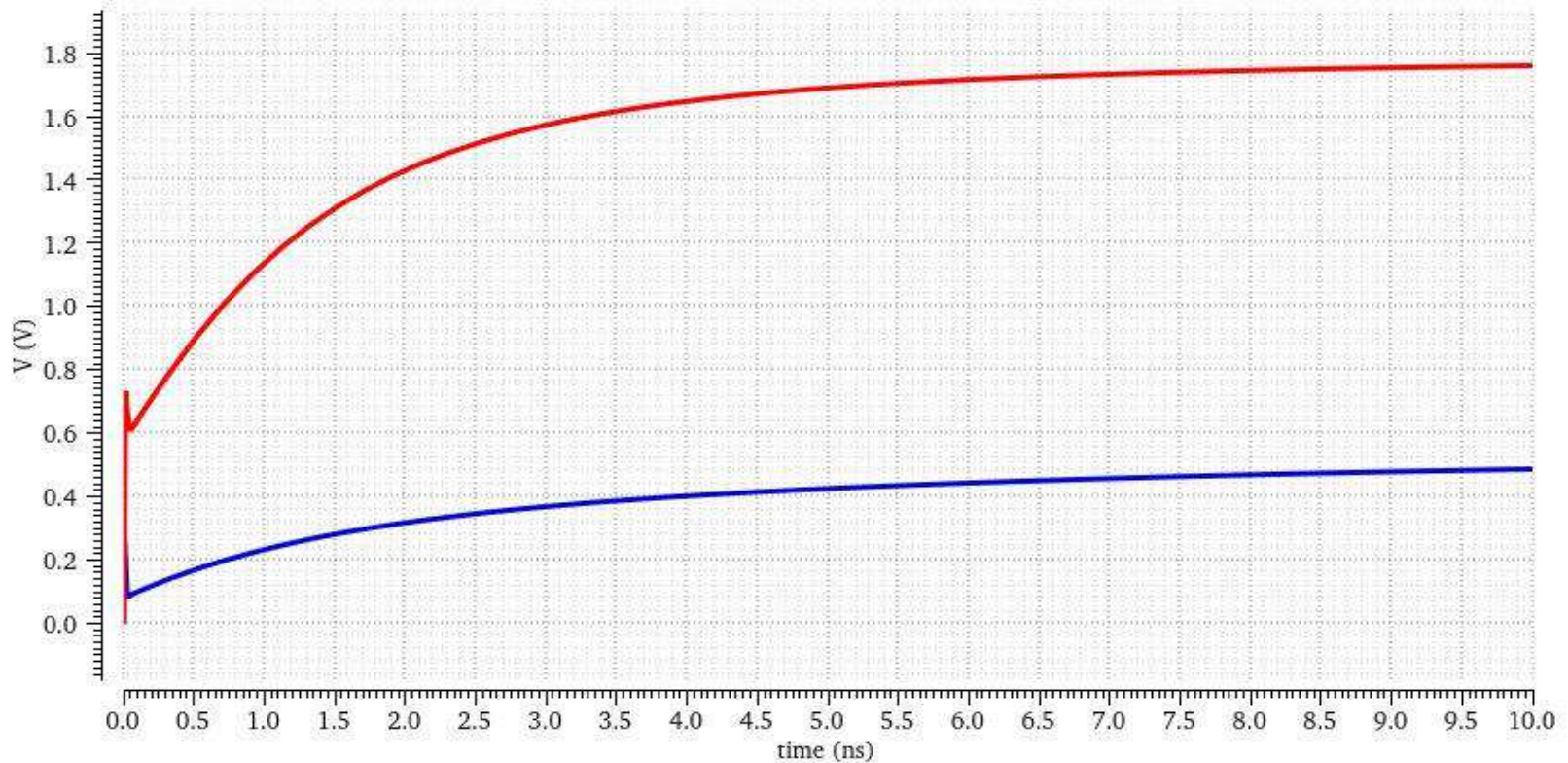
Transient Response

Thu Jun 25 16:51:18
2026

1

Name Vis

☒ /Vx
☒ /Vy





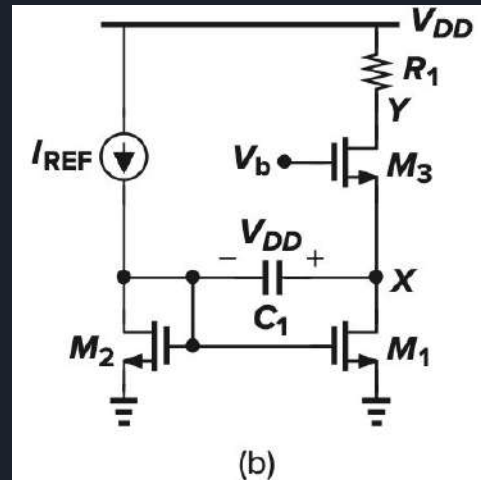
Observations

- At $t=0^+$, the uncharged capacitor keeps M1 and M3 strongly ON, creating a low-resistance path.
- The capacitor draws a large transient charging current through R1, causing a large IR drop and a sharp dip in V_y .
- As the capacitor charges, the branch current decreases, the voltage drop across R1 vanishes, and V_y reaches to VDD.

Problem 5.17-b

5.17. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.60. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

5.17.6

Initially $V_x > V_{DD}$ and M_3 is OFF, $\therefore V_y$ gets pulled up to V_{DD} .

As Capacitor discharges V_x decreases, and M_3 starts to turn ON.

Once $V_D - V_x > V_{th}$, M_3 turns on in saturation as V_y is still very high at that instant.

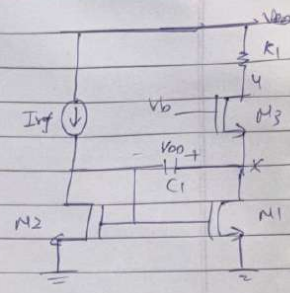
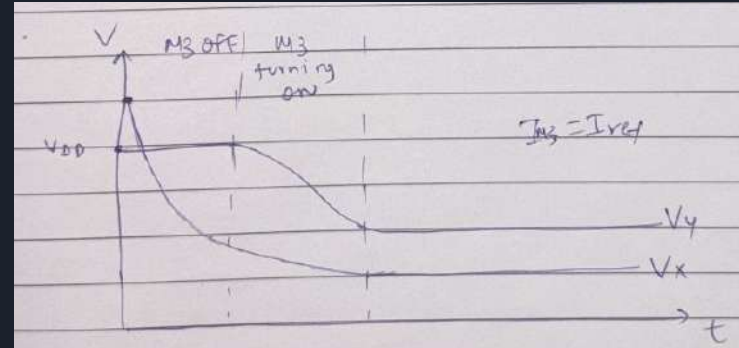
$\therefore M_3$ turns on in saturation, it pushes a large current into node X, trying to pull it up, but M_1 is trying to pull X to ground.

$\therefore$ slope of V_x decreases.

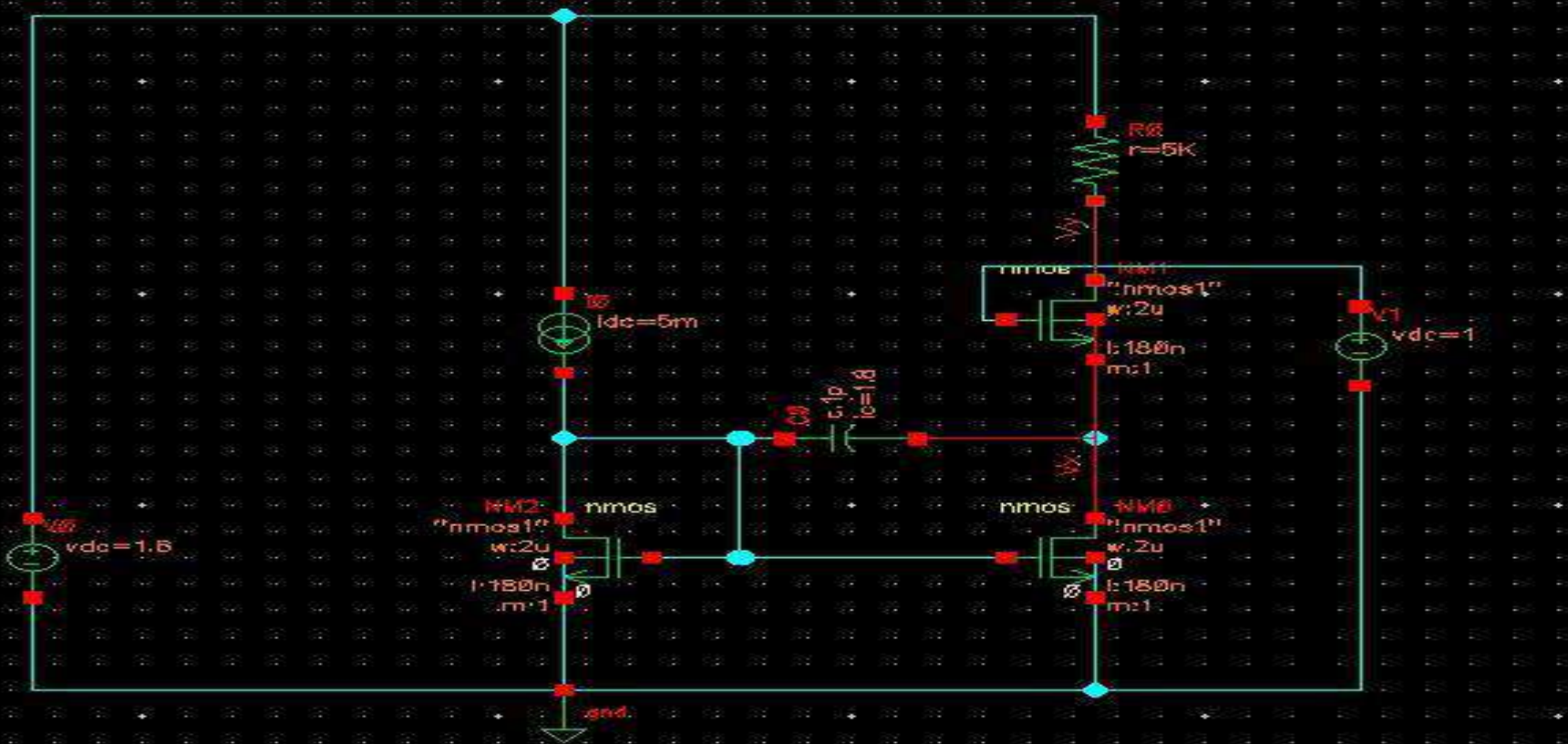
As V_x decreases $I_{M3} \uparrow$, $\therefore V_y \downarrow$.

Once V_y decreases enough as to put M_3 in triode region, I_{M3} reduces and matches I_{M1} . Capacitor stops discharging.

$I_{M3} = I_{M1} = I_{ref}$, and V_x and V_y become constant.

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for V_x and V_y

Thu Jun 25 16:42:56
2026

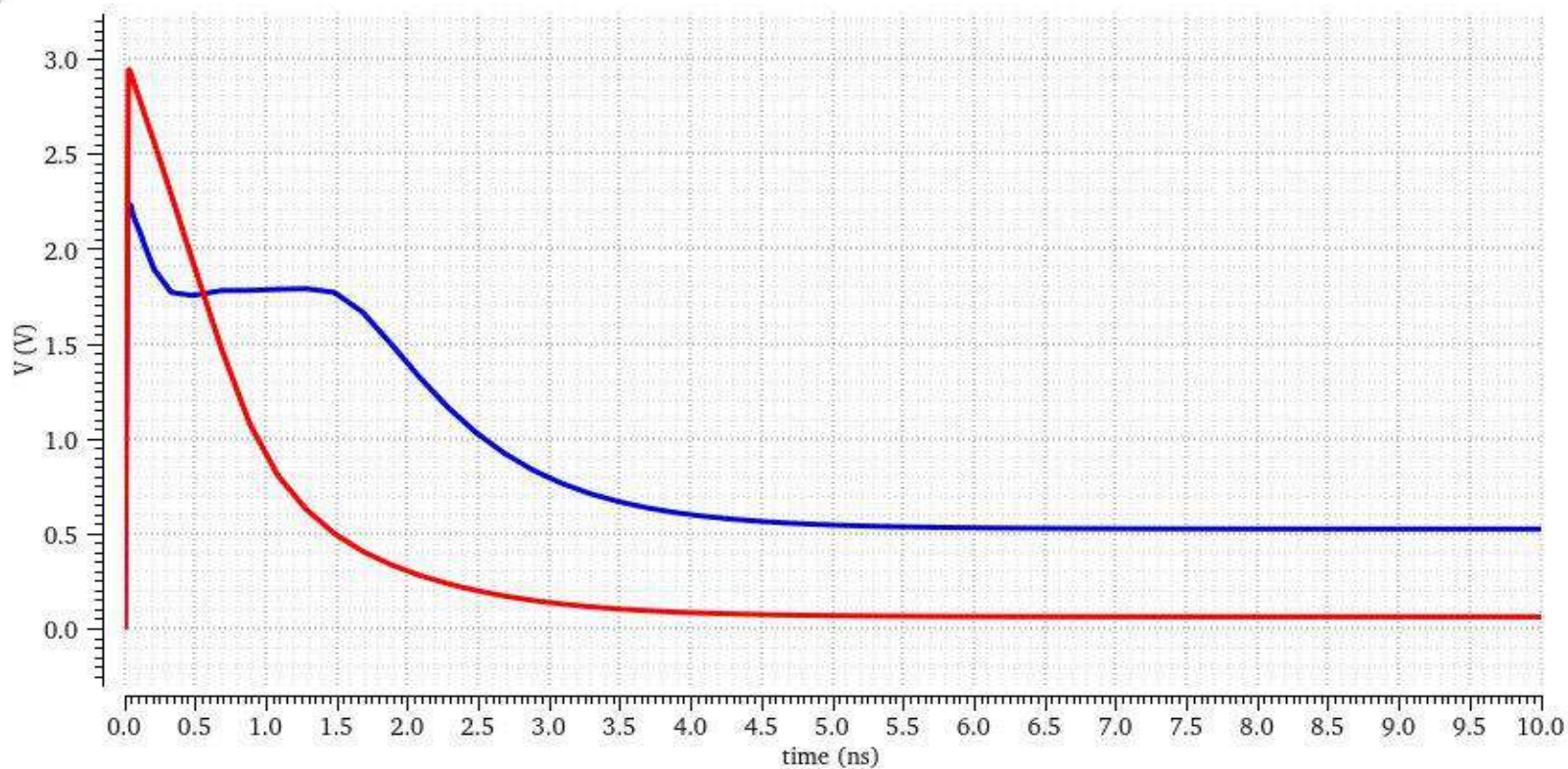
1

Transient Response

Name Vis

$/V_y$

$/V_x$





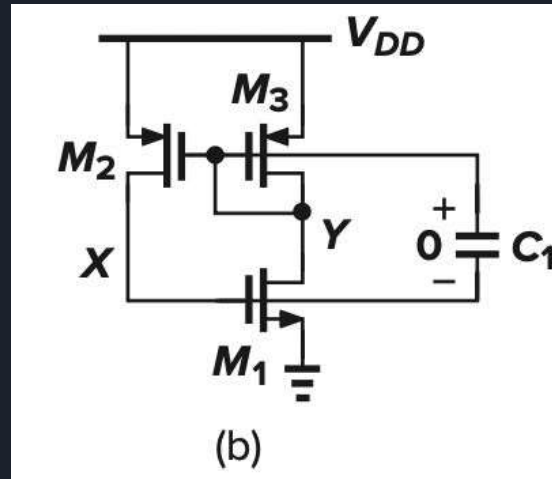
Observations

- **Node X:** The pre-charged capacitor C1 (initially charged to VDD) creates a bootstrap effect at startup, causing Node X to briefly rise above the supply voltage (≈ 3.0 V). The stored charge in C1 then gradually discharges through M1, bringing the voltage back to its steady-state value.
- **Node Y:** The rapid voltage increase at Node X is coupled to Node Y through the parasitic capacitances (CGD3 and CSD3) of M3, causing a brief overshoot (≈ 2.2 V). As M3 remains off and the parasitic capacitances store very little charge, the excess charge quickly discharges through R1, returning Node Y to VDD.

Problem 5.18-b

5.18. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.61. Assume the transistors in each circuit are identical.

Diagram -



Handwritten Solution

Initially, capacitor is short ckt

$$V_x = V_y$$

M_2 charges the capacitor

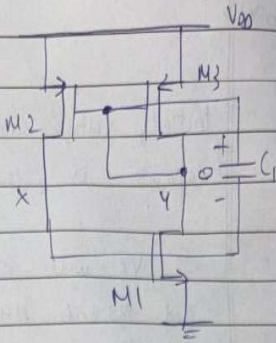
Once capacitor starts to

charge, V_x increases

exponentially which

increases V_{gs} , turning M_1 ON strongly

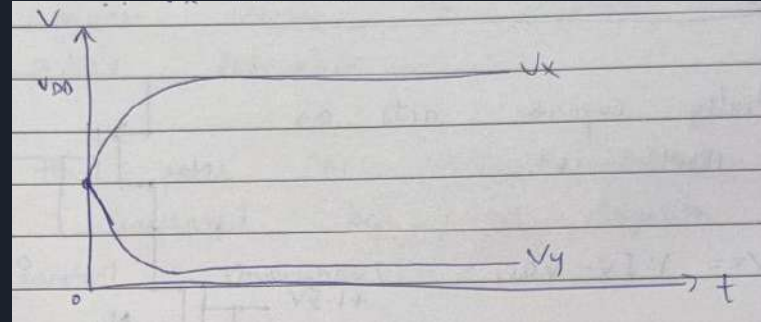
V_y gets pulled down



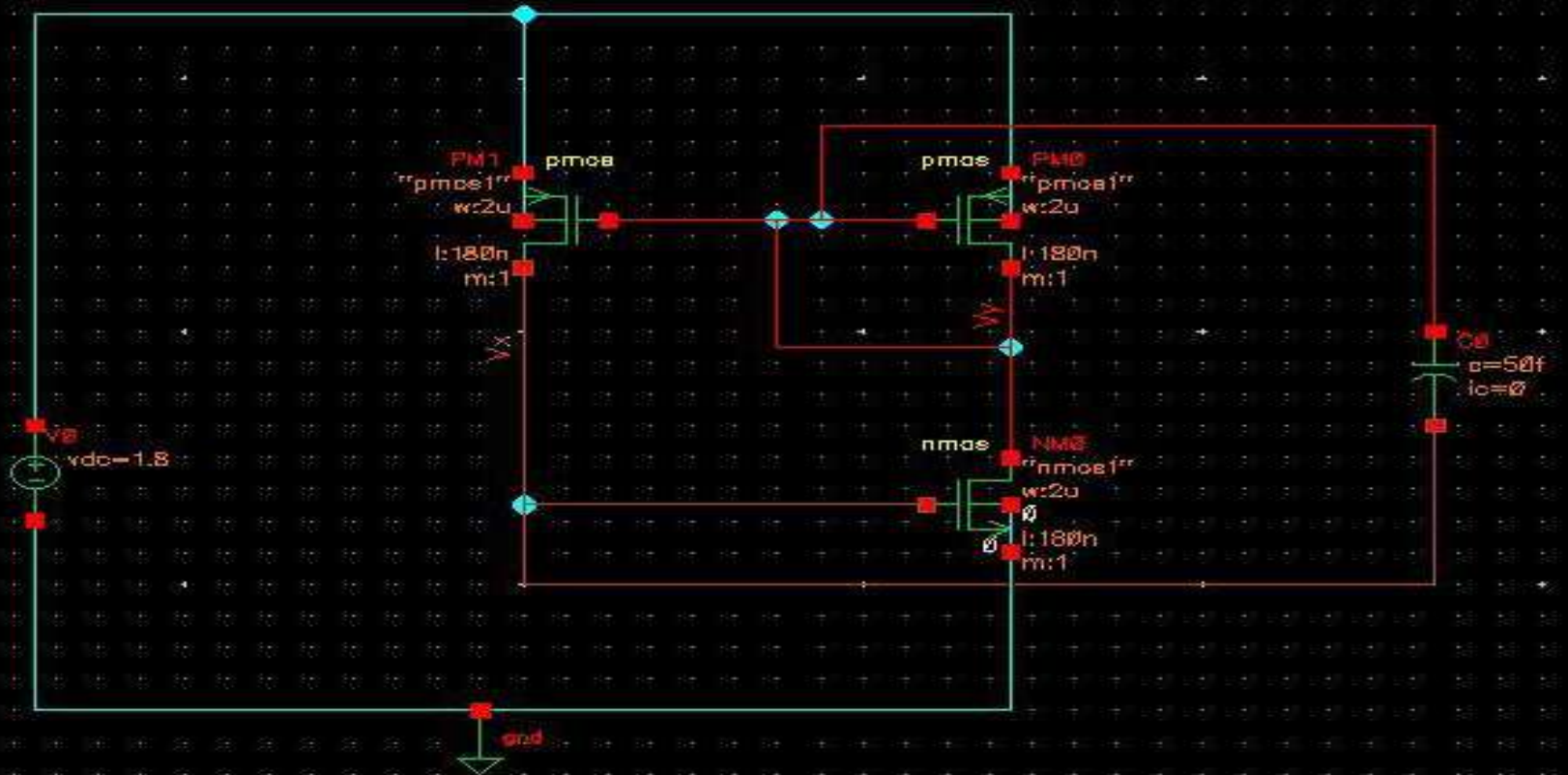
Once capacitor is completely charged

V_x gets pulled to V_{DD} (V_{gs} is very high)

and V_y gets pulled to a low potential
 V_x and V_y become constant.



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

Transient Response

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2026

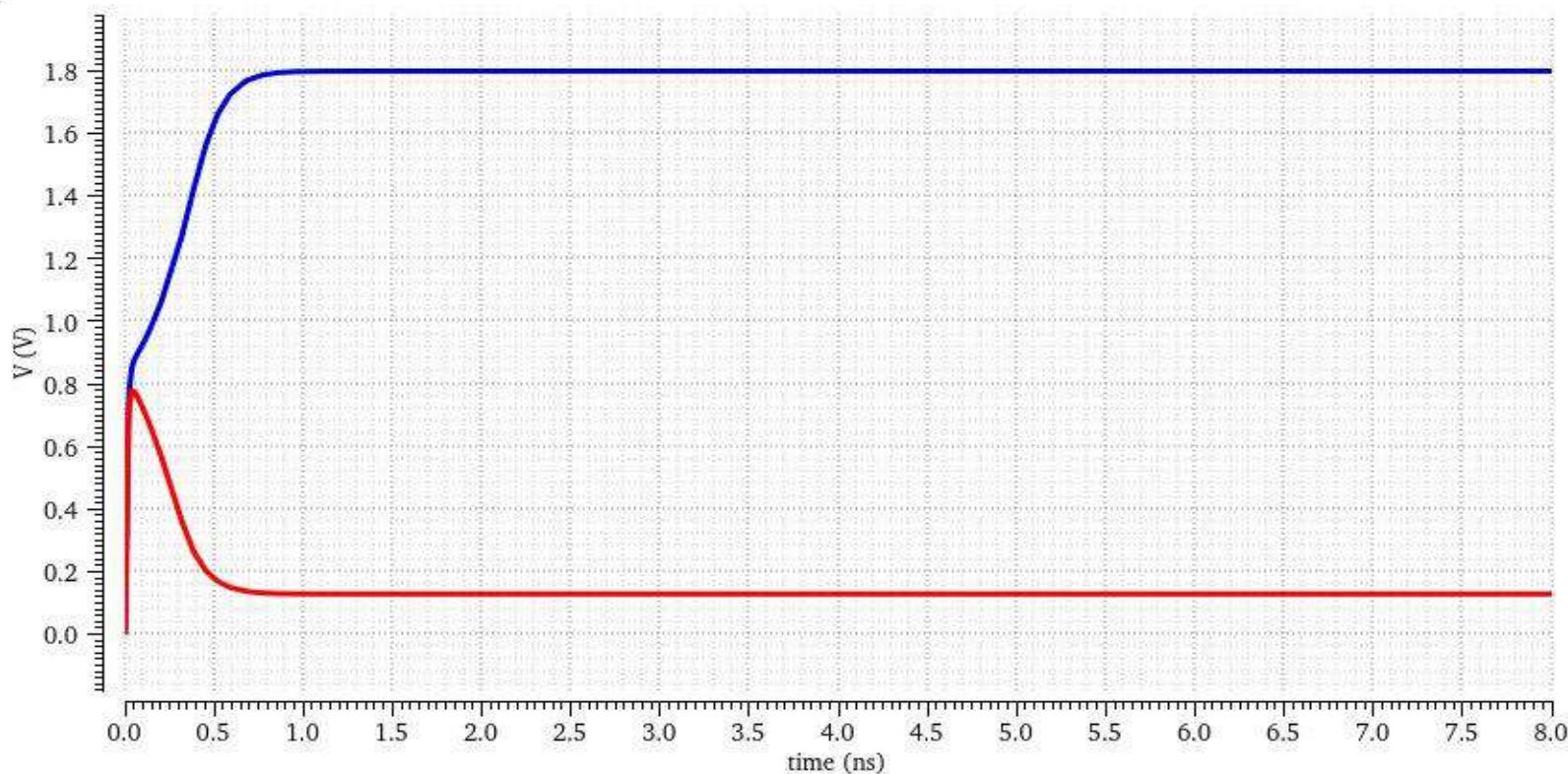
1

Name Vis

■ /Vx



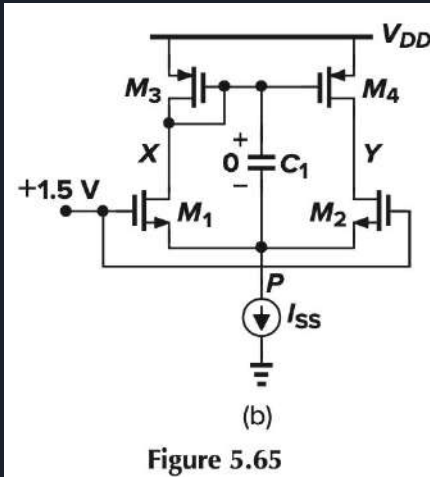
■ /Vy



Problem 5.22-b

5.22. Sketch V_X and V_Y for each circuit in Fig. 5.65 as a function of time. The initial voltage across C_1 is shown.

Diagram -



Handwritten Solution

522964

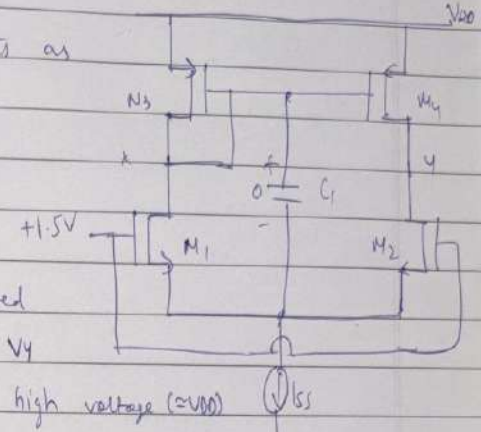
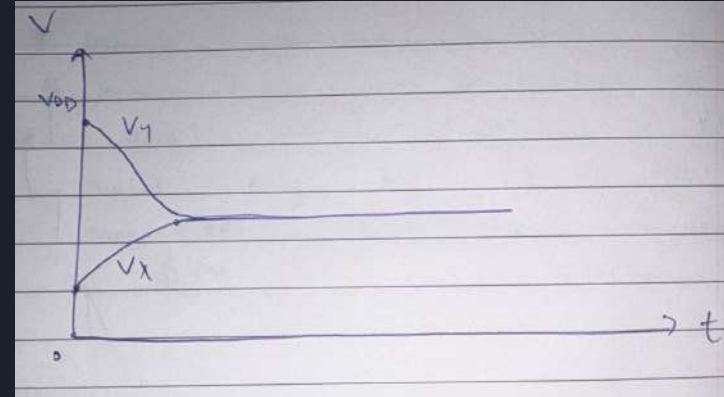
Initially capacitor acts as a short ckt.

$\therefore V_x = 1.5V - V_{GS1}$

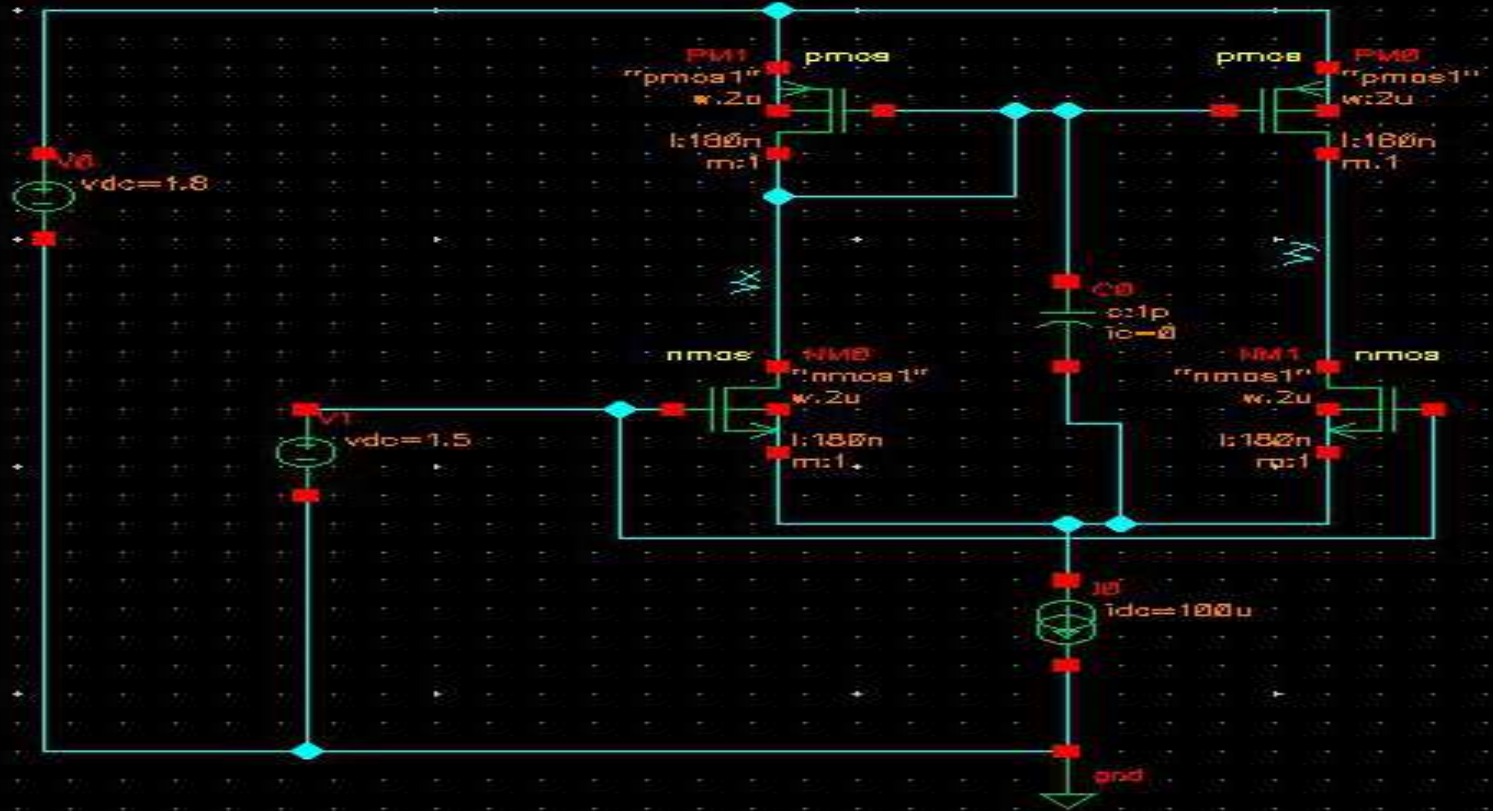
M_3 and M_1 are turned on strongly, $\therefore V_y$ is pulled up to a high voltage ($\approx V_{DD}$)

As capacitor charges, V_x will increase, $\therefore M_4$ will become less strong and V_y will start to decrease.

Once M_1 enters saturation V_x and V_y become constant

Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for V_x and V_y

Fri Jul 3 15:40:17
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1

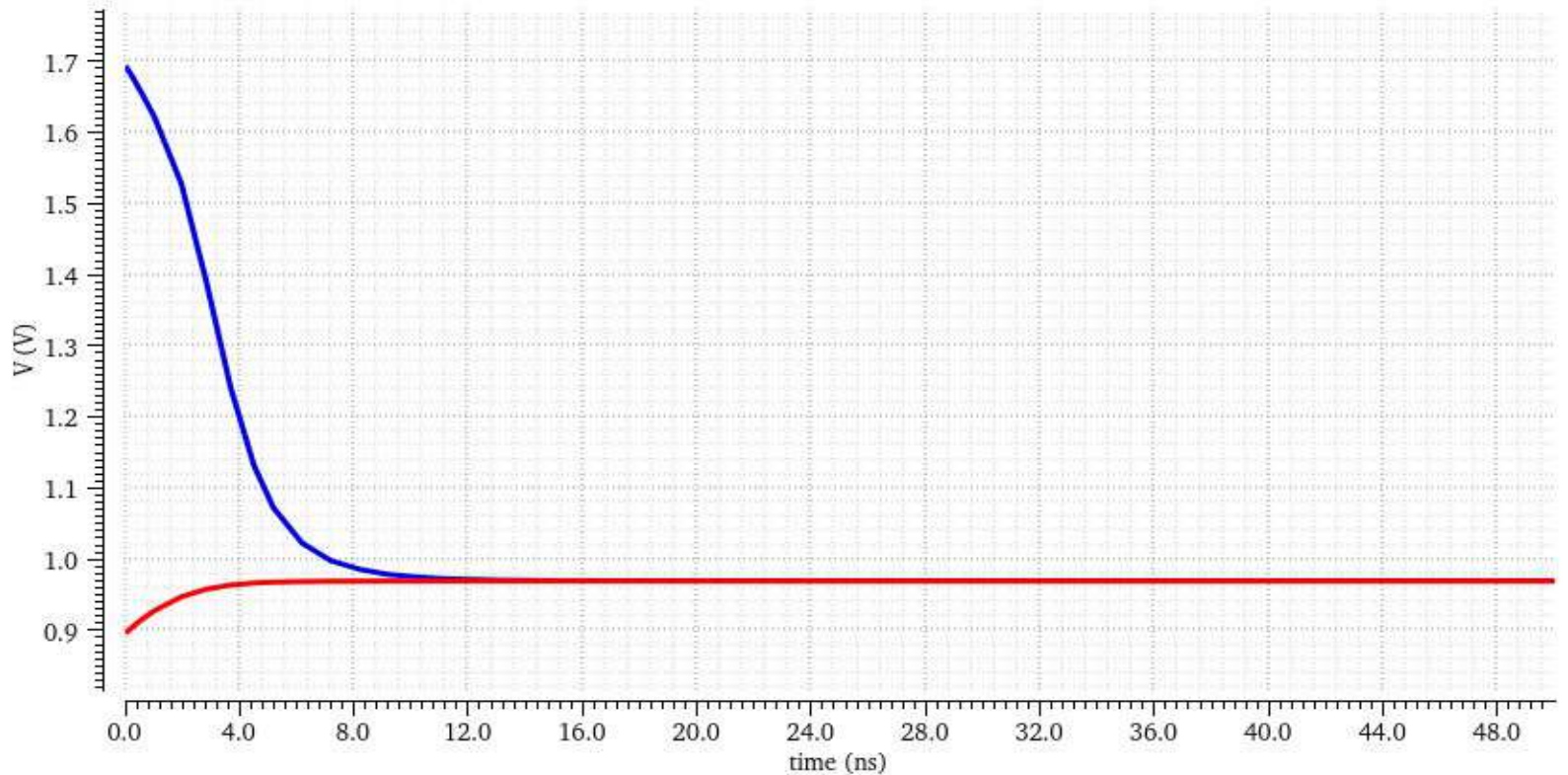
Transient Response

Name	Vis
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V_y



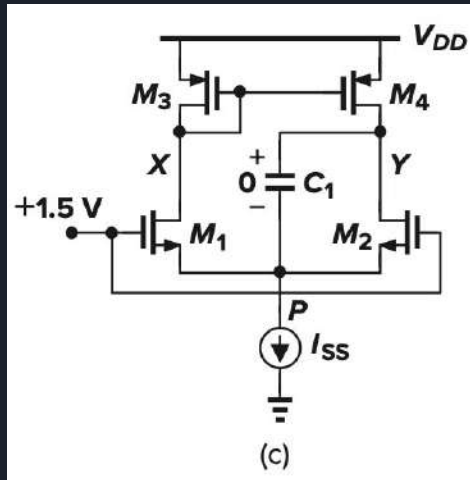
V_x



Problem 5.22-c

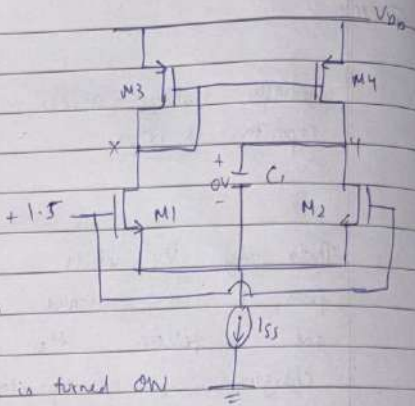
5.22. Sketch V_X and V_Y for each circuit in Fig. 5.65 as a function of time. The initial voltage across C_1 is shown.

Diagram -



Handwritten Solution

5.22/10/19

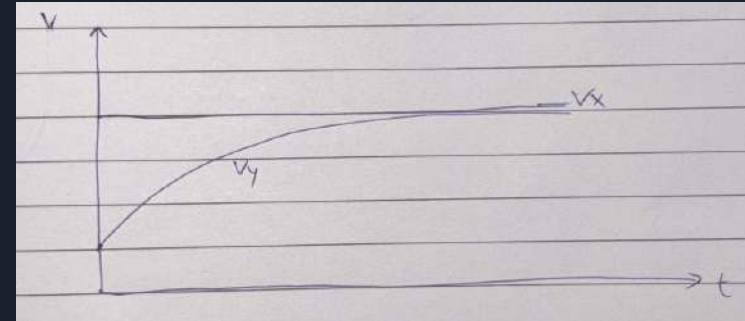


Initially the capacitor will act as a short circuit
 $\therefore V_{OS}$ is very low

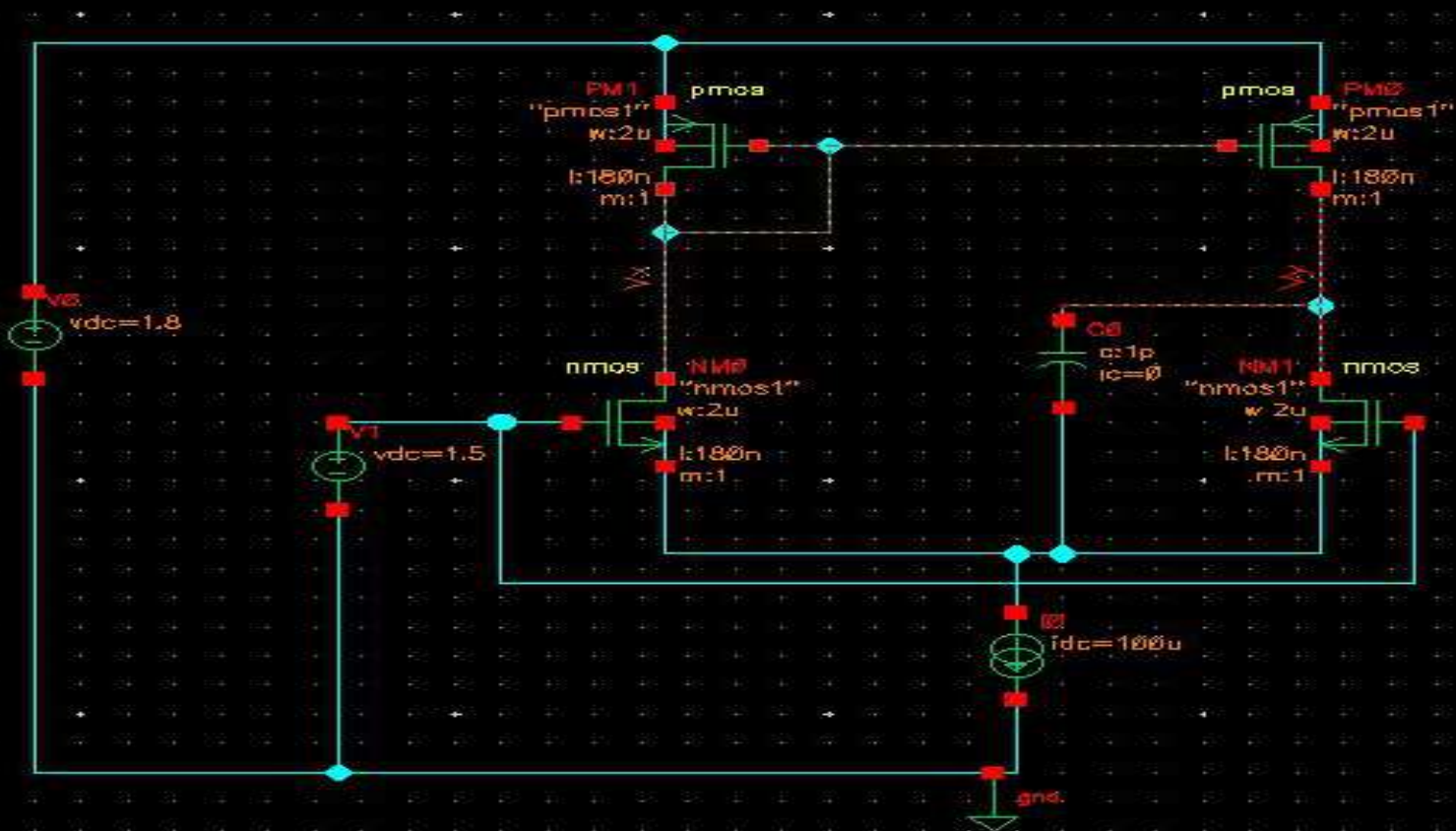
M_2 is in triode region
 I_{M2} is very low
 But V_{GS2} is high, $\therefore M_2$ is turned ON
 $\therefore V_y$ is pulled down

And I_{M4} is used to charge the capacitor
 As capacitor charges, V_y increases and M_2 enters saturation

After this point V_y becomes constant
 V_x is unaffected by the capacitor, $\therefore$ it stays constant throughout



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vx and Vy

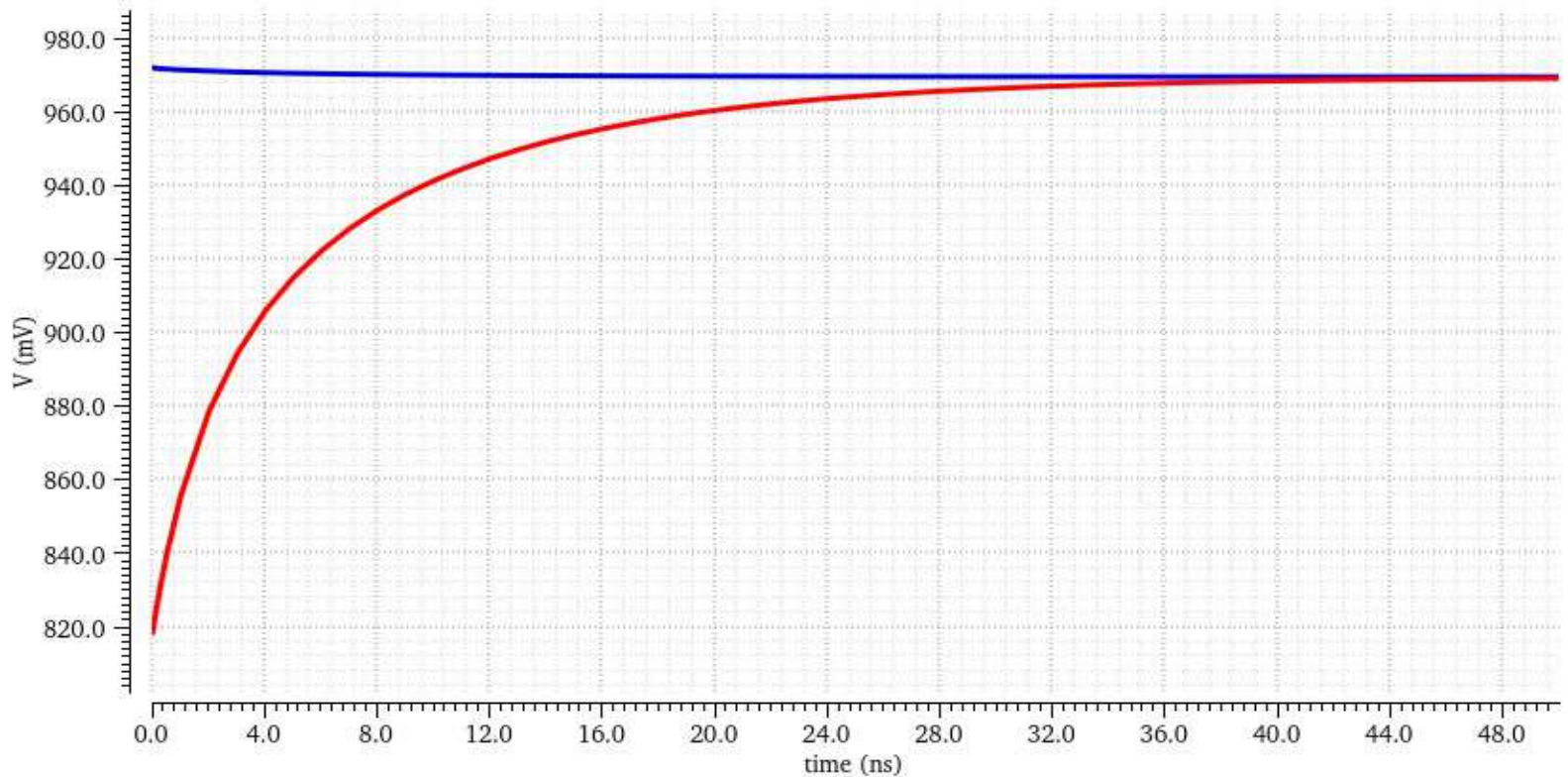
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2026

1

Transient Response

Name	Vis
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☒ /Vx	☒
☒ /Vy	☒





THANK YOU